

Resultados del Modelo

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CONVOLUTIONAL NEURAL NETWORK (CNN)

I. ENTRENAMIENTO

- El algoritmo de entrenamiento, `train.py`, permite los siguientes argumentos:

```
def main():
    parser = argparse.ArgumentParser(description="Entrena el modelo HD-only para detección de
    parser.add_argument("data_dir", help="Directorio raíz con subcarpetas de clases")
    parser.add_argument("-e", "--epochs", type=int, default=20, help="Número máximo de épocas")
    parser.add_argument("-a", "--alpha", type=float, default=0.50, help="Alpha")
    parser.add_argument("-g", "--gamma", type=float, default=3.0, help="Gamma")
    parser.add_argument("-lt", "--loss_type", type=str, choices=["focal_loss", "binary_crossentropy"])
    parser.add_argument("-rgb", "--rgb", action='store_true', default=False, help="Es RGB?")
    parser.add_argument("-mt", "--model_type", type=str, required=True, choices=["cnn", "rf"])
    parser.add_argument("-t", "--threshold", type=float, default=0.5, help="Umbral de decisión")
    args = parser.parse_args()
    run_training(args.data_dir, args.epochs, args.loss_type, args.rgb, args.alpha, args.gamma)
```

1. FOCAL LOSS MULTIESPECTRAL (FL MS)

1.1 Comando:

```
python src\pest_detection_models\train.py data\multiespectral\TTADDA-dataset\TTADDA_NARO_2026
```

1.2 Consola:

```
2026-03-24 12:35:57.871010: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-24 12:35:59.238525: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260324_1236] ---
```

```
--- [T=0.00s] ---
```

```
init. Iniciando entrenamiento MULTIESPECTRAL con perdida Focal Loss
```

```

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=0.00s] ---
1. 1. Extrayendo datos

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=0.01s] ---
1. Carga de datos cnn Multiespectral

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=0.07s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=0.08s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=0.08s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=57.09s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)
Shape train: (532, 224, 224, 5)
2026-03-24 12:36:58.234818: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with CPU instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op
Pesos de Clase Calculados: Plaga (0): 1.11, Sana (1): 1.55
-----

```

DEBUG - Conteo: Plaga: 360, Sana: 172

RESULTADO - Pesos Finales: Plaga (0): 1.11, Sana (1): 1.55

Epoch 1/80

2026-03-24 12:36:58.619713: E tensorflow/core/framework/node_def_util.cc:680] NodeDef menti

17/17 0s 1s/step - accuracy: 0.4025 - auc: 0.5092 - loss: 0.0907 - precision: 0.

Epoch 1: val_recall improved from None to 0.00000, saving model to C:\workspace\hd-image-pe

17/17 29s 1s/step - accuracy: 0.4925 - auc: 0.6282 - loss: 0.0722 - precision: 0

Epoch 2/80

17/17 0s 1s/step - accuracy: 0.7315 - auc: 0.8180 - loss: 0.0487 - precision: 0.

Epoch 2: val_recall did not improve from 0.00000

17/17 24s 1s/step - accuracy: 0.7274 - auc: 0.8069 - loss: 0.0461 - precision: 0

Epoch 3/80

17/17 0s 1s/step - accuracy: 0.7564 - auc: 0.8765 - loss: 0.0344 - precision: 0.

Epoch 3: val_recall did not improve from 0.00000

17/17 24s 1s/step - accuracy: 0.7500 - auc: 0.8622 - loss: 0.0340 - precision: 0

Epoch 4/80

17/17 0s 1s/step - accuracy: 0.7631 - auc: 0.8713 - loss: 0.0300 - precision: 0.

Epoch 4: val_recall did not improve from 0.00000

17/17 24s 1s/step - accuracy: 0.7575 - auc: 0.8522 - loss: 0.0306 - precision: 0

Epoch 5/80

17/17 0s 1s/step - accuracy: 0.7600 - auc: 0.8651 - loss: 0.0284 - precision: 0.

Epoch 5: val_recall did not improve from 0.00000

17/17 24s 1s/step - accuracy: 0.7538 - auc: 0.8616 - loss: 0.0286 - precision: 0

Epoch 6/80

17/17 0s 1s/step - accuracy: 0.7783 - auc: 0.8798 - loss: 0.0253 - precision: 0.

Epoch 6: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-05.

Epoch 6: val_recall did not improve from 0.00000

17/17 23s 1s/step - accuracy: 0.7575 - auc: 0.8602 - loss: 0.0275 - precision: 0

Epoch 7/80

17/17 0s 1s/step - accuracy: 0.7774 - auc: 0.8851 - loss: 0.0239 - precision: 0.

Epoch 7: val_recall did not improve from 0.00000

17/17 24s 1s/step - accuracy: 0.7744 - auc: 0.8642 - loss: 0.0269 - precision: 0

Epoch 8/80

17/17 0s 1s/step - accuracy: 0.7735 - auc: 0.8866 - loss: 0.0244 - precision: 0.

Epoch 8: val_recall did not improve from 0.00000

17/17 23s 1s/step - accuracy: 0.7707 - auc: 0.8668 - loss: 0.0259 - precision: 0

Epoch 9/80

17/17 0s 1s/step - accuracy: 0.7866 - auc: 0.8763 - loss: 0.0242 - precision: 0.

Epoch 9: val_recall did not improve from 0.00000

17/17 23s 1s/step - accuracy: 0.7763 - auc: 0.8655 - loss: 0.0258 - precision: 0

Epoch 10/80
17/17 0s 1s/step - accuracy: 0.7884 - auc: 0.8841 - loss: 0.0236 - precision: 0.
Epoch 10: val_recall improved from 0.00000 to 0.02326, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7707 - auc: 0.8591 - loss: 0.0265 - precision: 0.
Epoch 11/80
17/17 0s 1s/step - accuracy: 0.8054 - auc: 0.8806 - loss: 0.0233 - precision: 0.
Epoch 11: val_recall improved from 0.02326 to 0.06977, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7838 - auc: 0.8617 - loss: 0.0258 - precision: 0.
Epoch 12/80
17/17 0s 1s/step - accuracy: 0.8027 - auc: 0.8664 - loss: 0.0249 - precision: 0.
Epoch 12: val_recall improved from 0.06977 to 0.11628, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7838 - auc: 0.8621 - loss: 0.0255 - precision: 0.
Epoch 13/80
17/17 0s 1s/step - accuracy: 0.7941 - auc: 0.8981 - loss: 0.0212 - precision: 0.
Epoch 13: val_recall improved from 0.11628 to 0.23256, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7876 - auc: 0.8780 - loss: 0.0237 - precision: 0.
Epoch 14/80
17/17 0s 1s/step - accuracy: 0.8026 - auc: 0.9010 - loss: 0.0206 - precision: 0.
Epoch 14: val_recall improved from 0.23256 to 0.37209, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7895 - auc: 0.8760 - loss: 0.0235 - precision: 0.
Epoch 15/80
17/17 0s 1s/step - accuracy: 0.7863 - auc: 0.8893 - loss: 0.0221 - precision: 0.
Epoch 15: val_recall improved from 0.37209 to 0.51163, saving model to C:\workspace\hd-image
17/17 23s 1s/step - accuracy: 0.7801 - auc: 0.8626 - loss: 0.0252 - precision: 0.
Epoch 16/80
17/17 0s 1s/step - accuracy: 0.7813 - auc: 0.8911 - loss: 0.0216 - precision: 0.
Epoch 16: val_recall improved from 0.51163 to 0.58140, saving model to C:\workspace\hd-image
17/17 42s 1s/step - accuracy: 0.7707 - auc: 0.8651 - loss: 0.0244 - precision: 0.
Epoch 17/80
17/17 0s 1s/step - accuracy: 0.7800 - auc: 0.8784 - loss: 0.0232 - precision: 0.
Epoch 17: val_recall improved from 0.58140 to 0.62791, saving model to C:\workspace\hd-image
17/17 40s 1s/step - accuracy: 0.7782 - auc: 0.8656 - loss: 0.0247 - precision: 0.
Epoch 18/80
17/17 0s 1s/step - accuracy: 0.7817 - auc: 0.8847 - loss: 0.0221 - precision: 0.
Epoch 18: val_recall improved from 0.62791 to 0.65116, saving model to C:\workspace\hd-image
17/17 24s 1s/step - accuracy: 0.7782 - auc: 0.8591 - loss: 0.0256 - precision: 0.
Epoch 19/80
17/17 0s 1s/step - accuracy: 0.7991 - auc: 0.8801 - loss: 0.0229 - precision: 0.
Epoch 19: val_recall did not improve from 0.65116
17/17 23s 1s/step - accuracy: 0.7838 - auc: 0.8639 - loss: 0.0249 - precision: 0.
Epoch 20/80
17/17 0s 1s/step - accuracy: 0.7814 - auc: 0.8898 - loss: 0.0215 - precision: 0.
Epoch 20: val_recall improved from 0.65116 to 0.74419, saving model to C:\workspace\hd-image

```

17/17          24s 1s/step - accuracy: 0.7744 - auc: 0.8700 - loss: 0.0242 - precision: 0.
Epoch 21/80
17/17          0s 1s/step - accuracy: 0.7935 - auc: 0.8798 - loss: 0.0225 - precision: 0.
Epoch 21: val_recall improved from 0.74419 to 0.79070, saving model to C:\workspace\hd-image
17/17          23s 1s/step - accuracy: 0.7801 - auc: 0.8684 - loss: 0.0243 - precision: 0.
Epoch 22/80
17/17          0s 1s/step - accuracy: 0.7951 - auc: 0.8969 - loss: 0.0209 - precision: 0.
Epoch 22: val_recall improved from 0.79070 to 0.81395, saving model to C:\workspace\hd-image
17/17          23s 1s/step - accuracy: 0.7895 - auc: 0.8776 - loss: 0.0233 - precision: 0.
Epoch 23/80
17/17          0s 1s/step - accuracy: 0.7887 - auc: 0.8885 - loss: 0.0213 - precision: 0.
Epoch 23: val_recall improved from 0.81395 to 0.88372, saving model to C:\workspace\hd-image
17/17          23s 1s/step - accuracy: 0.7763 - auc: 0.8751 - loss: 0.0231 - precision: 0.
Epoch 24/80
17/17          0s 1s/step - accuracy: 0.7962 - auc: 0.8713 - loss: 0.0243 - precision: 0.
Epoch 24: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7838 - auc: 0.8548 - loss: 0.0258 - precision: 0.
Epoch 25/80
17/17          0s 1s/step - accuracy: 0.8035 - auc: 0.8910 - loss: 0.0209 - precision: 0.
Epoch 25: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7932 - auc: 0.8698 - loss: 0.0239 - precision: 0.
Epoch 26/80
17/17          0s 1s/step - accuracy: 0.7914 - auc: 0.8955 - loss: 0.0208 - precision: 0.
Epoch 26: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7820 - auc: 0.8696 - loss: 0.0239 - precision: 0.
Epoch 27/80
17/17          0s 1s/step - accuracy: 0.7943 - auc: 0.8936 - loss: 0.0219 - precision: 0.
Epoch 27: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7820 - auc: 0.8749 - loss: 0.0240 - precision: 0.
Epoch 28/80
17/17          0s 1s/step - accuracy: 0.7905 - auc: 0.8831 - loss: 0.0220 - precision: 0.
Epoch 28: val_recall did not improve from 0.88372
17/17          24s 1s/step - accuracy: 0.7820 - auc: 0.8656 - loss: 0.0240 - precision: 0.
Epoch 29/80
17/17          0s 1s/step - accuracy: 0.7847 - auc: 0.8970 - loss: 0.0215 - precision: 0.
Epoch 29: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7782 - auc: 0.8740 - loss: 0.0242 - precision: 0.
Epoch 30/80
17/17          0s 1s/step - accuracy: 0.7961 - auc: 0.8871 - loss: 0.0211 - precision: 0.
Epoch 30: val_recall did not improve from 0.88372
17/17          23s 1s/step - accuracy: 0.7838 - auc: 0.8700 - loss: 0.0232 - precision: 0.
Epoch 31/80
17/17          0s 1s/step - accuracy: 0.7682 - auc: 0.8823 - loss: 0.0231 - precision: 0.

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Epoch 31: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7707 - auc: 0.8676 - loss: 0.0248 - precision: 0.
Epoch 32/80
17/17 0s 1s/step - accuracy: 0.7951 - auc: 0.8979 - loss: 0.0210 - precision: 0.
Epoch 32: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7801 - auc: 0.8717 - loss: 0.0238 - precision: 0.
Epoch 33/80
17/17 0s 1s/step - accuracy: 0.7939 - auc: 0.8833 - loss: 0.0220 - precision: 0.
Epoch 33: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7801 - auc: 0.8647 - loss: 0.0245 - precision: 0.
Epoch 34/80
17/17 0s 1s/step - accuracy: 0.8045 - auc: 0.8868 - loss: 0.0218 - precision: 0.
Epoch 34: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7914 - auc: 0.8694 - loss: 0.0238 - precision: 0.
Epoch 35/80
17/17 0s 1s/step - accuracy: 0.7829 - auc: 0.8839 - loss: 0.0219 - precision: 0.
Epoch 35: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7744 - auc: 0.8650 - loss: 0.0241 - precision: 0.
Epoch 36/80
17/17 0s 1s/step - accuracy: 0.7835 - auc: 0.8920 - loss: 0.0213 - precision: 0.
Epoch 36: val_recall did not improve from 0.88372
17/17 26s 2s/step - accuracy: 0.7726 - auc: 0.8674 - loss: 0.0242 - precision: 0.
Epoch 37/80
17/17 0s 1s/step - accuracy: 0.7915 - auc: 0.8998 - loss: 0.0207 - precision: 0.
Epoch 37: val_recall did not improve from 0.88372
17/17 25s 1s/step - accuracy: 0.7801 - auc: 0.8690 - loss: 0.0240 - precision: 0.
Epoch 38/80
17/17 0s 1s/step - accuracy: 0.7782 - auc: 0.8980 - loss: 0.0209 - precision: 0.
Epoch 38: val_recall did not improve from 0.88372
17/17 24s 1s/step - accuracy: 0.7688 - auc: 0.8695 - loss: 0.0238 - precision: 0.
Epoch 39/80
17/17 0s 1s/step - accuracy: 0.7939 - auc: 0.8899 - loss: 0.0217 - precision: 0.
Epoch 39: val_recall did not improve from 0.88372
17/17 23s 1s/step - accuracy: 0.7838 - auc: 0.8755 - loss: 0.0232 - precision: 0.
Epoch 40/80
17/17 0s 1s/step - accuracy: 0.7953 - auc: 0.8901 - loss: 0.0216 - precision: 0.
Epoch 40: val_recall did not improve from 0.88372
17/17 24s 1s/step - accuracy: 0.7857 - auc: 0.8742 - loss: 0.0236 - precision: 0.
Epoch 41/80
17/17 0s 1s/step - accuracy: 0.7921 - auc: 0.8796 - loss: 0.0222 - precision: 0.
Epoch 41: val_recall did not improve from 0.88372
17/17 24s 1s/step - accuracy: 0.7820 - auc: 0.8663 - loss: 0.0241 - precision: 0.
Epoch 42/80

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17/17          0s 2s/step - accuracy: 0.7793 - auc: 0.8686 - loss: 0.0230 - precision: 0.
Epoch 42: val_recall did not improve from 0.88372
17/17          32s 2s/step - accuracy: 0.7744 - auc: 0.8595 - loss: 0.0242 - precision: 0.
Epoch 43/80
17/17          0s 2s/step - accuracy: 0.7955 - auc: 0.8921 - loss: 0.0206 - precision: 0.
Epoch 43: val_recall did not improve from 0.88372
17/17          41s 2s/step - accuracy: 0.7857 - auc: 0.8681 - loss: 0.0232 - precision: 0.
Epoch 44/80
17/17          0s 2s/step - accuracy: 0.7842 - auc: 0.8821 - loss: 0.0220 - precision: 0.
Epoch 44: val_recall improved from 0.88372 to 0.90698, saving model to C:\workspace\hd-image
17/17          28s 2s/step - accuracy: 0.7782 - auc: 0.8659 - loss: 0.0236 - precision: 0.
Epoch 45/80
17/17          0s 1s/step - accuracy: 0.7866 - auc: 0.8910 - loss: 0.0218 - precision: 0.
Epoch 45: val_recall did not improve from 0.90698
17/17          24s 1s/step - accuracy: 0.7744 - auc: 0.8710 - loss: 0.0237 - precision: 0.
Epoch 46/80
17/17          0s 2s/step - accuracy: 0.7836 - auc: 0.8935 - loss: 0.0217 - precision: 0.
Epoch 46: val_recall did not improve from 0.90698
17/17          29s 2s/step - accuracy: 0.7744 - auc: 0.8665 - loss: 0.0248 - precision: 0.
Epoch 47/80
17/17          0s 1s/step - accuracy: 0.7823 - auc: 0.8912 - loss: 0.0214 - precision: 0.
Epoch 47: val_recall did not improve from 0.90698
17/17          37s 1s/step - accuracy: 0.7726 - auc: 0.8742 - loss: 0.0234 - precision: 0.
Epoch 48/80
17/17          0s 1s/step - accuracy: 0.7873 - auc: 0.8991 - loss: 0.0213 - precision: 0.
Epoch 48: val_recall did not improve from 0.90698
17/17          24s 1s/step - accuracy: 0.7857 - auc: 0.8710 - loss: 0.0238 - precision: 0.
Epoch 49/80
17/17          0s 1s/step - accuracy: 0.7775 - auc: 0.8979 - loss: 0.0208 - precision: 0.
Epoch 49: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-05.

Epoch 49: val_recall did not improve from 0.90698
17/17          24s 1s/step - accuracy: 0.7820 - auc: 0.8800 - loss: 0.0224 - precision: 0.
Epoch 50/80
17/17          0s 1s/step - accuracy: 0.7809 - auc: 0.8840 - loss: 0.0220 - precision: 0.
Epoch 50: val_recall did not improve from 0.90698
17/17          24s 1s/step - accuracy: 0.7763 - auc: 0.8640 - loss: 0.0243 - precision: 0.
Epoch 51/80
17/17          0s 1s/step - accuracy: 0.7845 - auc: 0.8701 - loss: 0.0226 - precision: 0.
Epoch 51: val_recall did not improve from 0.90698
17/17          25s 1s/step - accuracy: 0.7801 - auc: 0.8621 - loss: 0.0241 - precision: 0.
Epoch 52/80
17/17          0s 1s/step - accuracy: 0.7787 - auc: 0.8864 - loss: 0.0214 - precision: 0.

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Epoch 52: val_recall did not improve from 0.90698
17/17 24s 1s/step - accuracy: 0.7744 - auc: 0.8735 - loss: 0.0230 - precision: 0.
Epoch 53/80
17/17 0s 1s/step - accuracy: 0.7743 - auc: 0.8842 - loss: 0.0218 - precision: 0.
Epoch 53: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7707 - auc: 0.8671 - loss: 0.0235 - precision: 0.
Epoch 54/80
17/17 0s 1s/step - accuracy: 0.7942 - auc: 0.8886 - loss: 0.0210 - precision: 0.
Epoch 54: ReduceLROnPlateau reducing learning rate to 1.249999968422344e-05.

Epoch 54: val_recall did not improve from 0.90698
17/17 23s 1s/step - accuracy: 0.7895 - auc: 0.8733 - loss: 0.0227 - precision: 0.
Epoch 55/80
17/17 0s 1s/step - accuracy: 0.7942 - auc: 0.8993 - loss: 0.0205 - precision: 0.
Epoch 55: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7838 - auc: 0.8773 - loss: 0.0226 - precision: 0.
Epoch 56/80
17/17 0s 1s/step - accuracy: 0.7851 - auc: 0.8850 - loss: 0.0226 - precision: 0.
Epoch 56: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7763 - auc: 0.8625 - loss: 0.0248 - precision: 0.
Epoch 57/80
17/17 0s 1s/step - accuracy: 0.7889 - auc: 0.8733 - loss: 0.0225 - precision: 0.
Epoch 57: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7838 - auc: 0.8589 - loss: 0.0241 - precision: 0.
Epoch 58/80
17/17 0s 1s/step - accuracy: 0.7893 - auc: 0.8864 - loss: 0.0213 - precision: 0.
Epoch 58: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7876 - auc: 0.8630 - loss: 0.0237 - precision: 0.
Epoch 59/80
17/17 0s 1s/step - accuracy: 0.7976 - auc: 0.8782 - loss: 0.0218 - precision: 0.
Epoch 59: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7857 - auc: 0.8625 - loss: 0.0235 - precision: 0.
Epoch 60/80
17/17 0s 1s/step - accuracy: 0.7950 - auc: 0.8841 - loss: 0.0211 - precision: 0.
Epoch 60: ReduceLROnPlateau reducing learning rate to 6.24999984211172e-06.

Epoch 60: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7876 - auc: 0.8628 - loss: 0.0236 - precision: 0.
Epoch 61/80
17/17 0s 1s/step - accuracy: 0.7794 - auc: 0.8923 - loss: 0.0212 - precision: 0.
Epoch 61: val_recall did not improve from 0.90698
17/17 22s 1s/step - accuracy: 0.7820 - auc: 0.8707 - loss: 0.0229 - precision: 0.
Epoch 62/80


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17/17          0s 1s/step - accuracy: 0.7794 - auc: 0.8874 - loss: 0.0223 - precision: 0.
Epoch 62: val_recall did not improve from 0.90698
17/17          22s 1s/step - accuracy: 0.7744 - auc: 0.8642 - loss: 0.0247 - precision: 0.
Epoch 63/80
17/17          0s 1s/step - accuracy: 0.7755 - auc: 0.8912 - loss: 0.0216 - precision: 0.
Epoch 63: val_recall did not improve from 0.90698
17/17          22s 1s/step - accuracy: 0.7688 - auc: 0.8713 - loss: 0.0237 - precision: 0.
Epoch 64/80
17/17          0s 1s/step - accuracy: 0.7866 - auc: 0.8807 - loss: 0.0213 - precision: 0.
Epoch 64: val_recall did not improve from 0.90698
17/17          22s 1s/step - accuracy: 0.7782 - auc: 0.8673 - loss: 0.0232 - precision: 0.
Epoch 65/80
17/17          0s 1s/step - accuracy: 0.7900 - auc: 0.8744 - loss: 0.0224 - precision: 0.
Epoch 65: ReduceLROnPlateau reducing learning rate to 3.12499992105586e-06.

Epoch 65: val_recall did not improve from 0.90698
17/17          23s 1s/step - accuracy: 0.7744 - auc: 0.8591 - loss: 0.0243 - precision: 0.
Epoch 65: early stopping
Restoring model weights from the end of the best epoch: 55.

```

--- [Fecha/hora inicio=20260324_1236] ---

--- [T=1662.99s] ---

5. Guardado y Ploteo (Usando utils_train)

```

Historial guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_m
2026-03-24 13:03:53.860803: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentio
17/17          4s 255ms/step

```

--- ANÁLISIS DE UMBRAL ÓPTIMO ---

Mejor Umbral detectado: 0.5214

Nueva Matriz de Confusión con Umbral Óptimo:

```

[[267  93]
 [ 15 157]]

```

Reporte de **Clasificación:**

	precision	recall	f1-score	support
0	0.95	0.74	0.83	360
1	0.63	0.91	0.74	172
accuracy			0.80	532

macro avg	0.79	0.83	0.79	532
weighted avg	0.84	0.80	0.80	532

Auc: 0.8892118863049095

Umbral mas optimo: 0.52143836

5/5 1s 208ms/step

--- MATRIZ DE CONFUSIÓN ---

```
[[64 26]
 [ 4 39]]
```

--- REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
0	0.94	0.71	0.81	90
1	0.60	0.91	0.72	43
accuracy			0.77	133
macro avg	0.77	0.81	0.77	133
weighted avg	0.83	0.77	0.78	133

AUC REAL: 0.8793

1.3 Resultado:

1.3.1 Model Fit History:

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[illegible]

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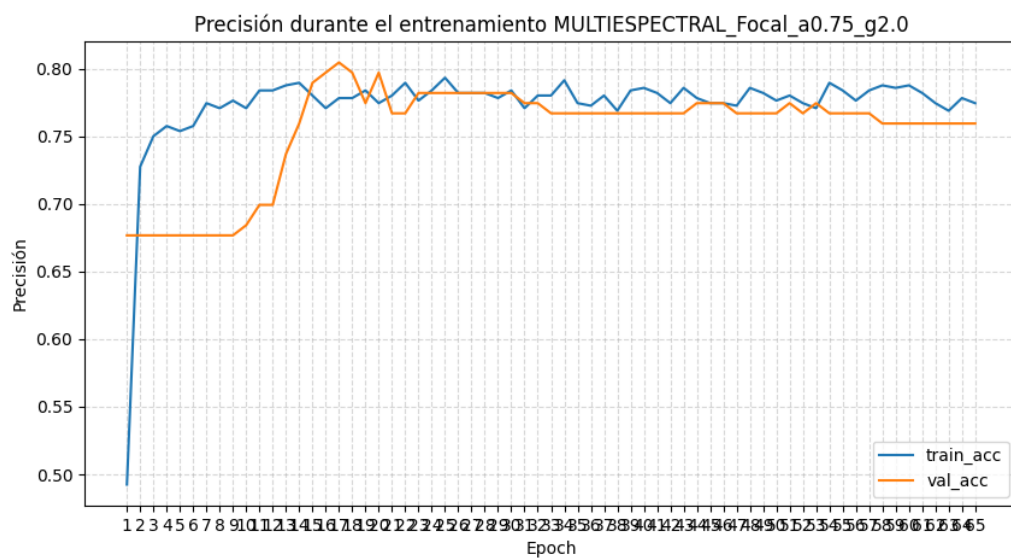


Figure 1: alt text

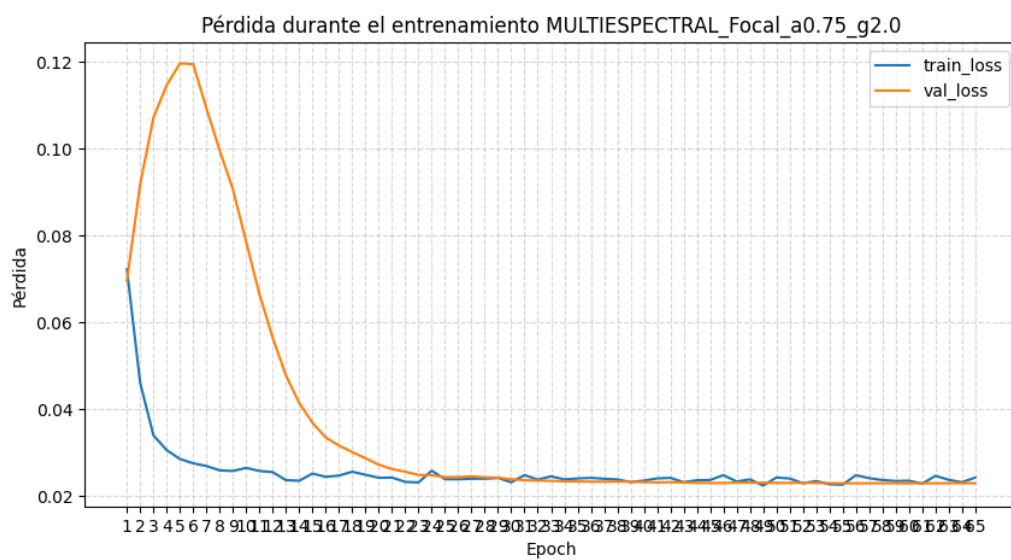


Figure 2: alt text

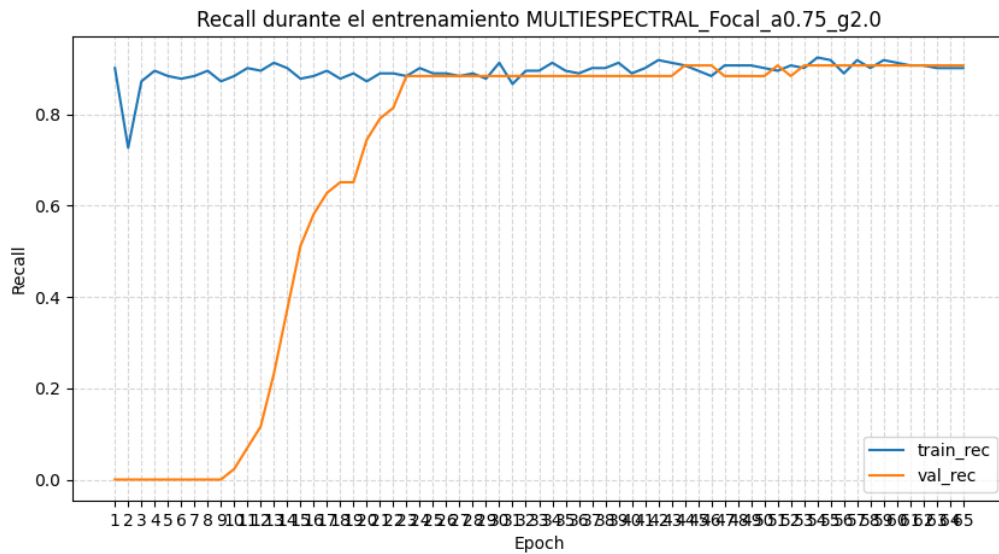


Figure 3: alt text

2. BINARY CROSS ENTROPY MULTIESPECTRAL (BCE MS)

2.1 Comando:

```
python src\pest_detection_models\train.py data\multispectral\TTADDA-dataset\TTADDA_NARO_20
```

2.2 Consola:

```
2026-03-24 13:50:23.446716: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-24 13:50:25.563229: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
Modelo cnn binary_crossentropy | RGB? False | alpha 0.25 | gamma 1.0
```

```
--- [Fecha/hora inicio=20260324_1350] ---
```

```
--- [T=0.00s] ---
```

```
init. Iniciando entrenamiento MULTIESPECTRAL con perdida Binary Crossentropy
```

```
--- [Fecha/hora inicio=20260324_1350] ---
```

```
--- [T=0.00s] ---
```

```
1. 1. Extrayendo datos
```

```
--- [Fecha/hora inicio=20260324_1350] ---
```

```
--- [T=0.02s] ---
```

```
1. Carga de datos cnn Multispectral
```

```

--- [Fecha/hora inicio=20260324_1350] ---

--- [T=0.09s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260324_1350] ---

--- [T=0.09s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260324_1350] ---

--- [T=0.09s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260324_1350] ---

--- [T=60.23s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)
Shape train: (532, 224, 224, 5)
2026-03-24 13:51:28.404398: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a variety of CPU instructions specific to the current CPU architecture.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations you may need to compile the binary with 'tensorflow/tools/pip_package/setup.py --cpu_flags=-march=native'.
Pesos de Clase Calculados: Plaga (0): 1.11, Sana (1): 1.55
-----
DEBUG - Conteo: Plaga: 360, Sana: 172
RESULTADO - Pesos Finales: Plaga (0): 1.11, Sana (1): 1.55
-----

Epoch 1/80
2026-03-24 13:51:28.831652: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op "Placeholder" which has not been registered: 2026-03-24 13:51:28.831652: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op "Placeholder" which has not been registered:
17/17          0s 1s/step - accuracy: 0.4152 - auc: 0.5052 - loss: 1.0127 - precision: 0.4152

Epoch 1: val_recall improved from None to 0.00000, saving model to C:\workspace\hd-image-pes
17/17          27s 2s/step - accuracy: 0.5094 - auc: 0.6175 - loss: 0.8935 - precision: 0.5094
Epoch 2/80

```

```

17/17          0s 1s/step - accuracy: 0.7215 - auc: 0.8141 - loss: 0.7294 - precision: 0.
Epoch 2: val_recall did not improve from 0.00000
17/17          24s 1s/step - accuracy: 0.7180 - auc: 0.7995 - loss: 0.7112 - precision: 0.
Epoch 3/80
17/17          0s 1s/step - accuracy: 0.7775 - auc: 0.8686 - loss: 0.6365 - precision: 0.
Epoch 3: val_recall did not improve from 0.00000
17/17          26s 2s/step - accuracy: 0.7669 - auc: 0.8526 - loss: 0.6288 - precision: 0.
Epoch 4/80
17/17          0s 1s/step - accuracy: 0.7946 - auc: 0.8703 - loss: 0.5822 - precision: 0.
Epoch 4: val_recall did not improve from 0.00000
17/17          25s 1s/step - accuracy: 0.7782 - auc: 0.8543 - loss: 0.5845 - precision: 0.
Epoch 5/80
17/17          0s 1s/step - accuracy: 0.7969 - auc: 0.8707 - loss: 0.5620 - precision: 0.
Epoch 5: val_recall did not improve from 0.00000
17/17          24s 1s/step - accuracy: 0.7895 - auc: 0.8612 - loss: 0.5651 - precision: 0.
Epoch 6/80
17/17          0s 2s/step - accuracy: 0.8029 - auc: 0.8856 - loss: 0.5287 - precision: 0.
Epoch 6: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-05.

Epoch 6: val_recall did not improve from 0.00000
17/17          28s 2s/step - accuracy: 0.7820 - auc: 0.8618 - loss: 0.5541 - precision: 0.
Epoch 7/80
17/17          0s 2s/step - accuracy: 0.8125 - auc: 0.8884 - loss: 0.5140 - precision: 0.
Epoch 7: val_recall did not improve from 0.00000
17/17          28s 2s/step - accuracy: 0.7876 - auc: 0.8664 - loss: 0.5434 - precision: 0.
Epoch 8/80
17/17          0s 2s/step - accuracy: 0.7705 - auc: 0.8846 - loss: 0.5266 - precision: 0.
Epoch 8: val_recall did not improve from 0.00000
17/17          28s 2s/step - accuracy: 0.7688 - auc: 0.8677 - loss: 0.5399 - precision: 0.
Epoch 9/80
17/17          0s 1s/step - accuracy: 0.7949 - auc: 0.8802 - loss: 0.5175 - precision: 0.
Epoch 9: val_recall did not improve from 0.00000
17/17          25s 1s/step - accuracy: 0.7857 - auc: 0.8662 - loss: 0.5366 - precision: 0.
Epoch 10/80
17/17          0s 2s/step - accuracy: 0.8149 - auc: 0.8846 - loss: 0.5180 - precision: 0.
Epoch 10: val_recall did not improve from 0.00000
17/17          28s 2s/step - accuracy: 0.7876 - auc: 0.8600 - loss: 0.5464 - precision: 0.
Epoch 11/80
17/17          0s 2s/step - accuracy: 0.8051 - auc: 0.8836 - loss: 0.5140 - precision: 0.
Epoch 11: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-05.

Epoch 11: val_recall improved from 0.00000 to 0.02326, saving model to C:\workspace\hd-image
17/17          28s 2s/step - accuracy: 0.7932 - auc: 0.8621 - loss: 0.5386 - precision: 0.

```

Epoch 11: early stopping

Restoring model weights from the end of the best epoch: 1.

--- [Fecha/hora inicio=20260324_1350] ---

--- [T=352.94s] ---

5. Guardado y Ploteo (Usando utils_train)

Historial guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_m
2026-03-24 13:56:30.153747: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentio
17/17 4s 234ms/step

--- ANÁLISIS DE UMBRAL ÓPTIMO ---

Mejor Umbral detectado: 0.2940

Nueva Matriz de Confusión con Umbral Óptimo:

```
[[275  85]
 [ 19 153]]
```

Reporte de **Clasificación**:

	precision	recall	f1-score	support
0	0.94	0.76	0.84	360
1	0.64	0.89	0.75	172
accuracy			0.80	532
macro avg	0.79	0.83	0.79	532
weighted avg	0.84	0.80	0.81	532

Auc: 0.8829780361757107

Umbral mas optimo: 0.29402164

5/5 1s 175ms/step

--- MATRIZ DE CONFUSIÓN ---

```
[[64 26]
 [ 7 36]]
```

--- REPORTE **DE** CLASIFICACIÓN ---

	precision	recall	f1-score	support
0	0.90	0.71	0.80	90
1	0.58	0.84	0.69	43

accuracy			0.75	133
macro avg	0.74	0.77	0.74	133
weighted avg	0.80	0.75	0.76	133

AUC REAL: 0.8708

2.3 Resultado:

2.3.1 Model Fit History:

```
{
  "accuracy": [
    0.5093985199928284,
    0.7180451154708862,
    0.7669172883033752,
    0.7781955003738403,
    0.7894737124443054,
    0.7819548845291138,
    0.7875939607620239,
    0.768796980381012,
    0.7857142686843872,
    0.7875939607620239,
    0.7932330965995789
  ],
  "auc": [
    0.6175065040588379,
    0.7995235919952393,
    0.8525597453117371,
    0.8542554974555969,
    0.8611595630645752,
    0.8617974519729614,
    0.8664324283599854,
    0.8676760792732239,
    0.8661822080612183,
    0.8600290417671204,
    0.8621123433113098
  ],
  "loss": [
    0.8934842944145203,
    0.7111803293228149,
    0.6287882328033447,
    0.5845021605491638,
```



```

    0.6766917109489441,
    0.6766917109489441,
    0.6842105388641357
],
"val_auc": [
    0.8713178634643555,
    0.8759689927101135,
    0.8722222447395325,
    0.868992269039154,
    0.8708009719848633,
    0.874289333820343,
    0.8726098537445068,
    0.8757106065750122,
    0.8759689927101135,
    0.8770025968551636,
    0.8784237504005432
],
"val_loss": [
    0.555253267288208,
    0.6082989573478699,
    0.6807368397712708,
    0.7361621260643005,
    0.7717756628990173,
    0.7748770713806152,
    0.7322303652763367,
    0.6953290104866028,
    0.6619917154312134,
    0.6171551942825317,
    0.5729148387908936
],
"val_precision": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    1.0
],

```

```

"val_recall": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.023255813866853714
],
"learning_rate": [
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05
]
}

```

2.3.2 Accuracy plot:

2.3.2 Loss plot:

2.3.2 Recall plot:

3. FOCAL LOSS RGB (FL RGB)

3.1 Comando:

```
python src\pest_detection_models\ttrain.py data\multispectral\TTADDA-dataset\TTADDA_NARO_202
```

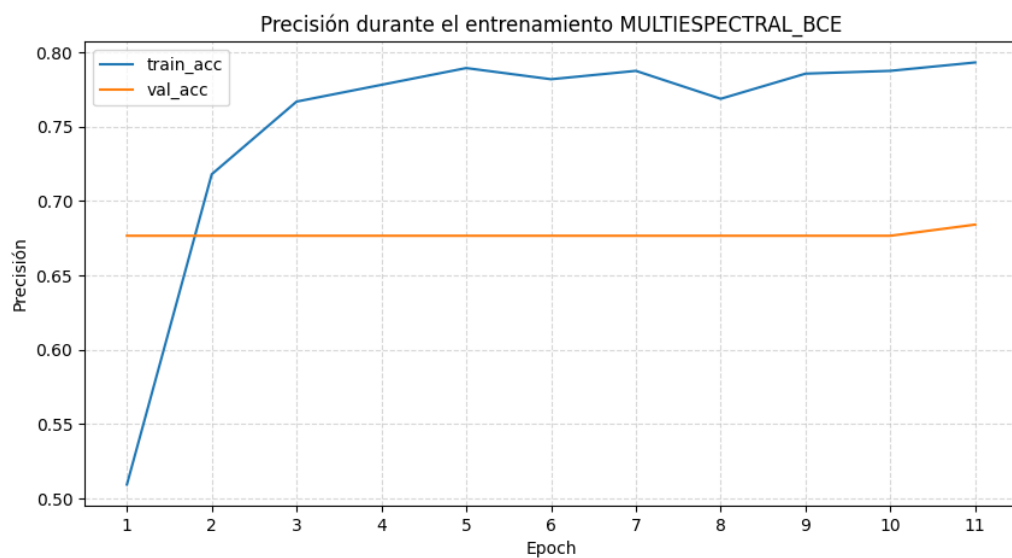


Figure 4: alt text

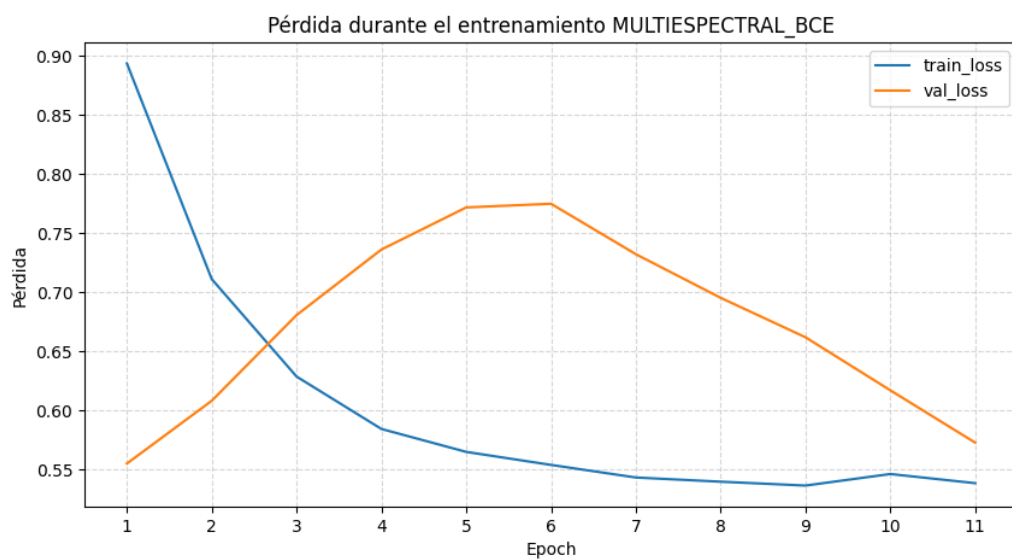


Figure 5: alt text

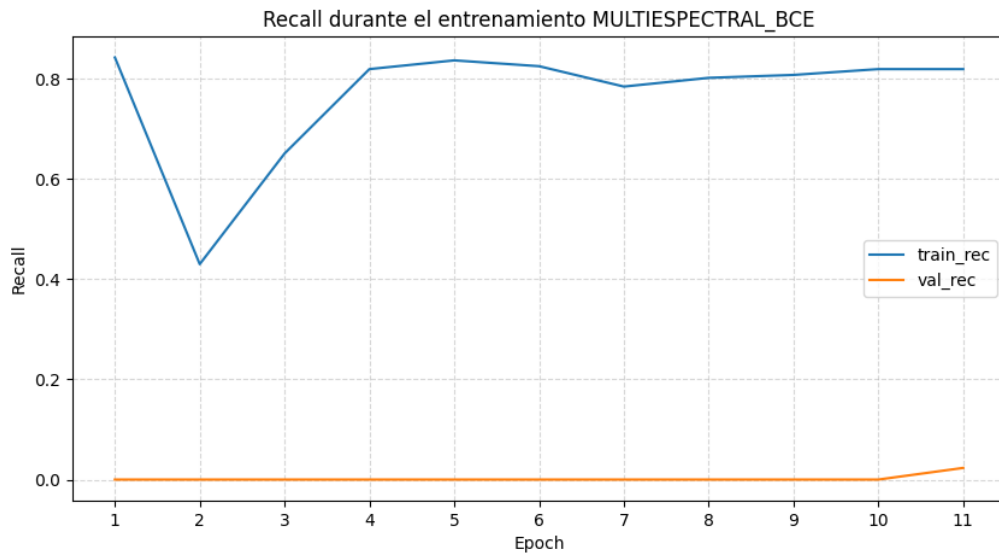


Figure 6: alt text

3.2 Consola:

```
2026-03-24 14:00:41.429132: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-24 14:00:42.805358: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
Modelo cnn focal_loss | RGB? True | alpha 0.75 | gamma 2.0
```

```
--- [Fecha/hora inicio=20260324_1400] ---
```

```
--- [T=0.00s] ---
```

```
init. Iniciando entrenamiento RGB con perdida Focal Loss
```

```
--- [Fecha/hora inicio=20260324_1400] ---
```

```
--- [T=0.00s] ---
```

```
1. 1. Extrayendo datos
```

```
--- [Fecha/hora inicio=20260324_1400] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn RGB
```

```
--- [Fecha/hora inicio=20260324_1400] ---
```

```
--- [T=0.07s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260324_1400] ---
```


--- [Fecha/hora inicio=20260324_1400] ---

--- [T=38.48s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Shape train: (504, 224, 224, 3)

2026-03-24 14:01:22.799115: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a NVIDIA GPU

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

Pesos de Clase Calculados: Plaga (0): 1.14, Sana (1): 1.47

DEBUG - Conteo: Plaga: 332, Sana: 172

RESULTADO - Pesos Finales: Plaga (0): 1.14, Sana (1): 1.47

Epoch 1/80

2026-03-24 14:01:23.054240: E tensorflow/core/framework/node_def_util.cc:680] NodeDef menti

16/16 0s 1s/step - accuracy: 0.4059 - auc: 0.5501 - loss: 0.0917 - precision: 0.

Epoch 1: val_recall improved from None to 1.00000, saving model to C:\workspace\hd-image-pe

16/16 22s 1s/step - accuracy: 0.4921 - auc: 0.6077 - loss: 0.0772 - precision: 0.

Epoch 2/80

16/16 0s 1s/step - accuracy: 0.6951 - auc: 0.7527 - loss: 0.0529 - precision: 0.

Epoch 2: val_recall did not improve from 1.00000

16/16 21s 1s/step - accuracy: 0.7262 - auc: 0.7916 - loss: 0.0474 - precision: 0.

Epoch 3/80

16/16 0s 1s/step - accuracy: 0.7411 - auc: 0.8227 - loss: 0.0428 - precision: 0.

Epoch 3: val_recall did not improve from 1.00000

16/16 21s 1s/step - accuracy: 0.7897 - auc: 0.8617 - loss: 0.0361 - precision: 0.

Epoch 4/80

16/16 0s 1s/step - accuracy: 0.7708 - auc: 0.8521 - loss: 0.0339 - precision: 0.

Epoch 4: val_recall did not improve from 1.00000

16/16 20s 1s/step - accuracy: 0.8095 - auc: 0.8827 - loss: 0.0291 - precision: 0.

Epoch 5/80

16/16 0s 1s/step - accuracy: 0.7657 - auc: 0.8442 - loss: 0.0360 - precision: 0.

Epoch 5: val_recall did not improve from 1.00000


```

16/16          21s 1s/step - accuracy: 0.8075 - auc: 0.8860 - loss: 0.0278 - precision: 0.
Epoch 6/80
16/16          0s 1s/step - accuracy: 0.7696 - auc: 0.8458 - loss: 0.0355 - precision: 0.
Epoch 6: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8194 - auc: 0.8888 - loss: 0.0270 - precision: 0.
Epoch 7/80
16/16          0s 1s/step - accuracy: 0.7984 - auc: 0.8815 - loss: 0.0283 - precision: 0.
Epoch 7: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-05.

Epoch 7: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8393 - auc: 0.9071 - loss: 0.0230 - precision: 0.
Epoch 8/80
16/16          0s 1s/step - accuracy: 0.7896 - auc: 0.8467 - loss: 0.0355 - precision: 0.
Epoch 8: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8254 - auc: 0.8893 - loss: 0.0259 - precision: 0.
Epoch 9/80
16/16          0s 1s/step - accuracy: 0.8053 - auc: 0.8817 - loss: 0.0267 - precision: 0.
Epoch 9: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8333 - auc: 0.9043 - loss: 0.0222 - precision: 0.
Epoch 10/80
16/16          0s 1s/step - accuracy: 0.7904 - auc: 0.8669 - loss: 0.0300 - precision: 0.
Epoch 10: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8393 - auc: 0.9075 - loss: 0.0219 - precision: 0.
Epoch 11/80
16/16          0s 1s/step - accuracy: 0.8148 - auc: 0.8696 - loss: 0.0285 - precision: 0.
Epoch 11: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.8552 - auc: 0.9029 - loss: 0.0216 - precision: 0.
Epoch 12/80
16/16          0s 1s/step - accuracy: 0.8101 - auc: 0.8702 - loss: 0.0271 - precision: 0.
Epoch 12: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-05.

Epoch 12: val_recall did not improve from 1.00000
16/16          20s 1s/step - accuracy: 0.8452 - auc: 0.9033 - loss: 0.0204 - precision: 0.
Epoch 12: early stopping
Restoring model weights from the end of the best epoch: 2.

```

```

--- [Fecha/hora inicio=20260324_1400] ---

```

```

--- [T=290.01s] ---

```

```

5. Guardado y Ploteo (Usando utils_train)

```

```

Historial guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_m
2026-03-24 14:06:03.627483: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentio

```

16/16 4s 209ms/step

--- ANÁLISIS DE UMBRAL ÓPTIMO ---

Mejor Umbral detectado: 0.5180

Nueva Matriz de Confusión con Umbral Óptimo:

```
[[230 102]
 [ 35 137]]
```

Reporte de **Clasificación**:

	precision	recall	f1-score	support
0	0.87	0.69	0.77	332
1	0.57	0.80	0.67	172
accuracy			0.73	504
macro avg	0.72	0.74	0.72	504
weighted avg	0.77	0.73	0.74	504

Auc: 0.7975798543009247

Umbral mas optimo: 0.5180124

4/4 1s 196ms/step

--- MATRIZ DE CONFUSIÓN ---

```
[[53 30]
 [10 33]]
```

--- REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
0	0.84	0.64	0.73	83
1	0.52	0.77	0.62	43
accuracy			0.68	126
macro avg	0.68	0.70	0.67	126
weighted avg	0.73	0.68	0.69	126

AUC REAL: 0.7504

3.3 Resultado:

3.3.1 Model Fit History:

```
{  
  "accuracy": [  
    0.4920634925365448,  
    0.726190447807312,  
    0.7896825671195984,  
    0.8095238208770752,  
    0.807539701461792,  
    0.8194444179534912,  
    0.8392857313156128,  
    0.8253968358039856,  
    0.8333333134651184,  
    0.8392857313156128,  
    0.8551587462425232,  
    0.8452380895614624  
  ],  
  "auc": [  
    0.6077420115470886,  
    0.7915557622909546,  
    0.8616734743118286,  
    0.8827490210533142,  
    0.886006236076355,  
    0.8887730836868286,  
    0.9071080684661865,  
    0.8892984986305237,  
    0.9043236970901489,  
    0.9075020551681519,  
    0.9029489755630493,  
    0.9032992124557495  
  ],  
  "loss": [  
    0.07717034965753555,  
    0.047411076724529266,  
    0.036136217415332794,  
    0.029069555923342705,  
    0.027779387310147285,  
    0.02703135274350643,  
    0.022950902581214905,  
    0.025886066257953644,  
    0.0221759881824255,  
    0.021886108443140984,  
  ]  
}
```



```

    0.341269850730896
],
"val_auc": [
    0.5361446142196655,
    0.7509106397628784,
    0.7573549747467041,
    0.7524516582489014,
    0.7548332810401917,
    0.7898570895195007,
    0.7906976938247681,
    0.7984029054641724,
    0.8135331869125366,
    0.8198374509811401,
    0.8241804242134094,
    0.838330090045929
],
"val_loss": [
    0.07264476269483566,
    0.07228100299835205,
    0.07247969508171082,
    0.0728490799665451,
    0.07326599210500717,
    0.0737476795911789,
    0.07423986494541168,
    0.07486511766910553,
    0.07537901401519775,
    0.07571977376937866,
    0.07568908482789993,
    0.07574506103992462
],
"val_precision": [
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896
]

```

```

],
"val_recall": [
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0
],
"learning_rate": [
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    9.999999747378752e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05,
    4.999999873689376e-05
]
}

```

3.3.2 Accuracy plot:

3.3.2 Loss plot:

3.3.2 Recall plot:

4. BINARY CROSS ENTROPY RGB (BCE RGB)

4.1 Comando:

```
python src\pest_detection_models\train.py data\multispectral\TTADDA-dataset\TTADDA_NARO_202
```

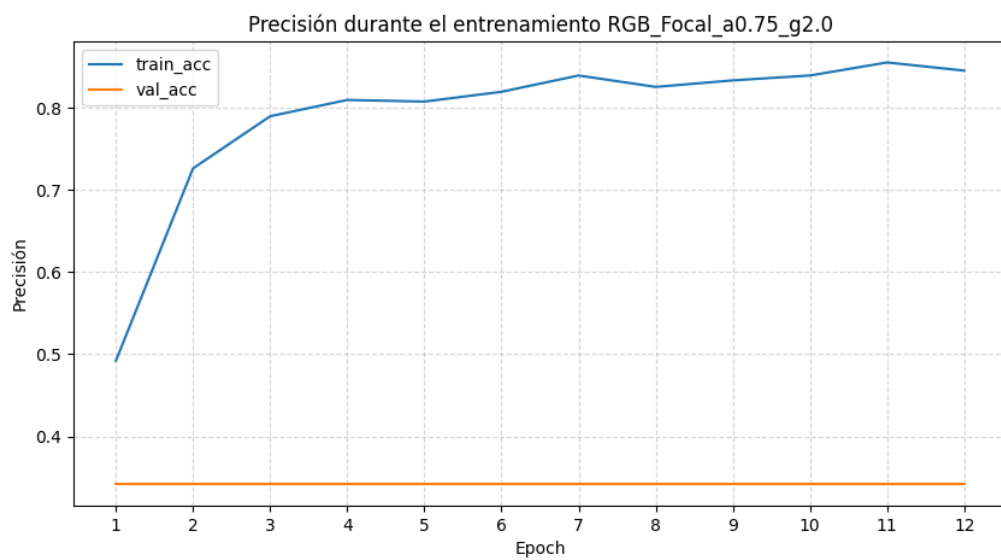


Figure 7: alt text

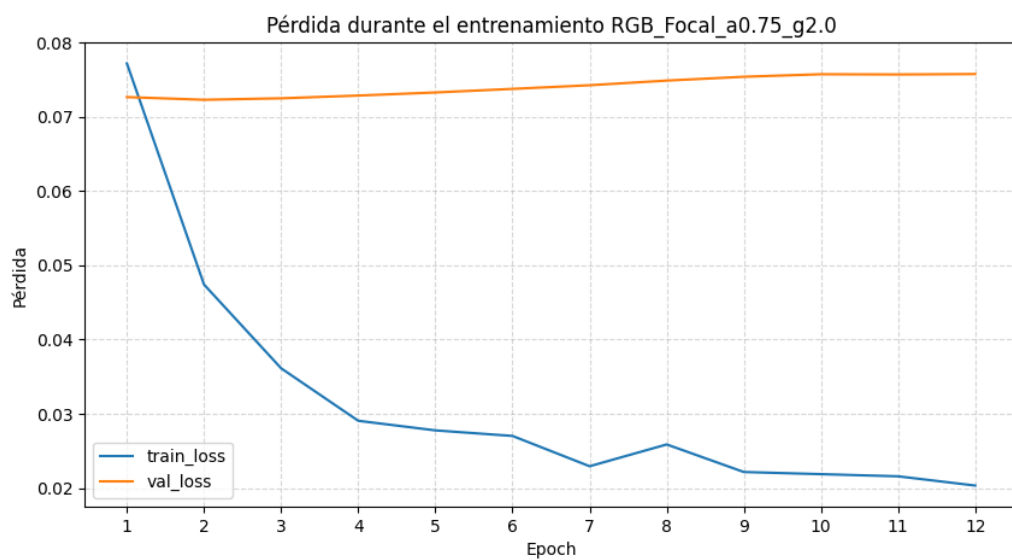


Figure 8: alt text

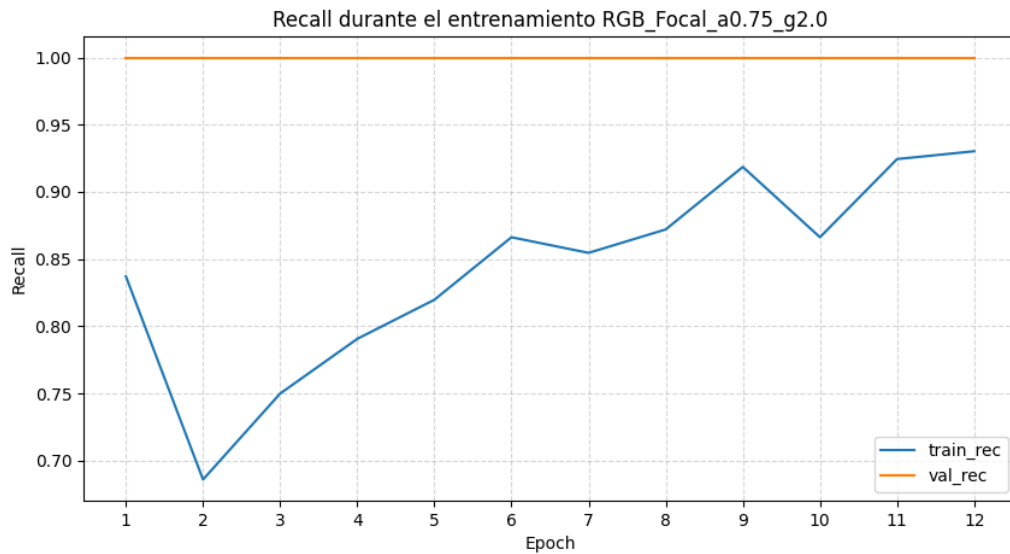


Figure 9: alt text

4.2 Consola:

2026-03-24 14:44:55.735693: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

2026-03-24 14:44:57.159805: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

Modelo cnn binary_crossentropy | RGB? True | alpha 0.75 | gamma 2.0

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.00s] ---

init. Iniciando entrenamiento RGB con perdida Binary Crossentropy

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.00s] ---

1. 1. Extrayendo datos

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.00s] ---

1. Carga de datos cnn RGB

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.04s] ---

2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.04s] ---

2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=0.04s] ---

2.2. Archivos seleccionados: ['RGB.tif']

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

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Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=31.18s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Shape train: (504, 224, 224, 3)

2026-03-24 14:45:29.822260: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

Pesos de Clase Calculados: Plaga (0): 1.14, Sana (1): 1.47

DEBUG - Conteo: Plaga: 332, Sana: 172

RESULTADO - Pesos Finales: Plaga (0): 1.14, Sana (1): 1.47

Epoch 1/80

2026-03-24 14:45:30.144247: E tensorflow/core/framework/node_def_util.cc:680] NodeDef menti

16/16 0s 1s/step - accuracy: 0.4086 - auc: 0.5339 - loss: 1.0425 - precision: 0.

Epoch 1: val_recall improved from None to 1.00000, saving model to C:\workspace\hd-image-pe

16/16 22s 1s/step - accuracy: 0.5000 - auc: 0.5745 - loss: 0.9334 - precision: 0

Epoch 2/80

16/16 0s 1s/step - accuracy: 0.7336 - auc: 0.7374 - loss: 0.7496 - precision: 0.

Epoch 2: val_recall did not improve from 1.00000

16/16 21s 1s/step - accuracy: 0.7560 - auc: 0.7753 - loss: 0.7126 - precision: 0

Epoch 3/80

16/16 0s 1s/step - accuracy: 0.7319 - auc: 0.8116 - loss: 0.6696 - precision: 0.

Epoch 3: val_recall did not improve from 1.00000

16/16 20s 1s/step - accuracy: 0.7718 - auc: 0.8477 - loss: 0.6269 - precision: 0

Epoch 4/80

16/16 0s 1s/step - accuracy: 0.7463 - auc: 0.8454 - loss: 0.6066 - precision: 0.

Epoch 4: val_recall did not improve from 1.00000

16/16 21s 1s/step - accuracy: 0.7798 - auc: 0.8735 - loss: 0.5691 - precision: 0

Epoch 5/80

16/16 0s 1s/step - accuracy: 0.7395 - auc: 0.8375 - loss: 0.6207 - precision: 0.

Epoch 5: val_recall did not improve from 1.00000

```

16/16          21s 1s/step - accuracy: 0.7956 - auc: 0.8799 - loss: 0.5476 - precision: 0.
Epoch 6/80
16/16          0s 1s/step - accuracy: 0.7437 - auc: 0.8417 - loss: 0.6037 - precision: 0.
Epoch 6: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-05.

Epoch 6: val_recall did not improve from 1.00000
16/16          21s 1s/step - accuracy: 0.7976 - auc: 0.8856 - loss: 0.5297 - precision: 0.
Epoch 7/80
16/16          0s 2s/step - accuracy: 0.7897 - auc: 0.8735 - loss: 0.5608 - precision: 0.
Epoch 7: val_recall did not improve from 1.00000
16/16          30s 2s/step - accuracy: 0.8313 - auc: 0.9012 - loss: 0.5032 - precision: 0.
Epoch 8/80
16/16          0s 1s/step - accuracy: 0.7593 - auc: 0.8451 - loss: 0.6159 - precision: 0.
Epoch 8: val_recall did not improve from 1.00000
16/16          23s 1s/step - accuracy: 0.8075 - auc: 0.8859 - loss: 0.5305 - precision: 0.
Epoch 9/80
16/16          0s 1s/step - accuracy: 0.7898 - auc: 0.8718 - loss: 0.5543 - precision: 0.
Epoch 9: val_recall did not improve from 1.00000
16/16          23s 1s/step - accuracy: 0.8313 - auc: 0.8961 - loss: 0.5101 - precision: 0.
Epoch 10/80
16/16          0s 1s/step - accuracy: 0.7688 - auc: 0.8611 - loss: 0.5763 - precision: 0.
Epoch 10: val_recall did not improve from 1.00000
16/16          23s 1s/step - accuracy: 0.8313 - auc: 0.9039 - loss: 0.4928 - precision: 0.
Epoch 11/80
16/16          0s 2s/step - accuracy: 0.7609 - auc: 0.8592 - loss: 0.5658 - precision: 0.
Epoch 11: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-05.

Epoch 11: val_recall did not improve from 1.00000
16/16          29s 2s/step - accuracy: 0.8274 - auc: 0.9010 - loss: 0.4896 - precision: 0.
Epoch 11: early stopping
Restoring model weights from the end of the best epoch: 1.

--- [Fecha/hora inicio=20260324_1444] ---

--- [T=285.25s] ---
5. Guardado y Ploteo (Usando utils_train)

Historial guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_m
2026-03-24 14:49:48.502642: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentio
16/16          4s 222ms/step

--- ANÁLISIS DE UMBRAL ÓPTIMO ---
Mejor Umbral detectado: 0.5037

```

Nueva Matriz de Confusión con Umbral Óptimo:

```
[[183 149]
 [ 27 145]]
```

Reporte de **Clasificación**:

	precision	recall	f1-score	support
0	0.87	0.55	0.68	332
1	0.49	0.84	0.62	172
accuracy			0.65	504
macro avg	0.68	0.70	0.65	504
weighted avg	0.74	0.65	0.66	504

Auc: 0.7228565424488652

Umbral mas optimo: 0.50366676

4/4 1s 230ms/step

--- MATRIZ DE CONFUSIÓN ---

```
[[43 40]
 [ 9 34]]
```

--- REPORTE **DE** CLASIFICACIÓN ---

	precision	recall	f1-score	support
0	0.83	0.52	0.64	83
1	0.46	0.79	0.58	43
accuracy			0.61	126
macro avg	0.64	0.65	0.61	126
weighted avg	0.70	0.61	0.62	126

AUC REAL: 0.6576

4.3 Resultado:

4.3.1 Model Fit History:

```
{
  "accuracy": [
    0.5,
```

```

0.7559523582458496,
0.77182537317276,
0.7797619104385376,
0.795634925365448,
0.7976190447807312,
0.8313491940498352,
0.807539701461792,
0.8313491940498352,
0.8313491940498352,
0.8273809552192688
],
"auc": [
0.5744781494140625,
0.775252103805542,
0.8476638793945312,
0.8734852075576782,
0.8798595666885376,
0.8856209516525269,
0.9012416005134583,
0.8859274387359619,
0.8960668444633484,
0.9039297103881836,
0.9009963870048523
],
"loss": [
0.9334239363670349,
0.7126238346099854,
0.6269177198410034,
0.5690836310386658,
0.5476221442222595,
0.5296936631202698,
0.5032134056091309,
0.5305377840995789,
0.5100593566894531,
0.49281516671180725,
0.4896389842033386
],
"precision": [
0.3863636255264282,
0.6689655184745789,
0.6887417435646057,
0.6942675113677979,
0.6927374005317688,

```

```

0.692307710647583,
0.7543859481811523,
0.7071823477745056,
0.7351351380348206,
0.7377049326896667,
0.7297297120094299
],
"recall": [
0.7906976938247681,
0.5639534592628479,
0.604651153087616,
0.6337209343910217,
0.7209302186965942,
0.7325581312179565,
0.75,
0.7441860437393188,
0.7906976938247681,
0.7848837375640869,
0.7848837375640869
],
"val_accuracy": [
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896,
0.341269850730896
],
"val_auc": [
0.608433723449707,
0.7046791911125183,
0.7237321138381958,
0.729756236076355,
0.7859344482421875,
0.8138133883476257,
0.8253012299537659,
0.8290837407112122,
0.8400112390518188,

```

```

    0.8367890119552612,
    0.8449145555496216
],
"val_loss": [
    0.6950236558914185,
    0.7021607756614685,
    0.7135727405548096,
    0.7252166271209717,
    0.7365506887435913,
    0.7462589144706726,
    0.7579497694969177,
    0.769260585308075,
    0.7797540426254272,
    0.7886399626731873,
    0.794927179813385
],
"val_precision": [
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896,
    0.341269850730896
],
"val_recall": [
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0
],
"learning_rate": [

```

```

9.999999747378752e-05,
9.999999747378752e-05,
9.999999747378752e-05,
9.999999747378752e-05,
9.999999747378752e-05,
9.999999747378752e-05,
4.999999873689376e-05,
4.999999873689376e-05,
4.999999873689376e-05,
4.999999873689376e-05,
4.999999873689376e-05
]
}

```

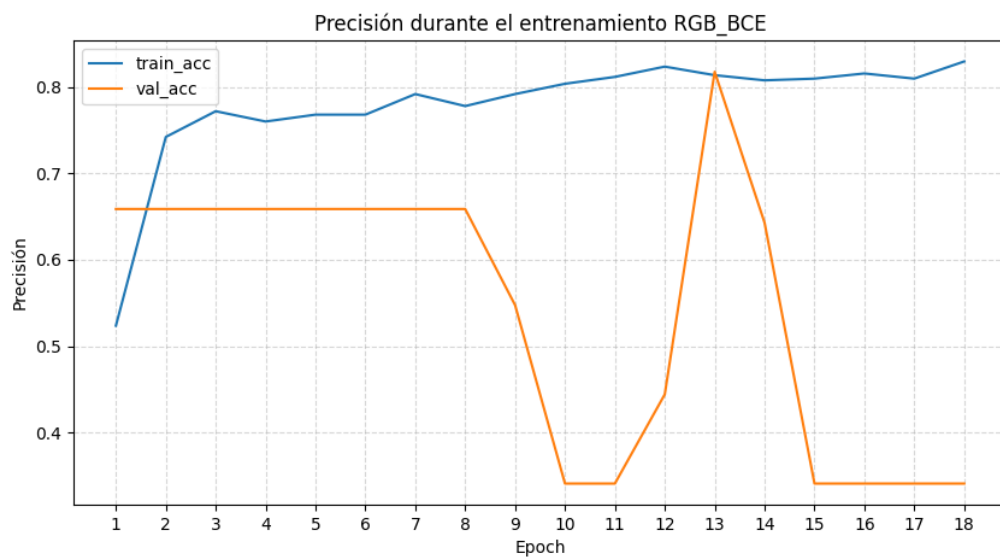


Figure 10: alt text

4.3.2 Accuracy plot:

4.3.2 Loss plot:

4.3.2 Recall plot:

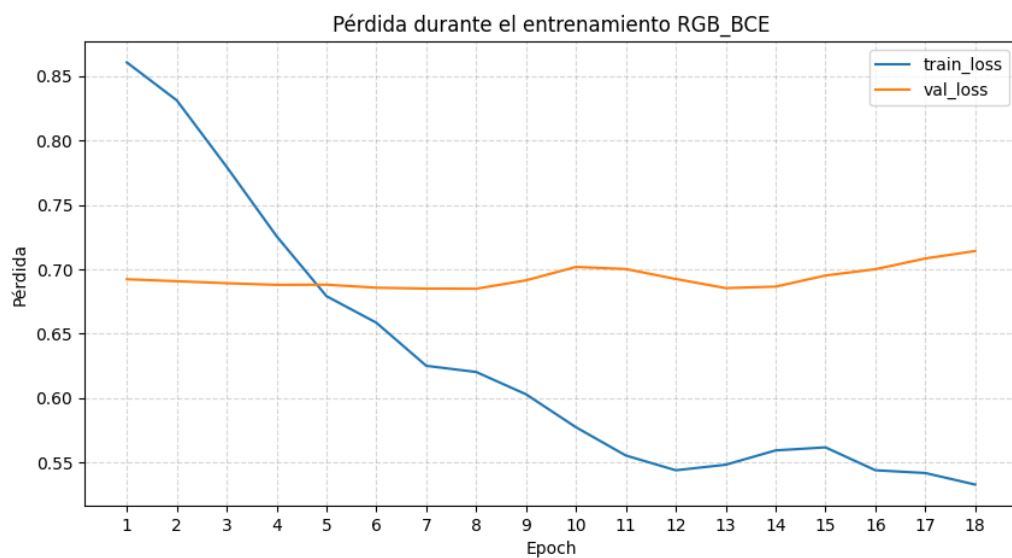


Figure 11: alt text

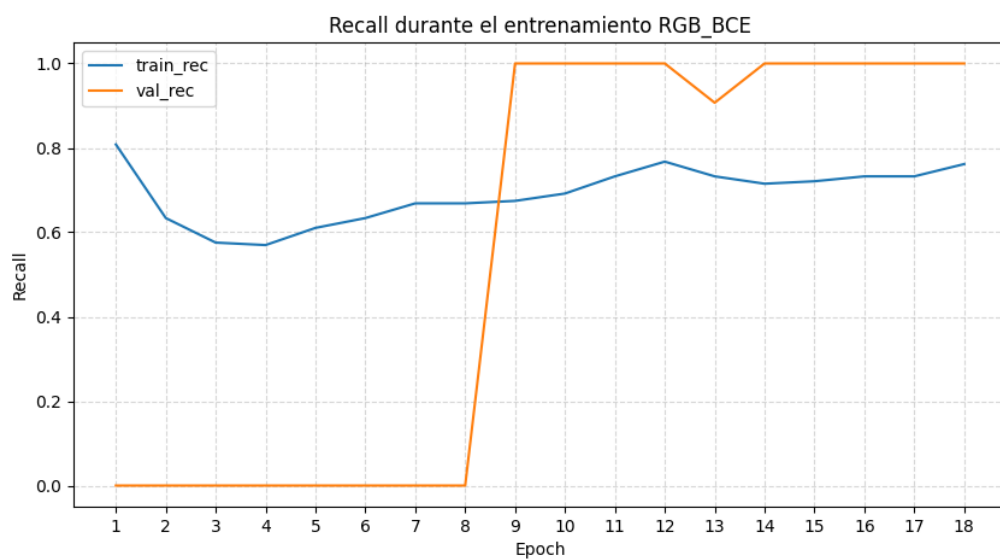


Figure 12: alt text

RANDOM FOREST (RF)

I. ENTRENAMIENTO

- El algoritmo de entrenamiento, `train.py`, permite los siguientes argumentos:

```
def main():
    parser = argparse.ArgumentParser(description="Entrena el modelo RGB o MULTIESPECTRAL para
    parser.add_argument("data_dir", help="Directorio raíz con subcarpetas de clases")
    parser.add_argument("-e", "--epochs", type=int, default=20, help="Número máximo de épocas")
    parser.add_argument("-a", "--alpha", type=float, default=0.50, help="Alpha")
    parser.add_argument("-g", "--gamma", type=float, default=3.0, help="Gamma")
    parser.add_argument("-lt", "--loss_type", type=str, choices=["focal_loss", "binary_crossentropy"])
    parser.add_argument("-rgb", "--rgb", action='store_true', default=False, help="Es RGB?")
    parser.add_argument("-mt", "--model_type", type=str, required=True, default='cnn', choices=
    parser.add_argument("-t", "--threshold", type=float, default=0.5, help="Umbral de decisión")
    args = parser.parse_args()
    run_training(
        data_dir=args.data_dir,
        epochs=args.epochs,
        loss_type=args.loss_type,
        isRgb=args.rgb,
        alpha=args.alpha,
        gamma=args.gamma,
        model_type=args.model_type,
        threshold=args.threshold
    )
```

1. MULTIESPECTRAL (MS)

1.1 Comando:

```
python src\pest_detection_models\train.py data\multiespectral\TTADDA-dataset\TTADDA_NARO_202
```

1.2 Consola:

```
2026-03-24 20:49:49.328473: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-24 20:49:52.983113: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
Modelo Random Forest con extracción CNN
```

```
--- [Fecha/hora inicio=20260324_2049] ---
```

```
--- [T=0.00s] ---
```

```
1. Iniciando entrenamiento Random Forest MULTIESPECTRAL
```

```

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=0.00s] ---
1. Cargando datos...

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=0.11s] ---
1. Carga de datos cnn Multiespectral

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=0.17s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=0.17s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=0.17s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=108.67s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)
Shape train: (532, 224, 224, 5)

--- [Fecha/hora inicio=20260324_2049] ---

```

```

--- [T=109.65s] ---
2. Cargando modelo CNN para extracción de features...
2026-03-24 20:51:44.888079: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a variety of CPU instructions specific to the CPU architecture (e.g. SSE, AVX) to enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, you will need to compile TensorFlow with a CPU architecture that supports SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA.
Feature extractor listo

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=109.96s] ---
3. Extrayendo features con CNN...
2026-03-24 20:51:45.746648: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op Placeholder which does not exist in the registry.
17/17          7s 361ms/step
2026-03-24 20:51:52.415924: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op Placeholder which does not exist in the registry.
5/5           2s 284ms/step
Shape features train: (532, 64)

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=118.78s] ---
4. Aplicando StandardScaler...
Pesos de Clase Calculados: Plaga (0): 1.11, Sana (1): 1.55
-----
DEBUG - Conteo: Plaga: 360, Sana: 172
RESULTADO - Pesos Finales: Plaga (0): 1.11, Sana (1): 1.55
-----

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=118.79s] ---
5. Entrenando Random Forest...

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=119.34s] ---
Finish. Entrenamiento completado

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=119.34s] ---
6. Evaluando modelo...

--- MATRIZ DE CONFUSIÓN ---
[[68 22]

```

[15 28]]

--- REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
0	0.82	0.76	0.79	90
1	0.56	0.65	0.60	43
accuracy			0.72	133
macro avg	0.69	0.70	0.69	133
weighted avg	0.74	0.72	0.73	133

AUC: 0.8443

--- [Fecha/hora inicio=20260324_2049] ---

--- [T=119.46s] ---

7. Guardando modelo...

Modelo RF + Scaler guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_det

2. RGB (RGB)

2.1 Comando:

```
python src\pest_detection_models\train.py data\multispectral\TTADDA-dataset\TTADDA_NARO_20
```

2.2 Consola:

2026-03-24 21:06:18.437540: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

2026-03-24 21:06:20.448869: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

Modelo Random Forest con extracción CNN

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=0.00s] ---

1. Iniciando entrenamiento Random Forest RGB

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=0.00s] ---

1. Cargando datos...

Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=92.08s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Shape train: (504, 224, 224, 3)

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=92.55s] ---

2. Cargando modelo CNN para extracción de features...

2026-03-24 21:07:54.996213: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a GPU enabled. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations you may need to compile the binary with the following flags: -xSSE3 -xSSE4.1 -xSSE4.2 -xAVX -xAVX2 -xAVX_VNNI -xFMA
Feature extractor listo

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=92.91s] ---

3. Extrayendo features con CNN...

2026-03-24 21:07:55.812984: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op `Placeholder` which was not found in the ops registry.
16/16 6s 355ms/step

2026-03-24 21:08:01.996477: E tensorflow/core/framework/node_def_util.cc:680] NodeDef mentions op `Placeholder` which was not found in the ops registry.

```

4/4          1s 351ms/step
Shape features train: (504, 64)

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=101.06s] ---
4. Aplicando StandardScaler...
Pesos de Clase Calculados: Plaga (0): 1.14, Sana (1): 1.47
-----
DEBUG - Conteo: Plaga: 332, Sana: 172
RESULTADO - Pesos Finales: Plaga (0): 1.14, Sana (1): 1.47
-----

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=101.07s] ---
5. Entrenando Random Forest...

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=101.59s] ---
Finish. Entrenamiento completado

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=101.59s] ---
6. Evaluando modelo...

--- MATRIZ DE CONFUSIÓN ---
[[69 14]
 [ 6 37]]

--- REPORTE DE CLASIFICACIÓN ---

```

	precision	recall	f1-score	support
0	0.92	0.83	0.87	83
1	0.73	0.86	0.79	43
accuracy			0.84	126
macro avg	0.82	0.85	0.83	126
weighted avg	0.85	0.84	0.84	126

AUC: 0.9229

--- [Fecha/hora inicio=20260324_2106] ---

--- [T=101.70s] ---

7. Guardando modelo...

Modelo RF + Scaler guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_de

II. EVALUACION

- El algoritmo de evaluación, `evaluate.py`, permite los siguientes argumentos:

```
if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Evalúa el modelo CNN con BCE/Focal RGB/MS")
    parser.add_argument("data_dir", help="Ruta al directorio de datos (raíz)")
    parser.add_argument("-m", "--model", required=True, help="Ruta al archivo del modelo Keras")
    parser.add_argument("-t", "--threshold", type=float, default=0.5, help="Umbral de decisión")
    parser.add_argument("-b", "--base_dir", default=BASE_DIR, help="Directorio base para guardar")
    parser.add_argument("-mt", "--model_type", type=str, required=True, choices=["cnn", "rf"])
    args = parser.parse_args()

    run_evaluation(args.data_dir, args.model, args.threshold, args.model_type, args.base_dir)
```

Curva ROC:

CNN MULTIESPECTRAL FOCAL LOSS

1 UMBRAL 0.4

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARQ
```

1.2 Consola:

2026-03-25 11:54:38.379352: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

2026-03-25 11:54:39.696820: I tensorflow/core/util/port.cc:153] oneDNN custom operations are

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.00s] ---

init. Evaluando CNN MULTIESPECTRAL - Loss: focal_loss - Threshold: 0.45

--- [Fecha/hora inicio=20260325_1154] ---

Curva ROC - best_model_final_MULTIESPECTRAL_focal_loss.keras (t=0.45)

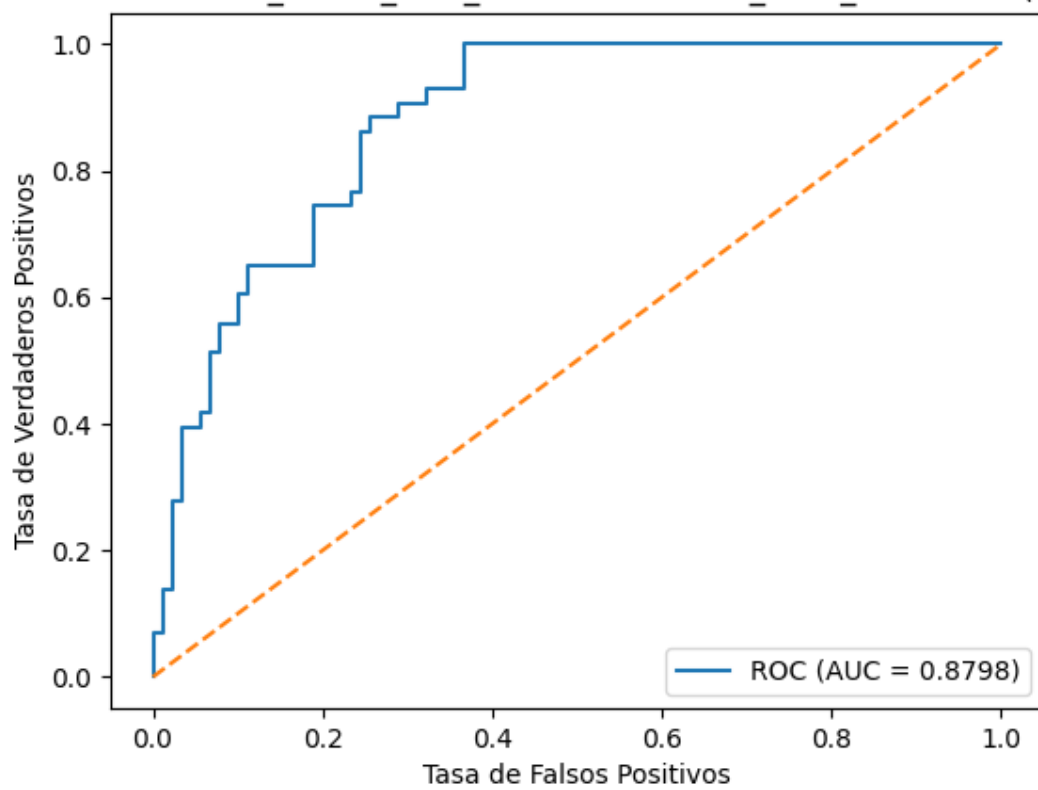


Figure 13: alt text

```

--- [T=0.00s] ---
1. Cargando datos...

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.00s] ---
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.00s] ---
1. Carga de datos cnn Multiespectral

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.03s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.03s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=0.03s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=54.56s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

--- [Fecha/hora inicio=20260325_1154] ---

```

```

--- [T=54.91s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=54.91s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=54.92s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=54.94s] ---
2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=54.98s] ---
1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:55:35.836245: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=55.13s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=55.08s] ---
3. Prediciendo...

--- [Fecha/hora inicio=20260325_1154] ---

--- [T=55.20s] ---
1. Iniciando predicción del modelo CNN...
5/5          1s 172ms/step

--- [Fecha/hora inicio=20260325_1154] ---

```

```
--- [T=56.19s] ---
```

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

```
[[62 28]
 [ 4 39]]
```

```
--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---
```

	precision	recall	f1-score	support
Plaga	0.94	0.69	0.79	90
Sana	0.58	0.91	0.71	43
accuracy			0.76	133
macro avg	0.76	0.80	0.75	133
weighted avg	0.82	0.76	0.77	133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.9393939393939394,
    "recall": 0.6888888888888889,
    "f1-score": 0.7948717948717948,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.582089552238806,
    "recall": 0.9069767441860465,
    "f1-score": 0.7090909090909091,
    "support": 43.0
  },
  "accuracy": 0.7593984962406015,
  "macro avg": {
```

```

        "precision": 0.7607417458163728,
        "recall": 0.7979328165374677,
        "f1-score": 0.751981351981352,
        "support": 133.0
    },
    "weighted avg": {
        "precision": 0.823874475877618,
        "recall": 0.7593984962406015,
        "f1-score": 0.7671381250328618,
        "support": 133.0
    }
}

```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_focal_loss.keras

- **Fecha:** 2026-03-25 11:55:39
- **Modelo:** best_model_final_MULTIESPECTRAL_focal_loss.keras
- **Umbral de decisión:** 0.45

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.94	0.69	0.79	90
Sana	0.58	0.91	0.71	43
accuracy			0.76	133
macro avg	0.76	0.80	0.75	133
weighted avg	0.82	0.76	0.77	133

Generado automáticamente por el sistema de detección de plagas.

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:55:39
- **Modelo:** report_best_model_final_MULTIESPECTRAL_focal_loss.keras_20260325_1155_t45_confusi

```

[[62 28]
 [ 4 39]]

```

Matriz de Confusión (best_model_final_MULTIESPECTRAL_focal_loss.keras

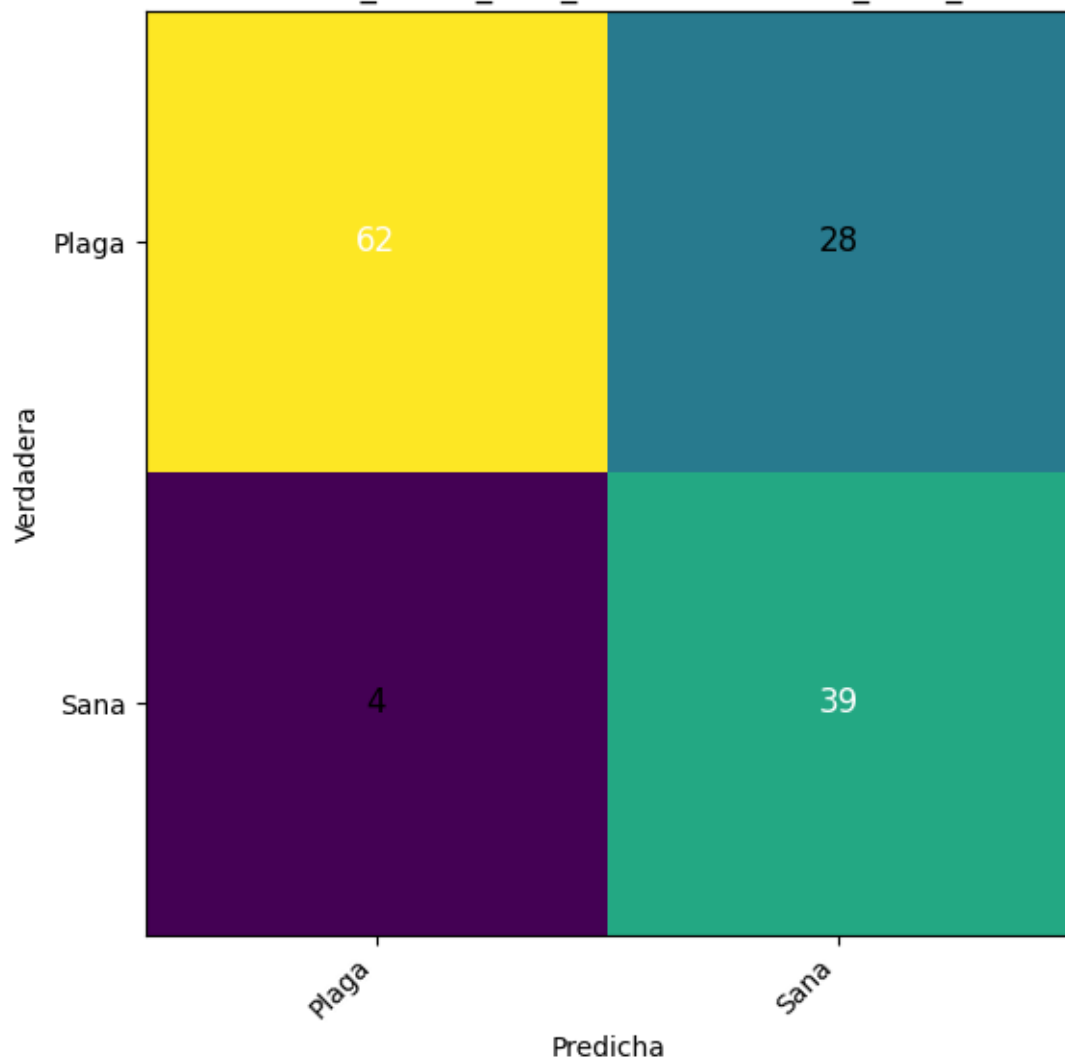


Figure 14: alt text

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.5214

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

2.2 Consola:

```
2026-03-25 11:56:17.500472: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 11:56:18.825798: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN MULTIESPECTRAL - Loss: focal_loss - Threshold: 0.5214
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.00s] ---
```

```
1. Cargando datos...
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.00s] ---
```

```
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cmn Multispectral
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.03s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```

```
--- [T=0.03s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```
--- [Fecha/hora inicio=20260325_1156] ---
```



```

--- [T=0.03s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=55.79s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.11s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.11s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.12s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.14s] ---
2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.18s] ---
1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:57:16.150520: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

```

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.32s] ---

2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.27s] ---

3. Prediciendo...

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=56.40s] ---

1. Iniciando predicción del modelo CNN...

5/5 1s 168ms/step

--- [Fecha/hora inicio=20260325_1156] ---

--- [T=57.37s] ---

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

[[64 26]

[5 38]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.93	0.71	0.81	90
Sana	0.59	0.88	0.71	43
accuracy			0.77	133
macro avg	0.76	0.80	0.76	133
weighted avg	0.82	0.77	0.77	133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.927536231884058,
    "recall": 0.7111111111111111,
    "f1-score": 0.8050314465408805,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.59375,
    "recall": 0.8837209302325582,
    "f1-score": 0.7102803738317757,
    "support": 43.0
  },
  "accuracy": 0.7669172932330827,
  "macro avg": {
    "precision": 0.760643115942029,
    "recall": 0.7974160206718346,
    "f1-score": 0.757655910186328,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.819620382477934,
    "recall": 0.7669172932330827,
    "f1-score": 0.7743976410785383,
    "support": 133.0
  }
}
```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_focal_loss.keras

- **Fecha:** 2026-03-25 11:57:24
- **Modelo:** best_model_final_MULTIESPECTRAL_focal_loss.keras
- **Umbral de decisión:** 0.5214

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.93	0.71	0.81	90
Sana	0.59	0.88	0.71	43
accuracy			0.77	133
macro avg	0.76	0.80	0.76	133
weighted avg	0.82	0.77	0.77	133

Generado automáticamente por el sistema de detección de plagas.

Matriz de Confusión (best_model_final_MULTIESPECTRAL_focal_loss.keras,

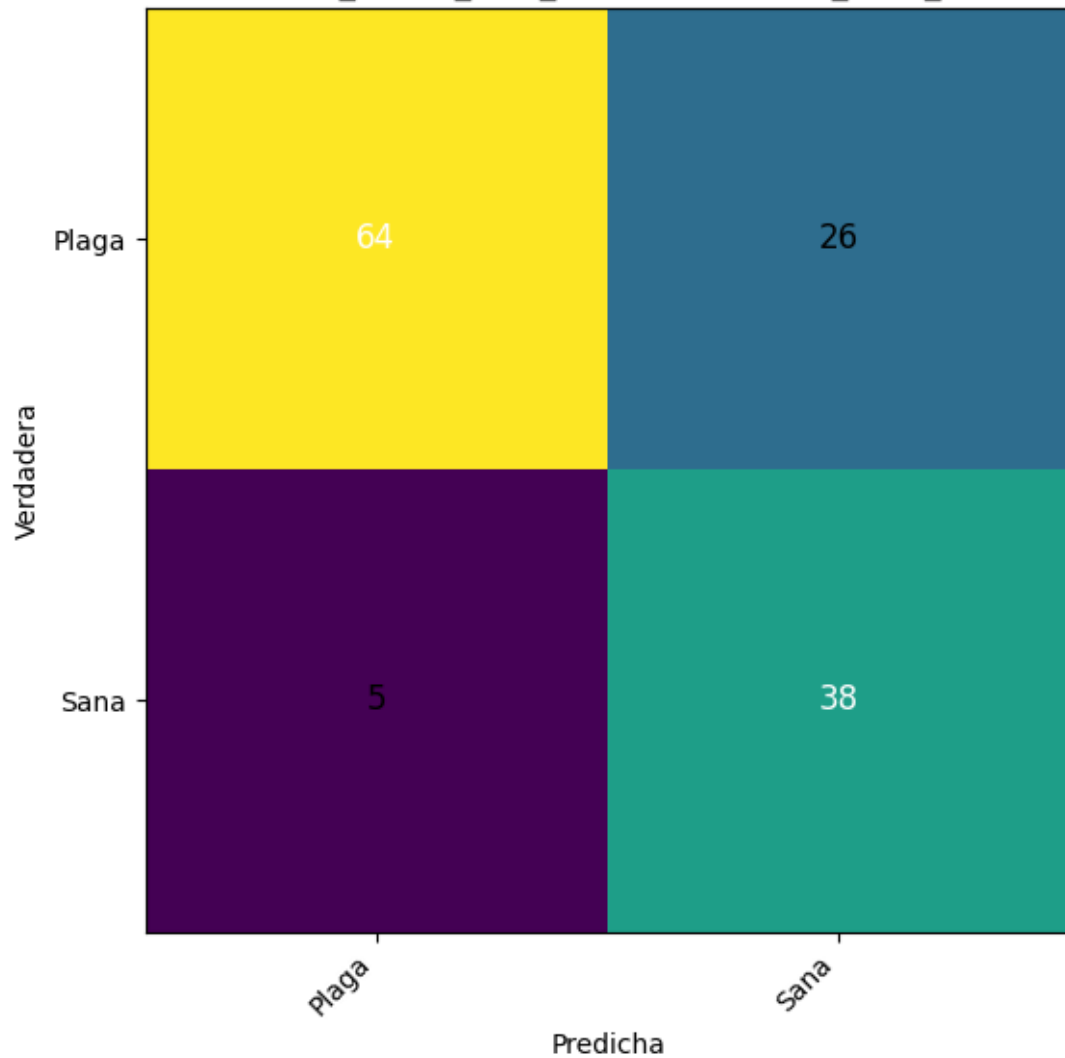


Figure 15: alt text

2.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:57:24
- **Modelo:** report_best_model_final_MULTIESPECTRAL_focal_loss.keras_20260325_1157_t52_confusi

```
[[64 26]
 [ 5 38]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

3.2 Consola:

```
2026-03-25 11:58:20.920530: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 11:58:22.393714: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1158] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN MULTIESPECTRAL - Loss: focal_loss - Threshold: 0.7
```

```
--- [Fecha/hora inicio=20260325_1158] ---
```

```
--- [T=0.00s] ---
```

```
1. Cargando datos...
```

```
--- [Fecha/hora inicio=20260325_1158] ---
```

```
--- [T=0.00s] ---
```

```
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)
```

```
--- [Fecha/hora inicio=20260325_1158] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn Multispectral
```

```
--- [Fecha/hora inicio=20260325_1158] ---
```

```
--- [T=0.04s] ---
```

```
2. Filtrado y extracción de datos
```

```

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=0.04s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=57.84s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.25s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.25s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.26s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.30s] ---
2. Cargando modelo...

```

```

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.34s] ---
1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:59:21.889262: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.55s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.51s] ---
3. Prediciendo...

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=58.63s] ---
1. Iniciando predicción del modelo CNN...
5/5          1s 207ms/step

--- [Fecha/hora inicio=20260325_1158] ---

--- [T=59.85s] ---
4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[80 10]
 [15 28]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

```

	precision	recall	f1-score	support
Plaga	0.84	0.89	0.86	90
Sana	0.74	0.65	0.69	43
accuracy			0.81	133
macro avg	0.79	0.77	0.78	133

weighted avg	0.81	0.81	0.81	133
--------------	------	------	------	-----

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.8421052631578947,
    "recall": 0.8888888888888888,
    "f1-score": 0.8648648648648649,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.7368421052631579,
    "recall": 0.6511627906976745,
    "f1-score": 0.691358024691358,
    "support": 43.0
  },
  "accuracy": 0.8120300751879699,
  "macro avg": {
    "precision": 0.7894736842105263,
    "recall": 0.7700258397932817,
    "f1-score": 0.7781114447781114,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.8080728136129797,
    "recall": 0.8120300751879699,
    "f1-score": 0.8087686684177912,
    "support": 133.0
  }
}
```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_focal_loss.ke

- Fecha: 2026-03-25 11:59:29

- **Modelo:** best_model_final_MULTIESPECTRAL_focal_loss.keras
- **Umbral de decisión:** 0.7

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.84	0.89	0.86	90
Sana	0.74	0.65	0.69	43
accuracy			0.81	133
macro avg	0.79	0.77	0.78	133
weighted avg	0.81	0.81	0.81	133

Generado automáticamente por el sistema de detección de plagas.

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:59:29
- **Modelo:** report_best_model_final_MULTIESPECTRAL_focal_loss.keras_20260325_1159_t70_confusi

```
[[80 10]
 [15 28]]
```

Generado automáticamente por el sistema de detección de plagas.

CNN MULTIESPECTRAL BINARY CROSS ENTROPY

1 UMBRAL 0.45

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multiespectral\TTADDA-dataset\TTADDA_NARO
```

1.2 Consola:

```
2026-03-25 11:36:24.426733: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 11:36:25.880040: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1136] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN MULTIESPECTRAL - Loss: binary_crossentropy - Threshold: 0.45
```

Matriz de Confusión (best_model_final_MULTIESPECTRAL_focal_loss.keras

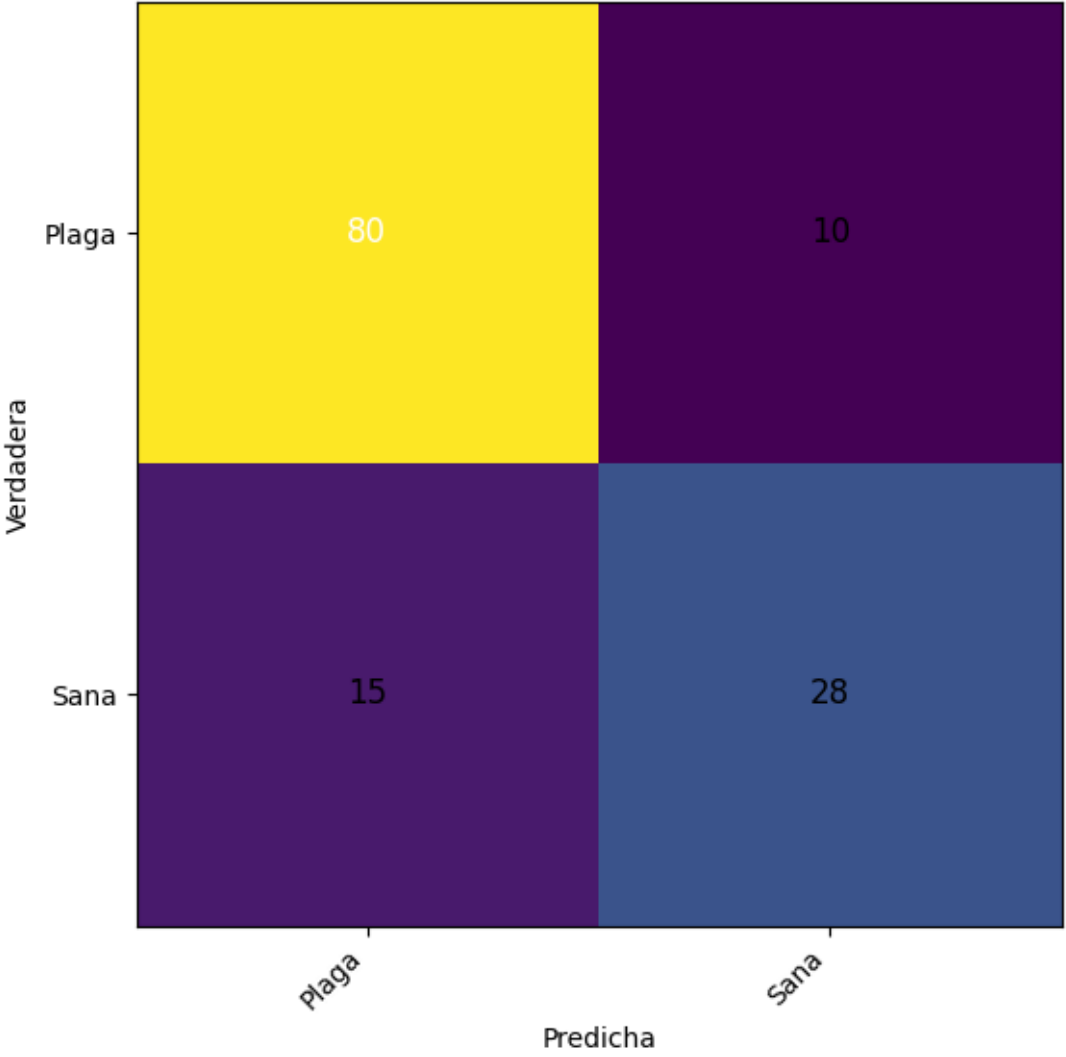


Figure 16: alt text

z de Confusión (best_model_final_MULTIESPECTRAL_binary_crossentropy.

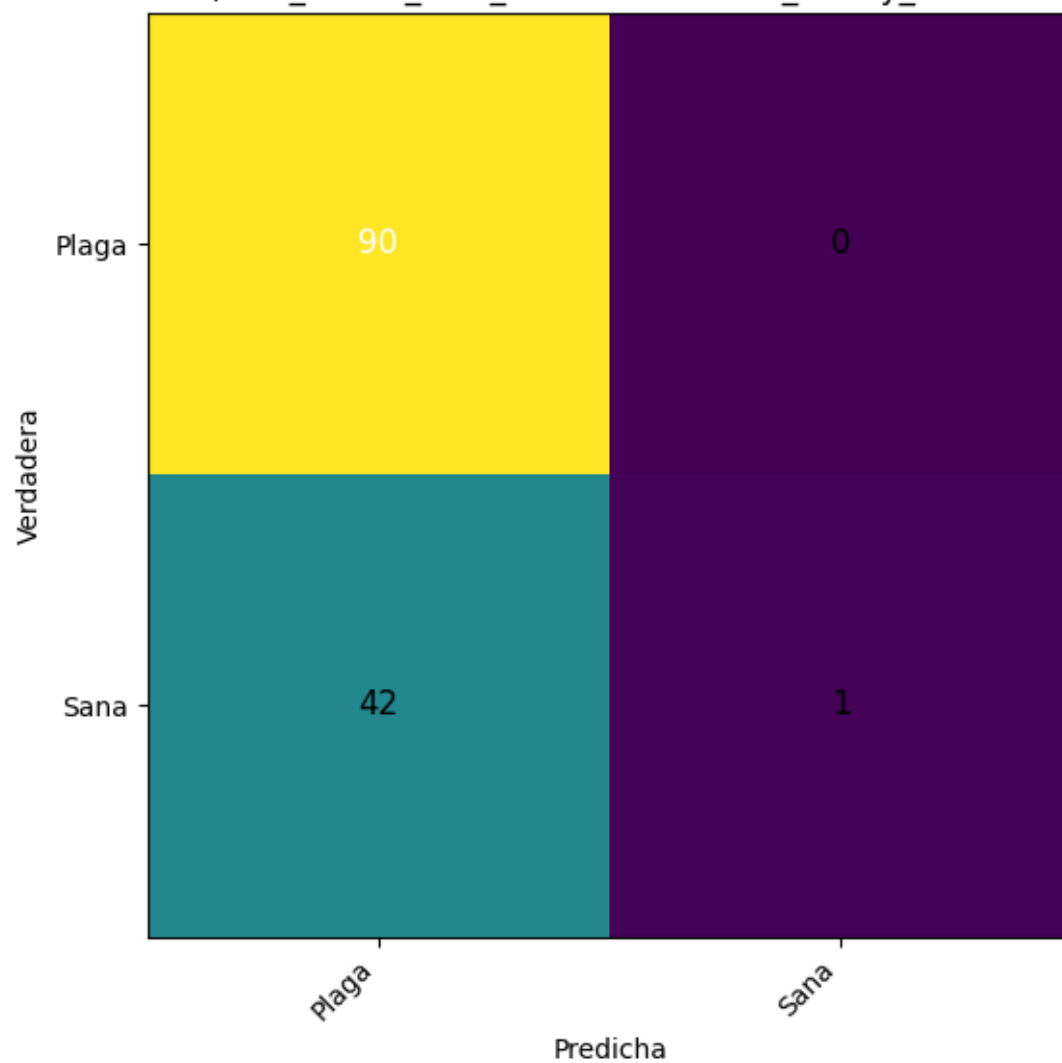


Figure 17: alt text

```

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.00s] ---
1. Cargando datos...

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.00s] ---
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.00s] ---
1. Carga de datos cmn Multispectral

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.04s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.04s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.40s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multispectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

```

```

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.77s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.77s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.77s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.80s] ---
2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.85s] ---
1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:37:21.897698: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.97s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=54.93s] ---
3. Prediciendo...

--- [Fecha/hora inicio=20260325_1136] ---

--- [T=55.05s] ---
1. Iniciando predicción del modelo CNN...
5/5          1s 169ms/step

```

```
--- [Fecha/hora inicio=20260325_1136] ---
```

```
--- [T=56.03s] ---
```

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

```
[[90  0]
 [42  1]]
```

```
--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---
```

	precision	recall	f1-score	support
Plaga	0.68	1.00	0.81	90
Sana	1.00	0.02	0.05	43
accuracy			0.68	133
macro avg	0.84	0.51	0.43	133
weighted avg	0.78	0.68	0.56	133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6818181818181818,
    "recall": 1.0,
    "f1-score": 0.8108108108108109,
    "support": 90.0
  },
  "Sana": {
    "precision": 1.0,
    "recall": 0.023255813953488372,
    "f1-score": 0.045454545454545456,
    "support": 43.0
  },
}
```

```

    "accuracy": 0.6842105263157895,
    "macro avg": {
        "precision": 0.8409090909090908,
        "recall": 0.5116279069767442,
        "f1-score": 0.42813267813267813,
        "support": 133.0
    },
    "weighted avg": {
        "precision": 0.784688995215311,
        "recall": 0.6842105263157895,
        "f1-score": 0.5633648002069055,
        "support": 133.0
    }
}

```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_binary_cross

- **Fecha:** 2026-03-25 11:51:26
- **Modelo:** best_model_final_MULTIESPECTRAL_binary_crossentropy.keras
- **Umbral de decisión:** 0.45

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.68	1.00	0.81	90
Sana	1.00	0.02	0.05	43
accuracy			0.68	133
macro avg	0.84	0.51	0.43	133
weighted avg	0.78	0.68	0.56	133

Generado automáticamente por el sistema de detección de plagas.

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:51:26
- **Modelo:** report_best_model_final_MULTIESPECTRAL_binary_crossentropy.keras_20260325_1137_t4

de Confusión (best_model_final_MULTIESPECTRAL_binary_crossentropy.l

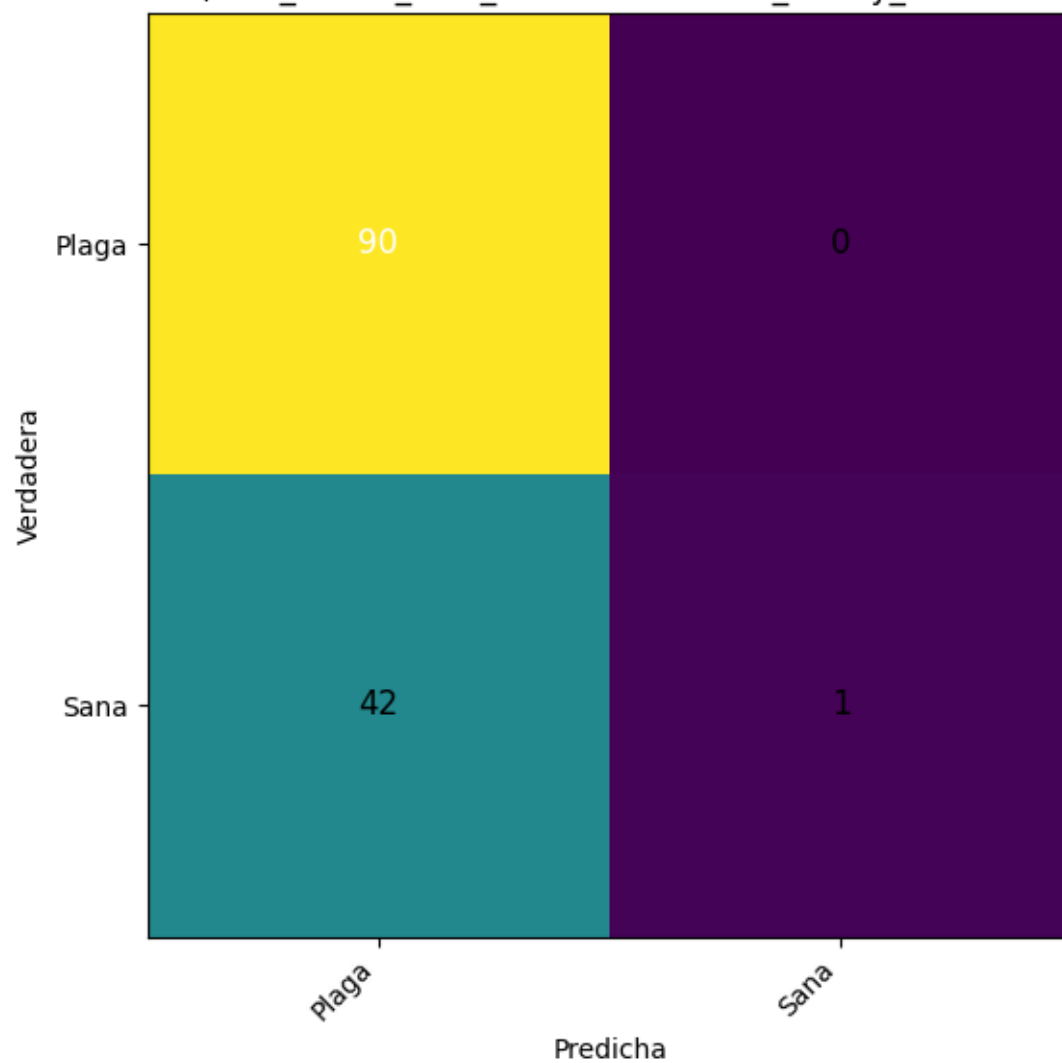


Figure 18: alt text


```
[[90  0]
 [42  1]]
```

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.50

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

2.2 Consola:

```
2026-03-25 11:51:41.194668: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 11:51:42.698857: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN MULTIESPECTRAL - Loss: binary_crossentropy - Threshold: 0.5
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.00s] ---
```

```
1. Cargando datos...
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.00s] ---
```

```
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn Multispectral
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.03s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1151] ---
```

```
--- [T=0.03s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=0.03s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=55.53s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=55.99s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=55.99s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=55.99s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=56.03s] ---
2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=56.08s] ---

```

```

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:52:39.938573: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=56.20s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=56.15s] ---
3. Prediciendo...

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=56.28s] ---
1. Iniciando predicción del modelo CNN...
5/5          1s 171ms/step

--- [Fecha/hora inicio=20260325_1151] ---

--- [T=57.24s] ---
4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[90  0]
 [42  1]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---
              precision    recall  f1-score   support

     Plaga         0.68        1.00        0.81         90
     Sana          1.00        0.02        0.05         43

 accuracy                   0.68         133
 macro avg              0.84        0.51        0.43         133
weighted avg              0.78        0.68        0.56         133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

```

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6818181818181818,
    "recall": 1.0,
    "f1-score": 0.8108108108108109,
    "support": 90.0
  },
  "Sana": {
    "precision": 1.0,
    "recall": 0.023255813953488372,
    "f1-score": 0.045454545454545456,
    "support": 43.0
  },
  "accuracy": 0.6842105263157895,
  "macro avg": {
    "precision": 0.8409090909090908,
    "recall": 0.5116279069767442,
    "f1-score": 0.42813267813267813,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.784688995215311,
    "recall": 0.6842105263157895,
    "f1-score": 0.5633648002069055,
    "support": 133.0
  }
}
```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_binary_cross

- **Fecha:** 2026-03-25 11:52:49
- **Modelo:** best_model_final_MULTIESPECTRAL_binary_crossentropy.keras
- **Umbral de decisión:** 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.68	1.00	0.81	90
Sana	1.00	0.02	0.05	43
accuracy			0.68	133
macro avg	0.84	0.51	0.43	133
weighted avg	0.78	0.68	0.56	133

Generado automáticamente por el sistema de detección de plagas.

z de Confusión (best_model_final_MULTIESPECTRAL_binary_crossentropy.

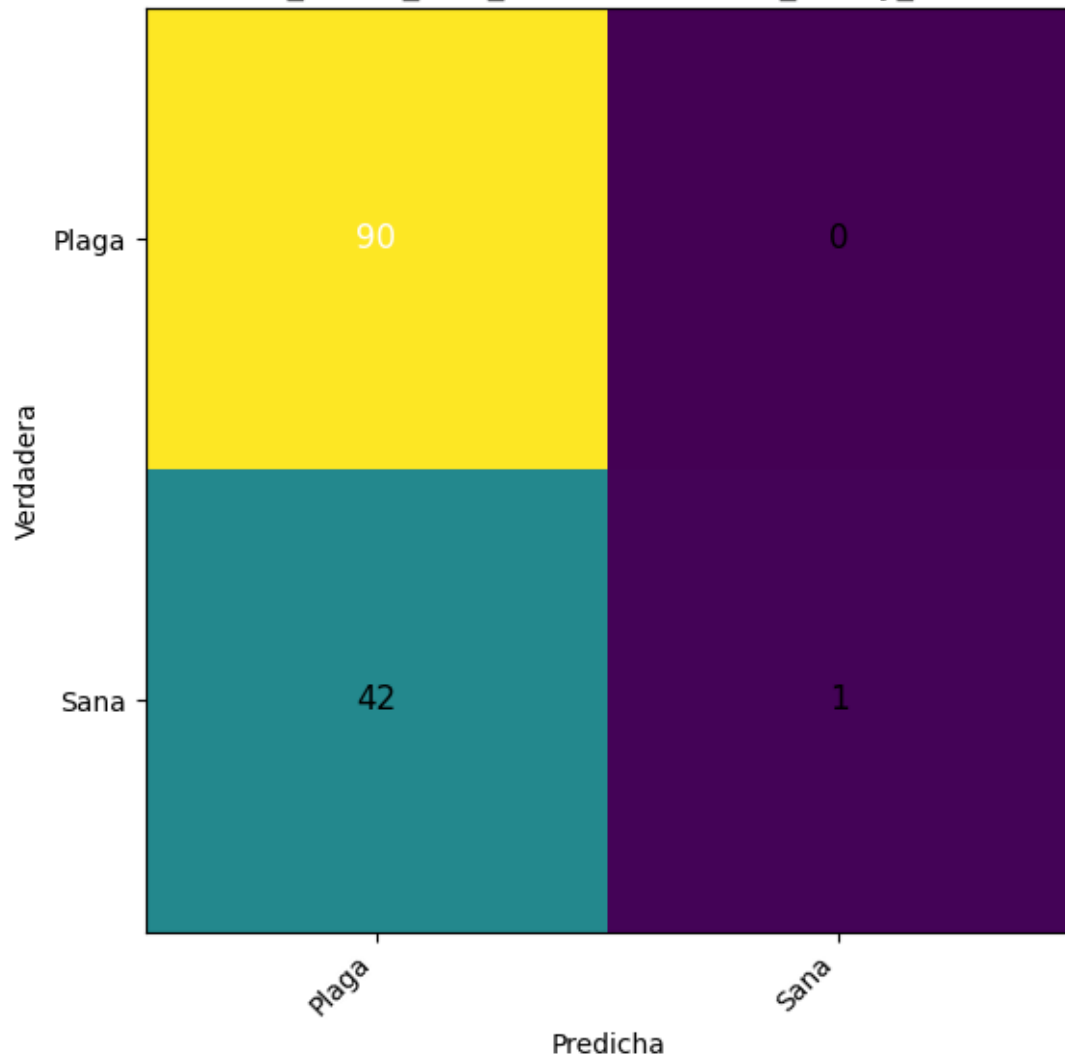


Figure 19: alt text

2.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:52:49
- **Modelo:** report_best_model_final_MULTIESPECTRAL_binary_crossentropy.keras_20260325_1152_t5

```
[[90  0]
 [42  1]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

3.2 Consola:

```
2026-03-25 11:53:12.165117: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 11:53:13.643264: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1153] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN MULTIESPECTRAL - Loss: binary_crossentropy - Threshold: 0.7
```

```
--- [Fecha/hora inicio=20260325_1153] ---
```

```
--- [T=0.00s] ---
```

```
1. Cargando datos...
```

```
--- [Fecha/hora inicio=20260325_1153] ---
```

```
--- [T=0.00s] ---
```

```
1. Ejecutando extracción y split de datos (extract_data_to_img_for_train)
```

```
--- [Fecha/hora inicio=20260325_1153] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn Multispectral
```

```
--- [Fecha/hora inicio=20260325_1153] ---
```

```
--- [T=0.04s] ---
```

```
2. Filtrado y extracción de datos
```

```

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=0.04s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.29s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.63s] ---
1.1. Clases detectadas: 0: Plaga, 1: Sana

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.63s] ---
2. Calculando pesos de clase (Estrategia: Class Weighting)

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.64s] ---
2.1. Pesos de Clase Calculados (train_counts): {np.int64(0): np.float64(0.7388888888888889)}

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.67s] ---
2. Cargando modelo...

```

```

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.73s] ---
1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_MULTIESPE
2026-03-25 11:54:10.608838: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.85s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.79s] ---
3. Prediciendo...

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=55.95s] ---
1. Iniciando predicción del modelo CNN...
5/5          1s 174ms/step

--- [Fecha/hora inicio=20260325_1153] ---

--- [T=56.91s] ---
4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[90  0]
 [43  0]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

```

	precision	recall	f1-score	support
Plaga	0.68	1.00	0.81	90
Sana	0.00	0.00	0.00	43
accuracy			0.68	133
macro avg	0.34	0.50	0.40	133

weighted avg	0.46	0.68	0.55	133
--------------	------	------	------	-----

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6766917293233082,
    "recall": 1.0,
    "f1-score": 0.8071748878923767,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.0,
    "recall": 0.0,
    "f1-score": 0.0,
    "support": 43.0
  },
  "accuracy": 0.6766917293233082,
  "macro avg": {
    "precision": 0.3383458646616541,
    "recall": 0.5,
    "f1-score": 0.40358744394618834,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.4579116965345695,
    "recall": 0.6766917293233082,
    "f1-score": 0.5462085707542399,
    "support": 133.0
  }
}
```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_MULTIESPECTRAL_binary_cross

- Fecha: 2026-03-25 11:54:15

- **Modelo:** best_model_final_MULTIESPECTRAL_binary_crossentropy.keras
- **Umbral de decisión:** 0.7

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.68	1.00	0.81	90
Sana	0.00	0.00	0.00	43
accuracy			0.68	133
macro avg	0.34	0.50	0.40	133
weighted avg	0.46	0.68	0.55	133

Generado automáticamente por el sistema de detección de plagas.

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 11:54:15
- **Modelo:** report_best_model_final_MULTIESPECTRAL_binary_crossentropy.keras_20260325_1154_t7

```
[[90  0]
 [43  0]]
```

Generado automáticamente por el sistema de detección de plagas.

CNN RGB FOCAL LOSS

1 UMBRAL 0.4

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multiespectral\TTADDA-dataset\TTADDA_NARO
```

1.2 Consola:

```
2026-03-25 14:25:08.962150: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 14:25:10.199941: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1425] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN RGB - Loss: focal_loss - Threshold: 0.45
```

Matriz de Confusión (best_model_final_MULTIESPECTRAL_binary_crossentropy.

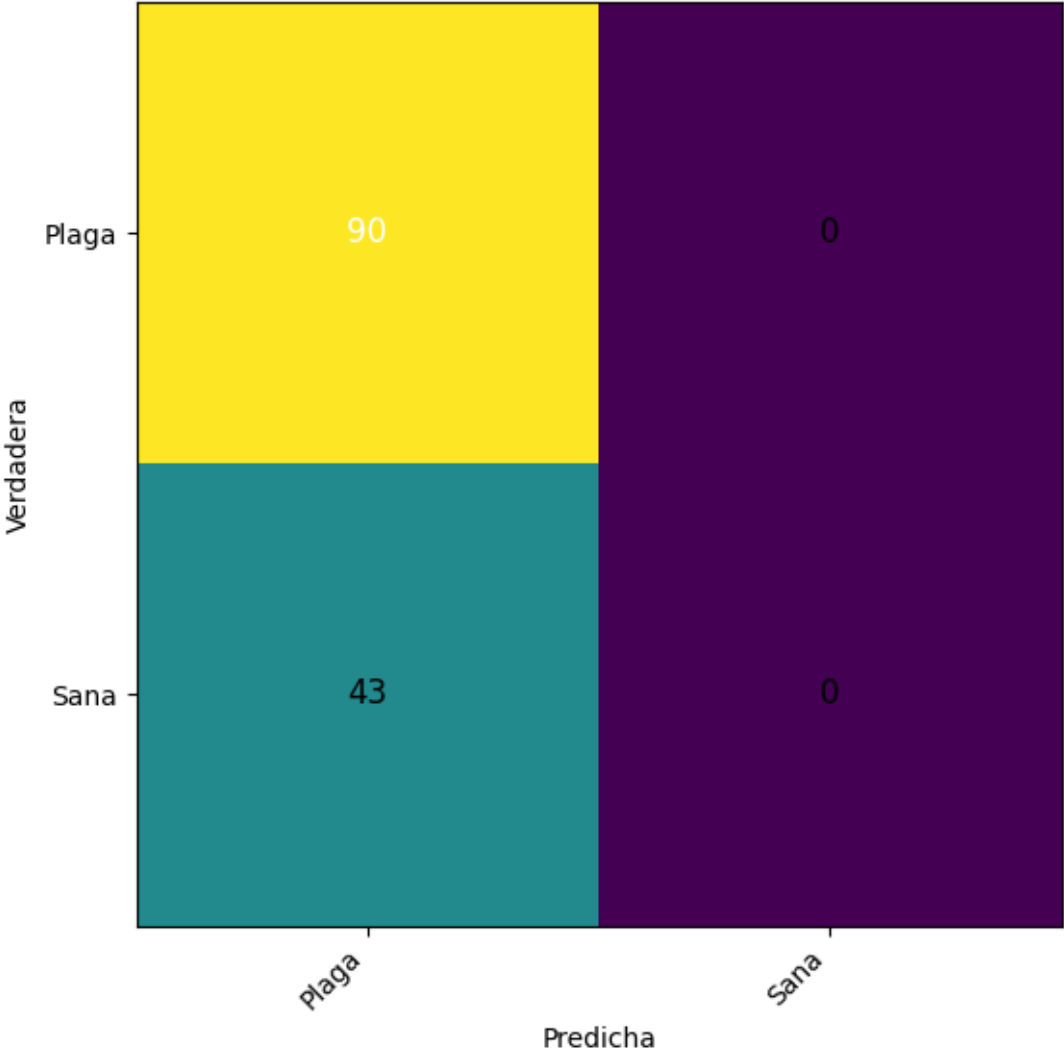


Figure 20: alt text

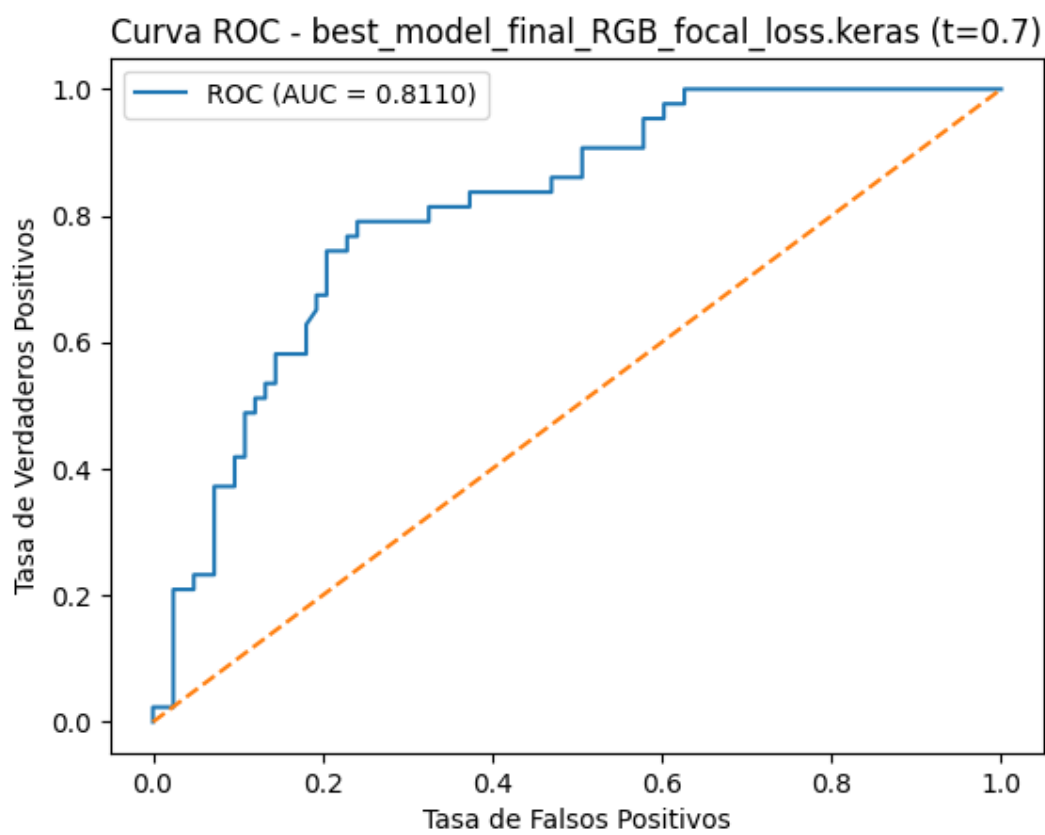


Figure 21: alt text

```

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=0.00s] ---
1. Extrayendo datos (MISMO pipeline que training)...

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=0.00s] ---
1. Carga de datos cnn RGB

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=0.03s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=0.03s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=0.03s] ---
2.2. Archivos seleccionados: ['RGB.tif']
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

```

Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=29.28s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=27.90s] ---

2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=27.95s] ---

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_focal
2026-03-25 14:25:38.524889: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a variety of CPU instructions specific to the current CPU (AMD Ryzen Threadripper 7970X). To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

```

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=28.07s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=28.02s] ---
3. Prediciendo...
4/4          1s 217ms/step

--- DEBUG PROBABILIDADES ---
MIN: 0.5013404
MAX: 0.50146043
MEAN: 0.50139904
PERCENTILES: [0.50134039 0.50136393 0.50139499 0.50143668 0.50146043]
X_val stats:
MIN: 0.0
MAX: 1.0
MEAN: 0.21663862
STD: 3.75954e-05

--- [Fecha/hora inicio=20260325_1425] ---

--- [T=29.11s] ---
4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[ 0 83]
 [ 0 43]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

```

	precision	recall	f1-score	support
Plaga	0.00	0.00	0.00	83
Sana	0.34	1.00	0.51	43
accuracy			0.34	126
macro avg	0.17	0.50	0.25	126
weighted avg	0.12	0.34	0.17	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.0,
    "recall": 0.0,
    "f1-score": 0.0,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.3412698412698413,
    "recall": 1.0,
    "f1-score": 0.5088757396449705,
    "support": 43.0
  },
  "accuracy": 0.3412698412698413,
  "macro avg": {
    "precision": 0.17063492063492064,
    "recall": 0.5,
    "f1-score": 0.25443786982248523,
    "support": 126.0
  },
  "weighted avg": {
    "precision": 0.11646510456034266,
    "recall": 0.3412698412698413,
    "f1-score": 0.17366394289471215,
    "support": 126.0
  }
}
```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_focal_loss.keras

- **Fecha:** 2026-03-25 14:29:54
- **Modelo:** best_model_final_RGB_focal_loss.keras

- Umbral de decisión: 0.45

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.00	0.00	0.00	83
Sana	0.34	1.00	0.51	43
accuracy			0.34	126
macro avg	0.17	0.50	0.25	126
weighted avg	0.12	0.34	0.17	126

Generado automáticamente por el sistema de detección de plagas.

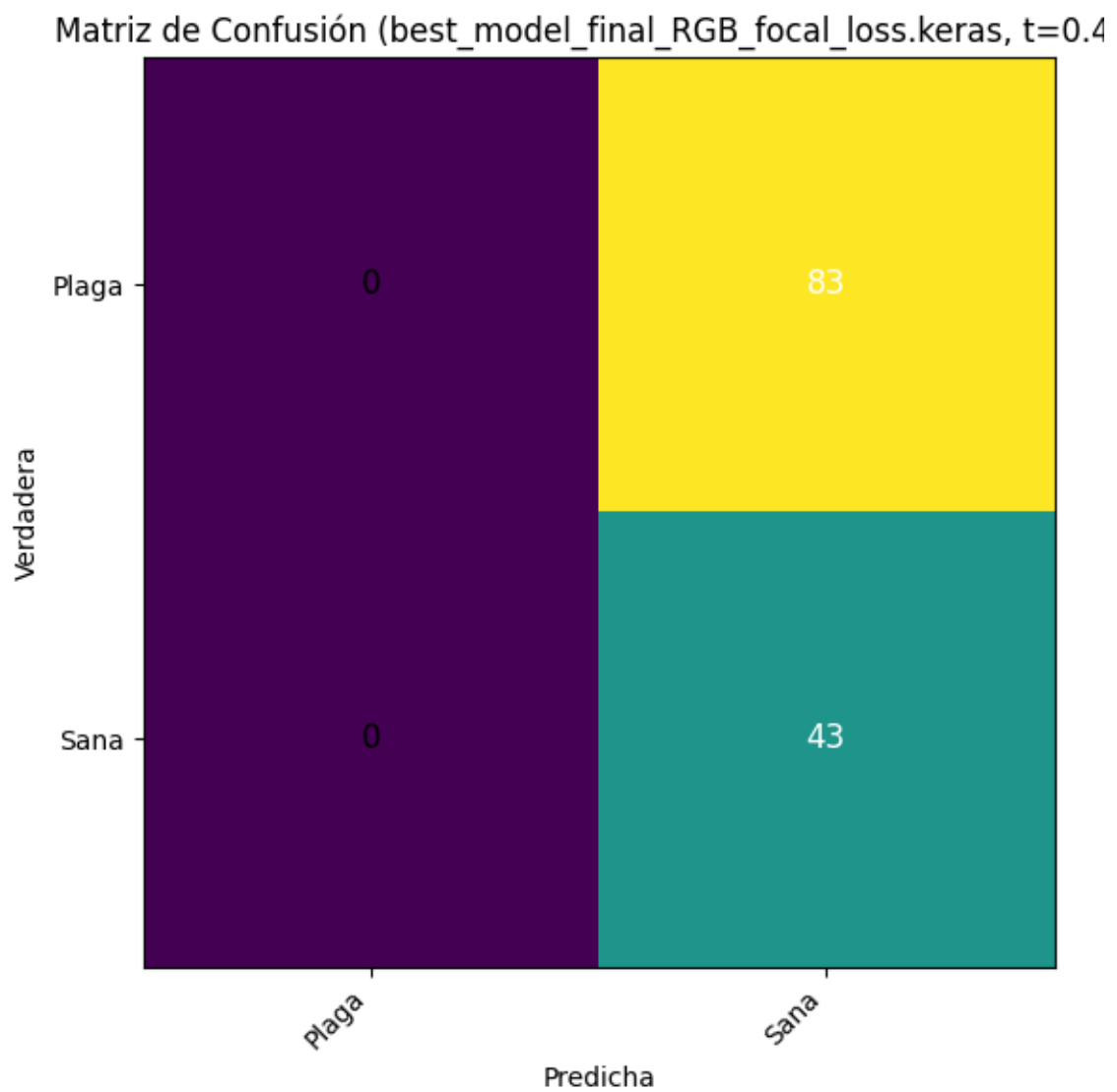


Figure 22: alt text

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 14:29:54
- **Modelo:** report_best_model_final_RGB_focal_loss.keras_20260325_1429_t45_confusion.md

```
[[ 0 83]
 [ 0 43]]
```

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.5214

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

2.2 Consola:

```
2026-03-25 14:40:57.244616: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 14:40:58.908067: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN RGB - Loss: focal_loss - Threshold: 0.5214
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```

```
--- [T=0.00s] ---
```

```
1. Extrayendo datos (MISMO pipeline que training)...
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn RGB
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```

```
--- [T=0.08s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```


--- [Fecha/hora inicio=20260325_1440] ---

--- [T=40.11s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1440] ---

--- [T=37.91s] ---

2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1440] ---

--- [T=38.05s] ---

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_focal

2026-03-25 14:41:37.603179: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1440] ---

--- [T=38.18s] ---

2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1440] ---

--- [T=38.05s] ---

3. Prediciendo...

4/4 1s 214ms/step

--- DEBUG PROBABILIDADES ---

MIN: 0.5013404

MAX: 0.50146043

MEAN: 0.50139904

PERCENTILES: [0.50134039 0.50136393 0.50139499 0.50143668 0.50146043]

```
X_val stats:
MIN: 0.0
MAX: 1.0
MEAN: 0.21663862
STD: 3.75954e-05
```

```
--- [Fecha/hora inicio=20260325_1440] ---
```

```
--- [T=39.10s] ---
```

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:

```
[[83  0]
 [43  0]]
```

```
--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---
```

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6587301587301587,
    "recall": 1.0,
    "f1-score": 0.7942583732057417,
    "support": 83.0
  },
  "Sana": {
```

```

        "precision": 0.0,
        "recall": 0.0,
        "f1-score": 0.0,
        "support": 43.0
    },
    "accuracy": 0.6587301587301587,
    "macro avg": {
        "precision": 0.32936507936507936,
        "recall": 0.5,
        "f1-score": 0.39712918660287083,
        "support": 126.0
    },
    "weighted avg": {
        "precision": 0.4339254220206601,
        "recall": 0.6587301587301587,
        "f1-score": 0.5232019442545758,
        "support": 126.0
    }
}

```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_focal_loss.keras

- **Fecha:** 2026-03-25 14:42:25
- **Modelo:** best_model_final_RGB_focal_loss.keras
- **Umbral de decisión:** 0.5214

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Generado automáticamente por el sistema de detección de plagas.

2.3.3 Matriz de confusión

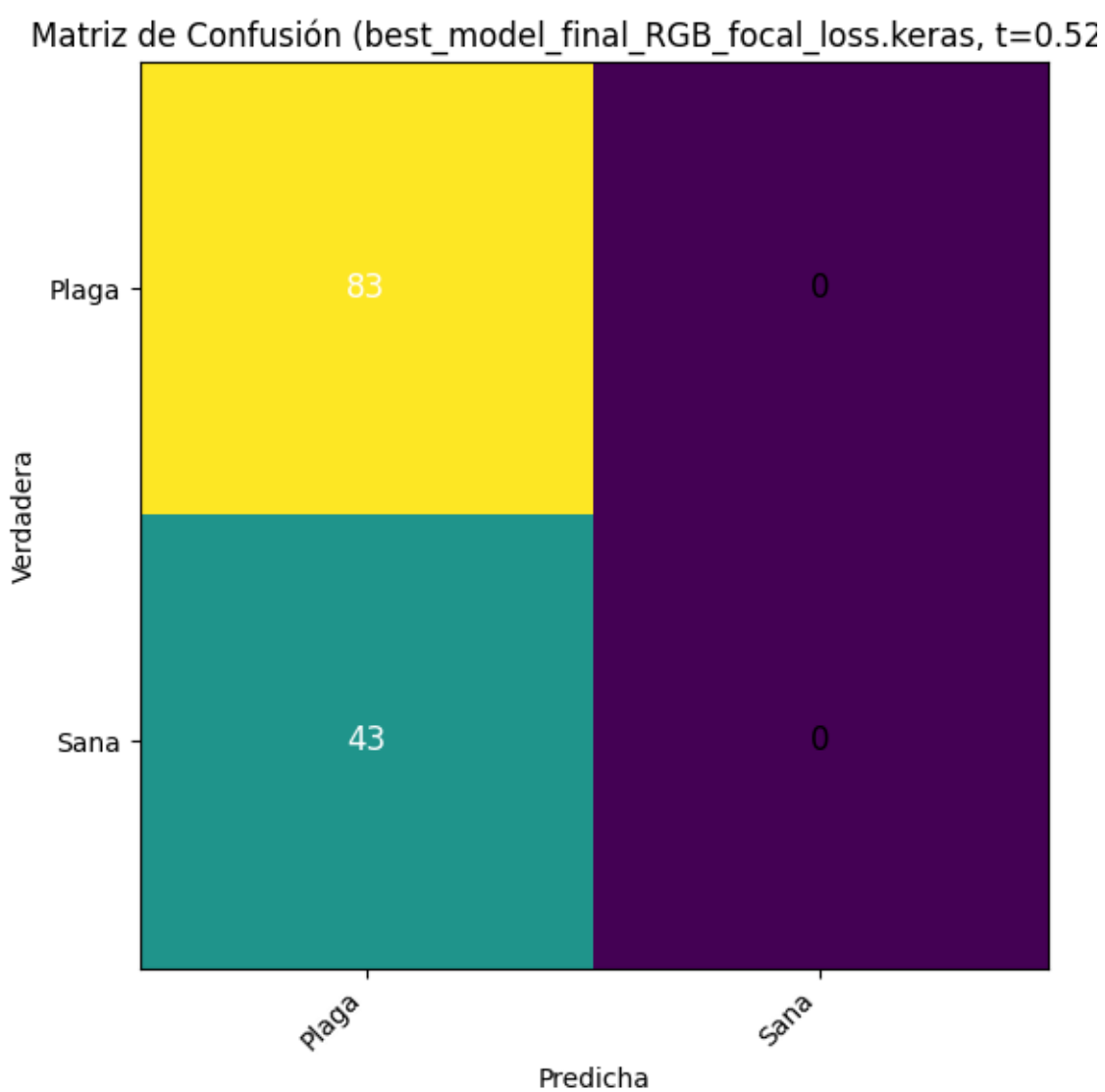


Figure 23: alt text

Matriz de confusión

- **Fecha:** 2026-03-25 14:42:25
- **Modelo:** report_best_model_final_RGB_focal_loss.keras_20260325_1441_t52_confusion.md

```
[[83  0]
 [43  0]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

3.2 Consola:

```
2026-03-25 14:43:29.855790: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 14:43:31.164865: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1443] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN RGB - Loss: focal_loss - Threshold: 0.7
```

```
--- [Fecha/hora inicio=20260325_1443] ---
```

```
--- [T=0.00s] ---
```

```
1. Extrayendo datos (MISMO pipeline que training)...
```

```
--- [Fecha/hora inicio=20260325_1443] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn RGB
```

```
--- [Fecha/hora inicio=20260325_1443] ---
```

```
--- [T=0.03s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1443] ---
```

```
--- [T=0.03s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```


--- [Fecha/hora inicio=20260325_1443] ---

--- [T=0.03s] ---

2.2. Archivos seleccionados: ['RGB.tif']

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

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Imagen inválida: 4 canales (esperado 3)

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Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=31.00s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=29.44s] ---

2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=29.51s] ---

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_focal
2026-03-25 14:44:01.062842: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a variety of CPU instructions specific to the current CPU architecture. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=29.59s] ---

2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=29.52s] ---

3. Prediciendo...

4/4 1s 206ms/step

--- DEBUG PROBABILIDADES ---

MIN: 0.5013404

MAX: 0.50146043

MEAN: 0.50139904

PERCENTILES: [0.50134039 0.50136393 0.50139499 0.50143668 0.50146043]

X_val stats:

MIN: 0.0

MAX: 1.0

MEAN: 0.21663862

STD: 3.75954e-05

--- [Fecha/hora inicio=20260325_1443] ---

--- [T=30.51s] ---

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

```
[[83  0]
 [43  0]]
```

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6587301587301587,
    "recall": 1.0,
    "f1-score": 0.7942583732057417,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.0,
    "recall": 0.0,
    "f1-score": 0.0,
```

```

        "support": 43.0
    },
    "accuracy": 0.6587301587301587,
    "macro avg": {
        "precision": 0.32936507936507936,
        "recall": 0.5,
        "f1-score": 0.39712918660287083,
        "support": 126.0
    },
    "weighted avg": {
        "precision": 0.4339254220206601,
        "recall": 0.6587301587301587,
        "f1-score": 0.5232019442545758,
        "support": 126.0
    }
}

```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_focal_loss.keras

- **Fecha:** 2026-03-25 14:45:13
- **Modelo:** best_model_final_RGB_focal_loss.keras
- **Umbral de decisión:** 0.7

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Generado automáticamente por el sistema de detección de plagas.

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 14:45:13

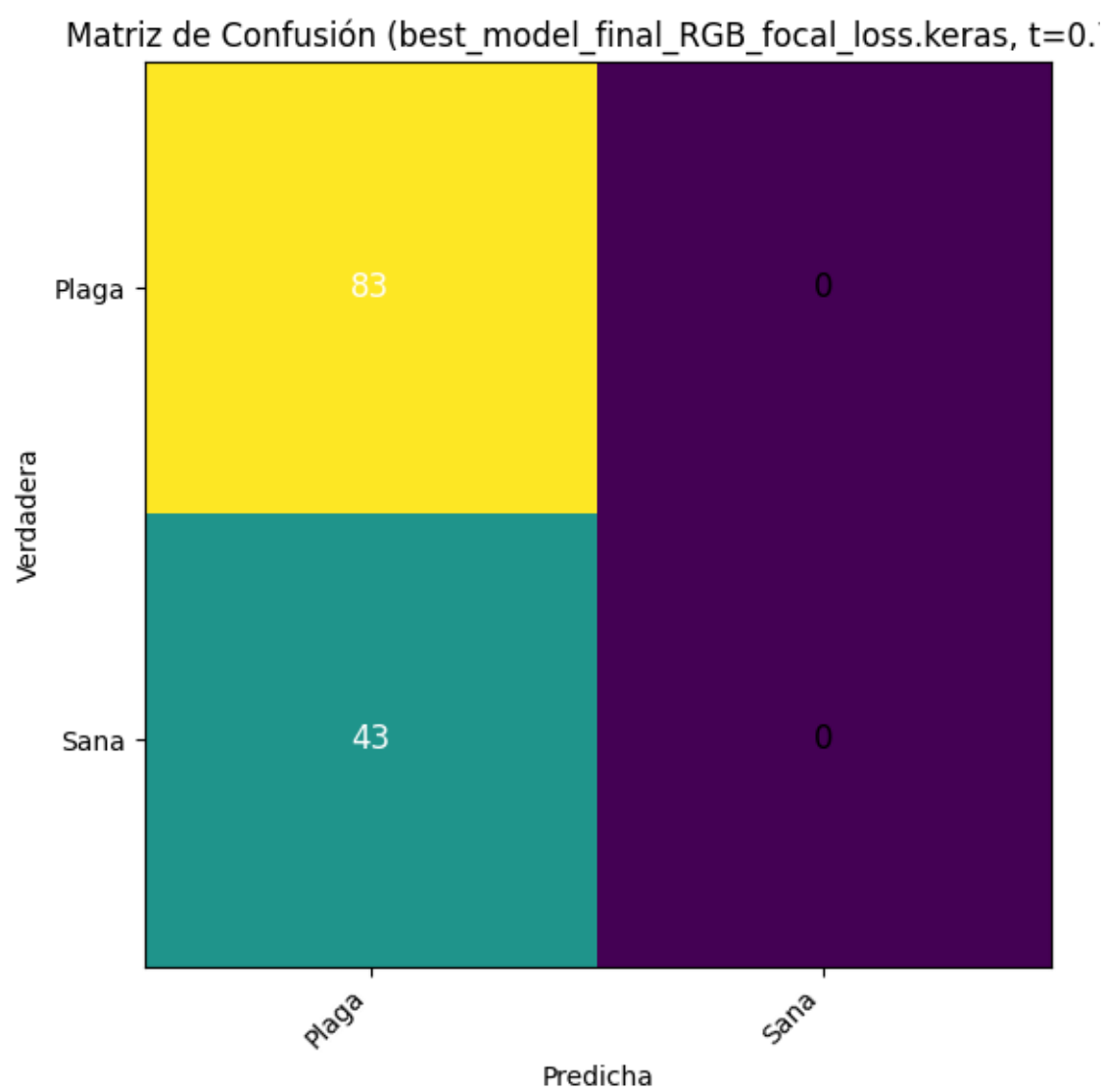


Figure 24: alt text

- **Modelo:** report_best_model_final_RGB_focal_loss.keras_20260325_1444_t70_confusion.md

```
[[83  0]
 [43  0]]
```

Generado automáticamente por el sistema de detección de plagas.

CNN RGB BINARY CROSS ENTROPY

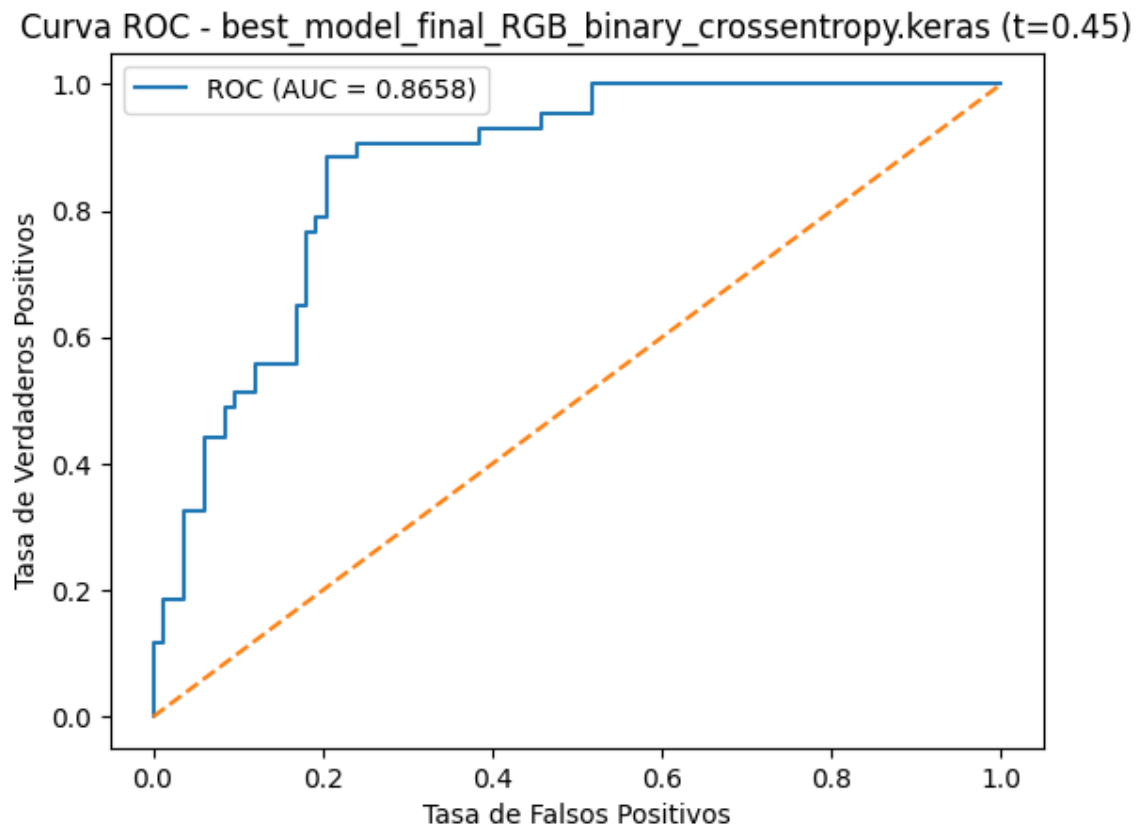


Figure 25: alt text

1 UMBRAL 0.4

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

1.2 Consola:

```
2026-03-25 15:03:37.136602: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 15:03:38.368497: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1503] ---
```

```

--- [T=0.00s] ---
init. Evaluando CNN RGB - Loss: binary_crossentropy - Threshold: 0.45

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=0.00s] ---
1. Extrayendo datos (MISMO pipeline que training)...

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=0.00s] ---
1. Carga de datos cnn RGB

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=0.03s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=0.03s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=0.03s] ---
2.2. Archivos seleccionados: ['RGB.tif']
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

```

Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=38.60s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=37.16s] ---

2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=37.21s] ---

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_binar
2026-03-25 15:04:15.984252: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=37.34s] ---

2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_mode

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=37.29s] ---

3. Prediciendo...

4/4 2s 361ms/step

--- DEBUG PROBABILIDADES ---

MIN: 0.49760857

MAX: 0.50683403

MEAN: 0.502261

PERCENTILES: [0.49760857 0.5005665 0.50172877 0.50449479 0.50683403]

X_val stats:

MIN: 0.0

MAX: 1.0

MEAN: 0.21663862

STD: 0.002675977

--- [Fecha/hora inicio=20260325_1503] ---

--- [T=38.98s] ---

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

```
[[ 0 83]
 [ 0 43]]
```

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.00	0.00	0.00	83
Sana	0.34	1.00	0.51	43

accuracy			0.34	126
macro avg	0.17	0.50	0.25	126
weighted avg	0.12	0.34	0.17	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
 Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
 ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.0,
    "recall": 0.0,
    "f1-score": 0.0,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.3412698412698413,
    "recall": 1.0,
    "f1-score": 0.5088757396449705,
    "support": 43.0
  },
  "accuracy": 0.3412698412698413,
  "macro avg": {
    "precision": 0.17063492063492064,
    "recall": 0.5,
    "f1-score": 0.25443786982248523,
    "support": 126.0
  },
  "weighted avg": {
    "precision": 0.11646510456034266,
    "recall": 0.3412698412698413,
    "f1-score": 0.17366394289471215,
    "support": 126.0
  }
}
```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_binary_crossentropy.keras

- **Fecha:** 2026-03-25 15:04:57
- **Modelo:** best_model_final_RGB_binary_crossentropy.keras
- **Umbral de decisión:** 0.45

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.00	0.00	0.00	83
Sana	0.34	1.00	0.51	43
accuracy			0.34	126
macro avg	0.17	0.50	0.25	126
weighted avg	0.12	0.34	0.17	126

Generado automáticamente por el sistema de detección de plagas.

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 15:04:57
- **Modelo:** report_best_model_final_RGB_binary_crossentropy.keras_20260325_1504_t45_confusion.m

```
[[ 0 83]
 [ 0 43]]
```

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.5214

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

2.2 Consola:

```
2026-03-25 15:07:29.572147: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 15:07:30.740660: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1507] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN RGB - Loss: binary_crossentropy - Threshold: 0.5214
```

Matriz de Confusión (best_model_final_RGB_binary_crossentropy.keras,

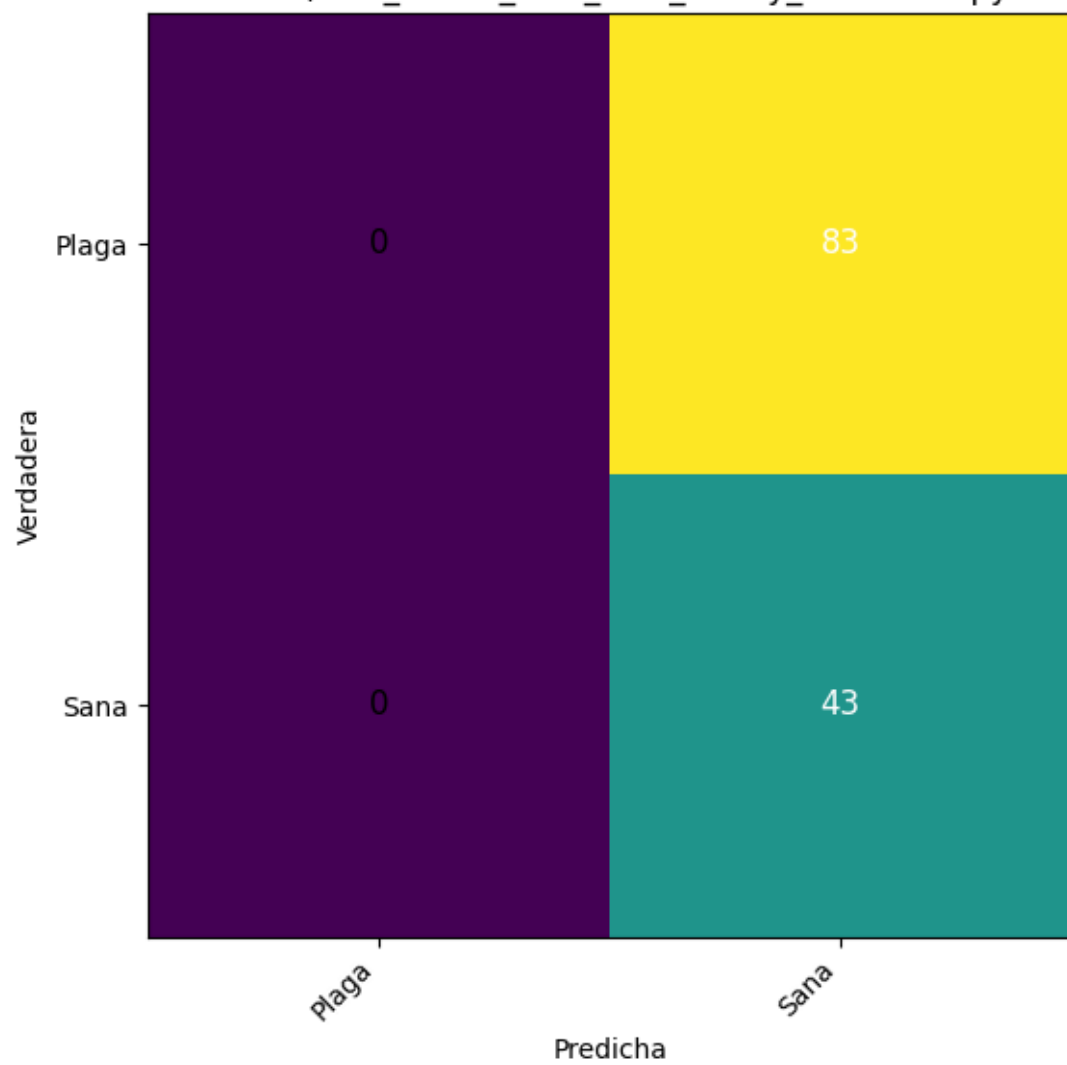


Figure 26: alt text

```

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=0.00s] ---
1. Extrayendo datos (MISMO pipeline que training)...

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=0.00s] ---
1. Carga de datos cnn RGB

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=0.03s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=0.04s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['RGB.tif']
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
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Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

```

Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
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Imagen inválida: 4 canales (esperado 3)
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Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=29.33s] ---

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=27.98s] ---

2. Cargando modelo...

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=28.03s] ---

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_binar
2026-03-25 15:07:59.121480: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=28.15s] ---

2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=28.10s] ---

3. Prediciendo...

4/4 1s 235ms/step

--- DEBUG PROBABILIDADES ---

MIN: 0.49760857

MAX: 0.50683403

MEAN: 0.502261

PERCENTILES: [0.49760857 0.5005665 0.50172877 0.50449479 0.50683403]

X_val stats:

MIN: 0.0

MAX: 1.0

MEAN: 0.21663862

STD: 0.002675977

--- [Fecha/hora inicio=20260325_1507] ---

--- [T=29.34s] ---

4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

[[83 0]

[43 0]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126

weighted avg	0.43	0.66	0.52	126
--------------	------	------	------	-----

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.6587301587301587,
    "recall": 1.0,
    "f1-score": 0.7942583732057417,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.0,
    "recall": 0.0,
    "f1-score": 0.0,
    "support": 43.0
  },
  "accuracy": 0.6587301587301587,
  "macro avg": {
    "precision": 0.32936507936507936,
    "recall": 0.5,
    "f1-score": 0.39712918660287083,
    "support": 126.0
  },
  "weighted avg": {
    "precision": 0.4339254220206601,
    "recall": 0.6587301587301587,
    "f1-score": 0.5232019442545758,
    "support": 126.0
  }
}
```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_binary_crossentropy.keras

- Fecha: 2026-03-25 15:08:02

- **Modelo:** best_model_final_RGB_binary_crossentropy.keras
- **Umbral de decisión:** 0.5214

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Generado automáticamente por el sistema de detección de plagas.

2.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 15:08:02
- **Modelo:** report_best_model_final_RGB_binary_crossentropy.keras_20260325_1508_t52_confusion.m

```
[[83  0]
 [43  0]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

3.2 Consola:

```
2026-03-25 15:09:03.388378: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 15:09:04.587774: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1509] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando CNN RGB - Loss: binary_crossentropy - Threshold: 0.7
```

```
--- [Fecha/hora inicio=20260325_1509] ---
```

Matriz de Confusión (best_model_final_RGB_binary_crossentropy.keras, t=

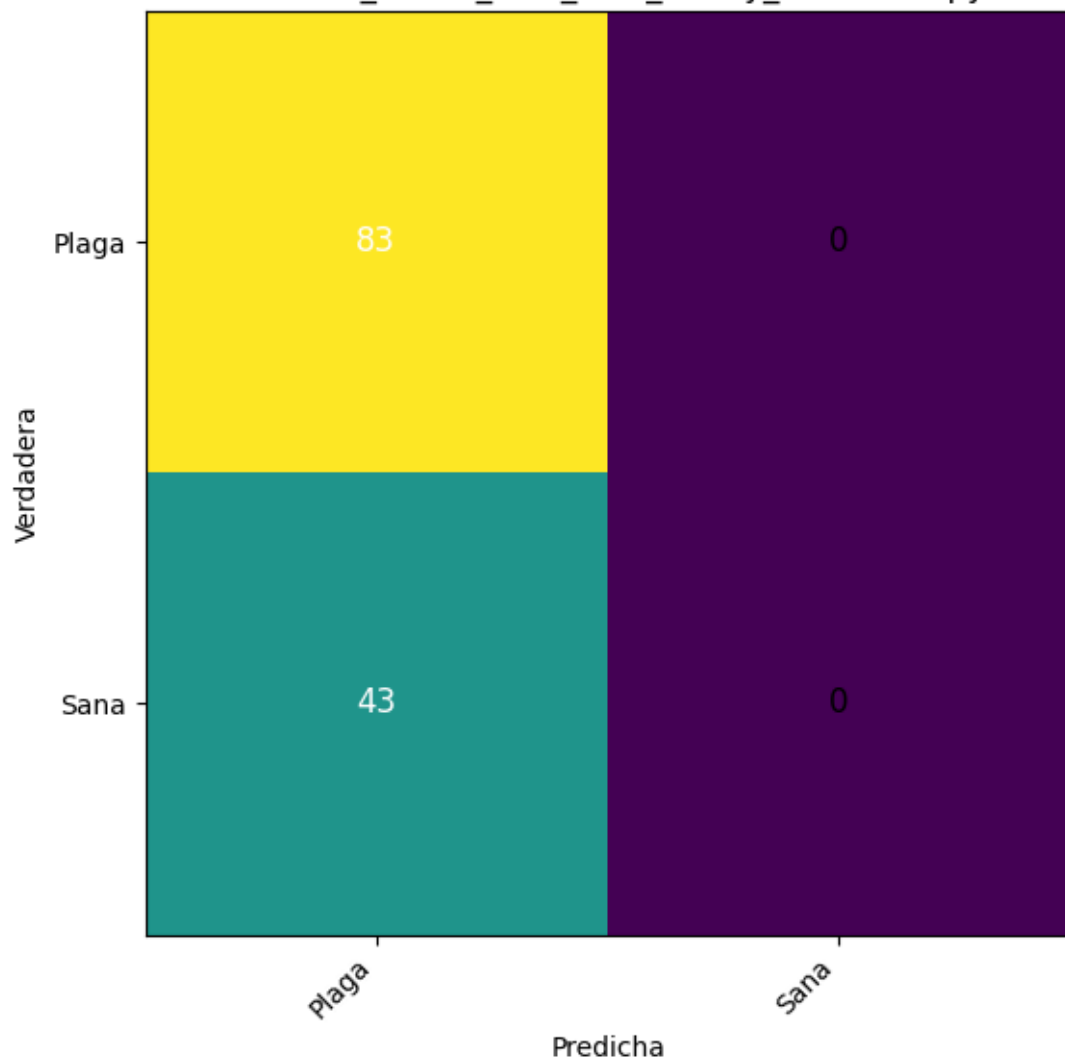


Figure 27: alt text


```
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
```

```
--- [Fecha/hora inicio=20260325_1509] ---
```

```
--- [T=30.83s] ---
```

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

```
--- [Fecha/hora inicio=20260325_1509] ---
```

```
--- [T=29.46s] ---
```

2. Cargando modelo...

```
--- [Fecha/hora inicio=20260325_1509] ---
```

```
--- [T=29.51s] ---
```

1. Cargando modelo desde: src\pest_detection_models\best_models\best_model_final_RGB_binar
2026-03-25 15:09:34.463606: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

```
--- [Fecha/hora inicio=20260325_1509] ---
```

```

--- [T=29.59s] ---
2. Modelo Keras cargado exitosamente desde src\pest_detection_models\best_models\best_model

--- [Fecha/hora inicio=20260325_1509] ---

--- [T=29.55s] ---
3. Prediciendo...
4/4          1s 233ms/step

--- DEBUG PROBABILIDADES ---
MIN: 0.49760857
MAX: 0.50683403
MEAN: 0.502261
PERCENTILES: [0.49760857 0.5005665 0.50172877 0.50449479 0.50683403]
X_val stats:
MIN: 0.0
MAX: 1.0
MEAN: 0.21663862
STD: 0.002675977

--- [Fecha/hora inicio=20260325_1509] ---

--- [T=30.66s] ---
4. Generando métricas...

Reporte de Clasificación guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[83  0]
 [43  0]]

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---
          precision    recall  f1-score   support

     Plaga         0.66         1.00         0.79         83
      Sana         0.00         0.00         0.00         43

 accuracy                   0.66         126
 macro avg         0.33         0.50         0.40         126
weighted avg         0.43         0.66         0.52         126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

```

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```
{  
  "Plaga": {  
    "precision": 0.6587301587301587,  
    "recall": 1.0,  
    "f1-score": 0.7942583732057417,  
    "support": 83.0  
  },  
  "Sana": {  
    "precision": 0.0,  
    "recall": 0.0,  
    "f1-score": 0.0,  
    "support": 43.0  
  },  
  "accuracy": 0.6587301587301587,  
  "macro avg": {  
    "precision": 0.32936507936507936,  
    "recall": 0.5,  
    "f1-score": 0.39712918660287083,  
    "support": 126.0  
  },  
  "weighted avg": {  
    "precision": 0.4339254220206601,  
    "recall": 0.6587301587301587,  
    "f1-score": 0.5232019442545758,  
    "support": 126.0  
  }  
}
```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - best_model_final_RGB_binary_crossentropy.keras

- **Fecha:** 2026-03-25 15:09:37
- **Modelo:** best_model_final_RGB_binary_crossentropy.keras
- **Umbral de decisión:** 0.7

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.66	1.00	0.79	83
Sana	0.00	0.00	0.00	43
accuracy			0.66	126
macro avg	0.33	0.50	0.40	126
weighted avg	0.43	0.66	0.52	126

Generado automáticamente por el sistema de detección de plagas.

Matriz de Confusión (best_model_final_RGB_binary_crossentropy.keras,

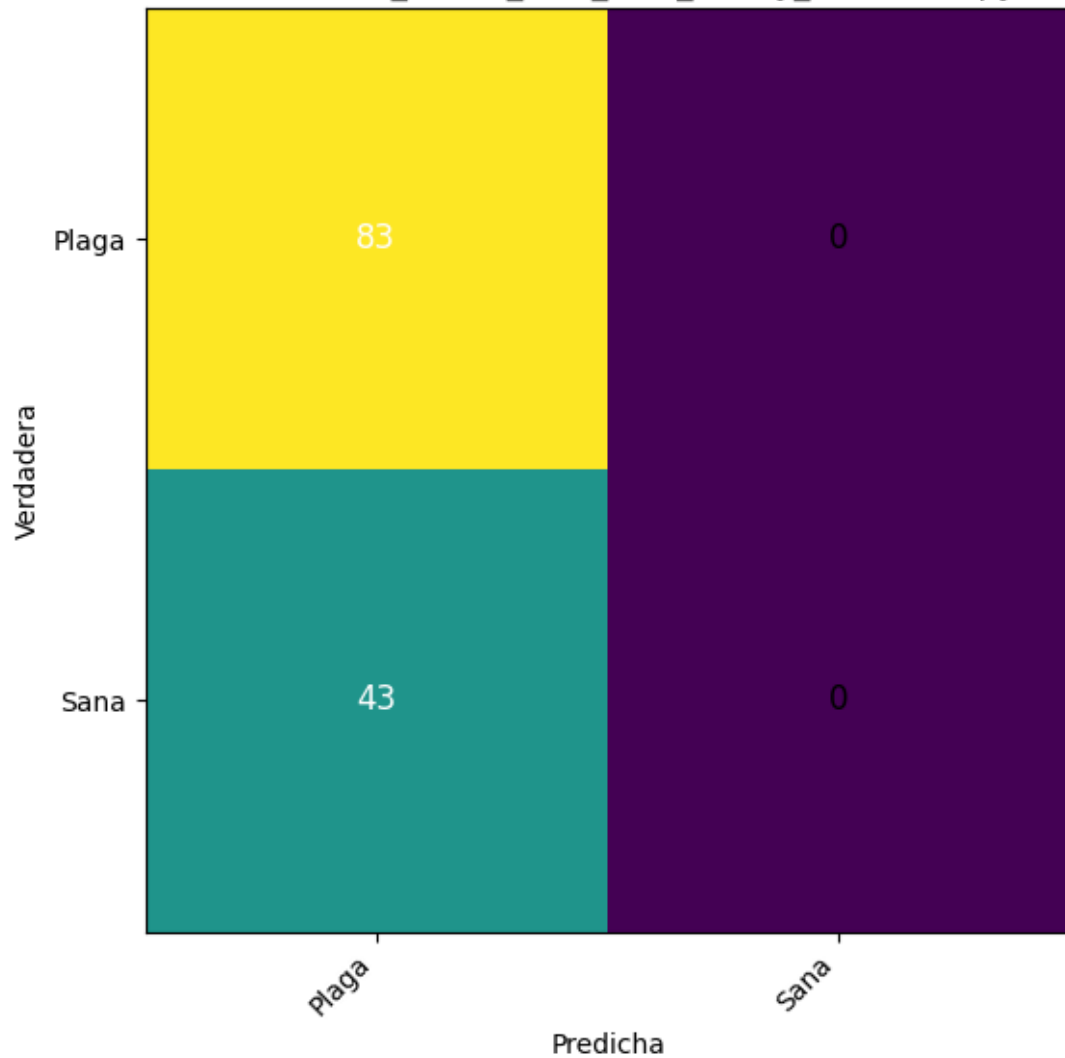


Figure 28: alt text

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 15:09:37
- **Modelo:** report_best_model_final_RGB_binary_crossentropy.keras_20260325_1509_t70_confusion.m

```
[[83  0]  
 [43  0]]
```

Generado automáticamente por el sistema de detección de plagas.

CNN + RANDOM FOREST MULTIESPECTRAL

irva ROC - report_table_best_model_RANDOM_FOREST_MULTIESPECTRAL (t=C

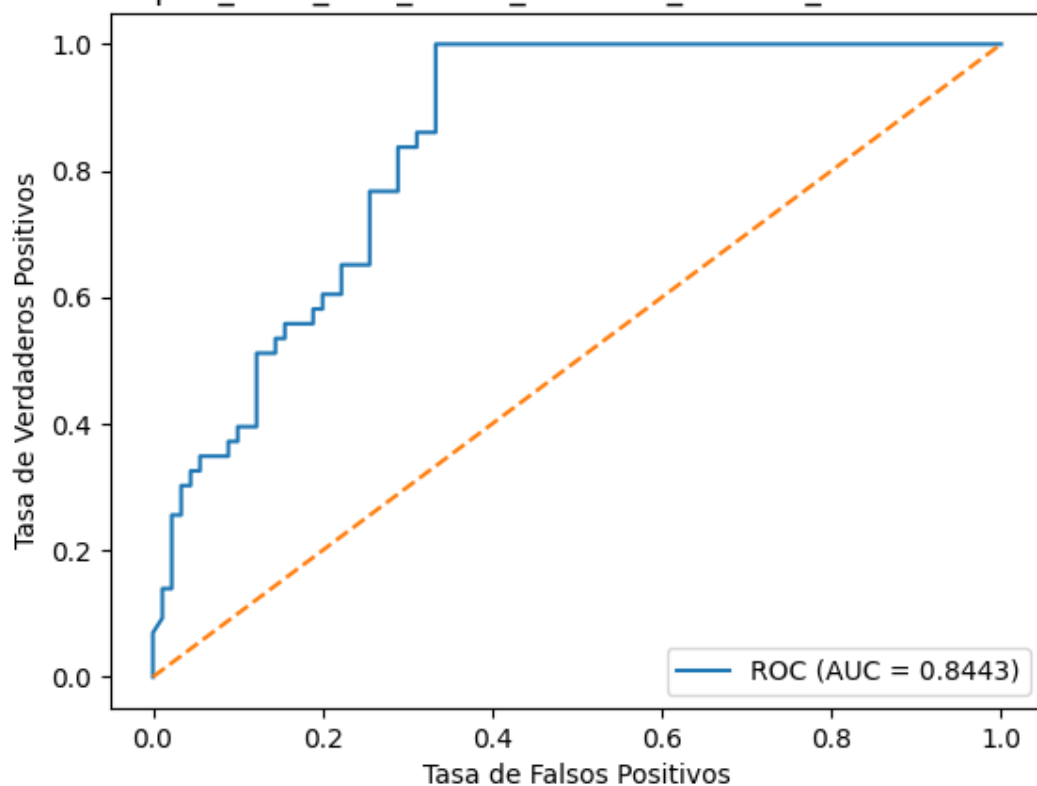


Figure 29: alt text

1 UMBRAL 0.4

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multiespectral\TTADDA-dataset\TTADDA_NARO
```

1.2 Consola:

```
2026-03-25 16:14:24.537635: I tensorflow/core/util/port.cc:153] oneDNN custom operations are  
2026-03-25 16:14:25.706544: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```



```

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=0.00s] ---
init. Evaluando RANDOM FOREST MULTIESPECTRAL

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=0.00s] ---
1. Carga de datos cnn Multiespectral

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=0.03s] ---
2. Filtrado y extracción de datos

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=0.04s] ---
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=57.48s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)
Distribución y_val: {np.int64(0): np.int64(90), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1614] ---

```

```

--- [T=56.58s] ---
2. Cargando modelo...
2026-03-25 16:15:22.847390: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a GPU enabled only on certain device architectures. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- DEBUG RF ---
X_val shape: (133, 224, 224, 5)
X_val min: -3020207.8
X_val max: 1.0

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=56.83s] ---
3. Extrayendo features...
5/5          1s 187ms/step
Features shape: (133, 64)

--- [Fecha/hora inicio=20260325_1614] ---

--- [T=57.97s] ---
4. Prediciendo...

--- DEBUG PROB ---
MIN: 0.0
MAX: 1.0
STD: 0.3479576397721457
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[67 23]
 [13 30]]

--- REPORTE ---

```

	precision	recall	f1-score	support
Plaga	0.84	0.74	0.79	90
Sana	0.57	0.70	0.62	43
accuracy			0.73	133
macro avg	0.70	0.72	0.71	133
weighted avg	0.75	0.73	0.74	133

```

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

```

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.8375,
    "recall": 0.7444444444444445,
    "f1-score": 0.788235294117647,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.5660377358490566,
    "recall": 0.6976744186046512,
    "f1-score": 0.625,
    "support": 43.0
  },
  "accuracy": 0.7293233082706767,
  "macro avg": {
    "precision": 0.7017688679245283,
    "recall": 0.7210594315245478,
    "f1-score": 0.7066176470588235,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.7497340048233793,
    "recall": 0.7293233082706767,
    "f1-score": 0.7354599734630695,
    "support": 133.0
  }
}
```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_MULTIESPECTRAL_20260325_

- **Fecha:** 2026-03-25 16:27:04
- **Modelo:** RANDOM_FOREST_MULTIESPECTRAL_20260325_1625
- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.84	0.74	0.79	90
Sana	0.57	0.70	0.62	43
accuracy			0.73	133
macro avg	0.70	0.72	0.71	133
weighted avg	0.75	0.73	0.74	133

Generado automáticamente por el sistema de detección de plagas.

Irva ROC - report_table_best_model_RANDOM_FOREST_MULTIESPECTRAL (t=C

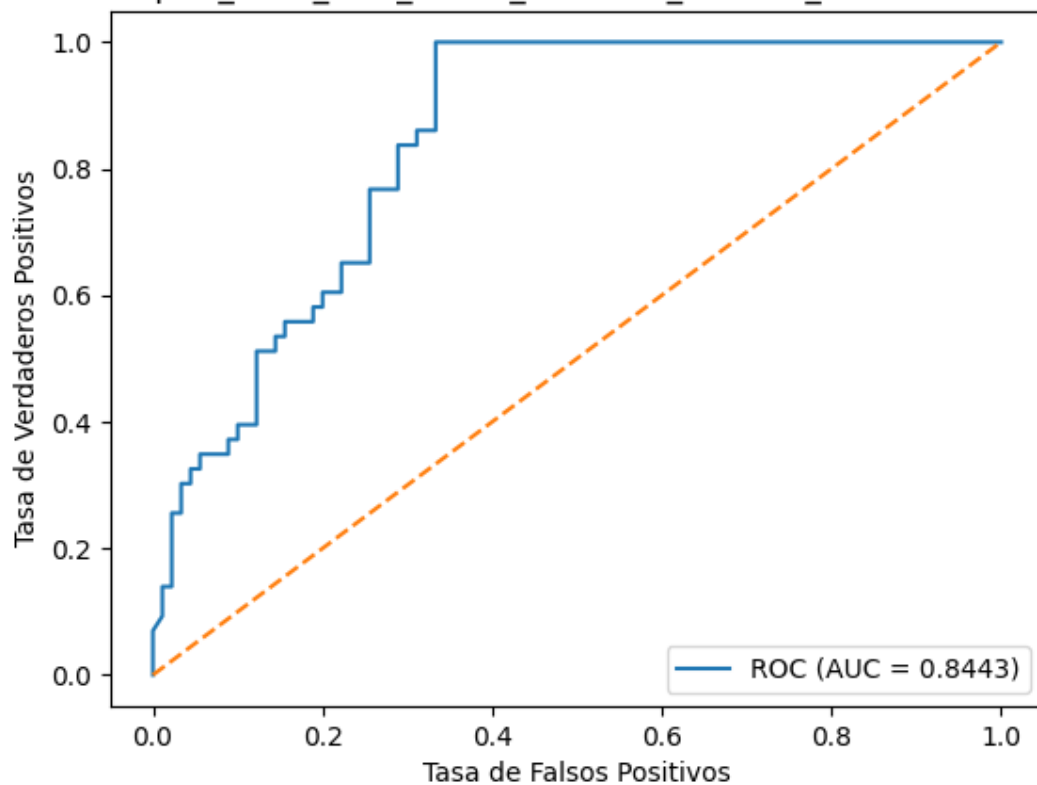


Figure 30: alt text

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 16:27:04
- **Modelo:** report_best_model_RANDOM_FOREST_MULTIESPECTRAL_20260325_1625.md

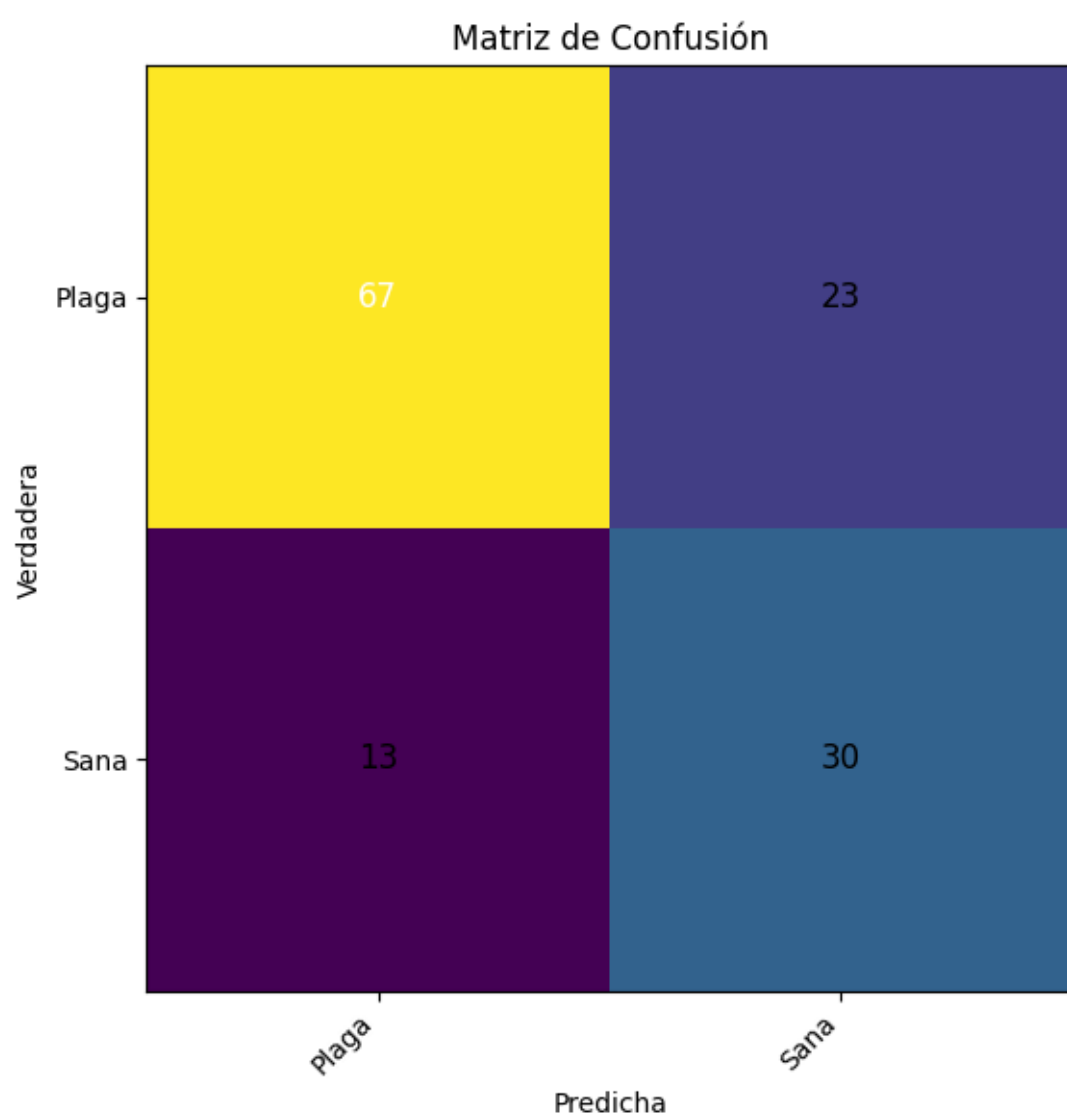


Figure 31: alt text

```
[[67 23]
 [13 30]]
```

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.5214

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

2.2 Consola:

```
2026-03-25 16:30:09.467949: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 16:30:10.768347: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando RANDOM FOREST MULTIESPECTRAL
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn Multispectral
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=0.06s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=0.07s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=0.07s] ---
```

```
2.2. Archivos seleccionados: ['red.tif', 'red edge.tif', 'nir.tif', 'blue.tif', 'green.tif']
```

```
--- [Fecha/hora inicio=20260325_1630] ---
```

```
--- [T=59.86s] ---
```

```
7. Extracción de imágenes completada.
```

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:

Total de parches extraídos: 665

X Train/Val Split: 532 / 133

Y Train/Val Split: 532 / 133

Forma X_train: (532, 224, 224, 5)

Forma Y_train: (532,)

Distribución y_val: {np.int64(0): np.int64(90), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1630] ---

--- [T=58.51s] ---

2. Cargando modelo...

2026-03-25 16:31:10.022663: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a NVIDIA GPU

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- DEBUG RF ---

X_val shape: (133, 224, 224, 5)

X_val min: -3020207.8

X_val max: 1.0

--- [Fecha/hora inicio=20260325_1630] ---

--- [T=58.83s] ---

3. Extrayendo features...

5/5 1s 187ms/step

Features shape: (133, 64)

--- [Fecha/hora inicio=20260325_1630] ---

--- [T=60.13s] ---

4. Prediciendo...

--- DEBUG PROB ---

MIN: 0.0

MAX: 1.0

STD: 0.3479576397721457

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

[[70 20]

[16 27]]

--- REPORTE ---

	precision	recall	f1-score	support
Plaga	0.81	0.78	0.80	90
Sana	0.57	0.63	0.60	43
accuracy			0.73	133
macro avg	0.69	0.70	0.70	133
weighted avg	0.74	0.73	0.73	133

Reporte guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_model\reporte.json'

--- [Fecha/hora inicio=20260325_1630] ---

--- [T=68.81s] ---

4. Evaluación y Reporte de Random Forest...

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.81	0.78	0.80	90
Sana	0.57	0.63	0.60	43
accuracy			0.73	133
macro avg	0.69	0.70	0.70	133
weighted avg	0.74	0.73	0.73	133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\reporte.md

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\roc_curve.png

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\roc.png

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.813953488372093,
    "recall": 0.7777777777777778,
    "f1-score": 0.7954545454545454,
    "support": 90.0
  }
}
```



```

},
"Sana": {
  "precision": 0.574468085106383,
  "recall": 0.627906976744186,
  "f1-score": 0.6,
  "support": 43.0
},
"accuracy": 0.7293233082706767,
"macro avg": {
  "precision": 0.694210786739238,
  "recall": 0.702842377260982,
  "f1-score": 0.6977272727272728,
  "support": 133.0
},
"weighted avg": {
  "precision": 0.7365258767899462,
  "recall": 0.7293233082706767,
  "f1-score": 0.7322624743677375,
  "support": 133.0
}
}

```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_MULTIESPECTRAL_20260325_

- **Fecha:** 2026-03-25 16:31:20
- **Modelo:** RANDOM_FOREST_MULTIESPECTRAL_20260325_1630
- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.81	0.78	0.80	90
Sana	0.57	0.63	0.60	43
accuracy			0.73	133
macro avg	0.69	0.70	0.70	133
weighted avg	0.74	0.73	0.73	133

Generado automáticamente por el sistema de detección de plagas.

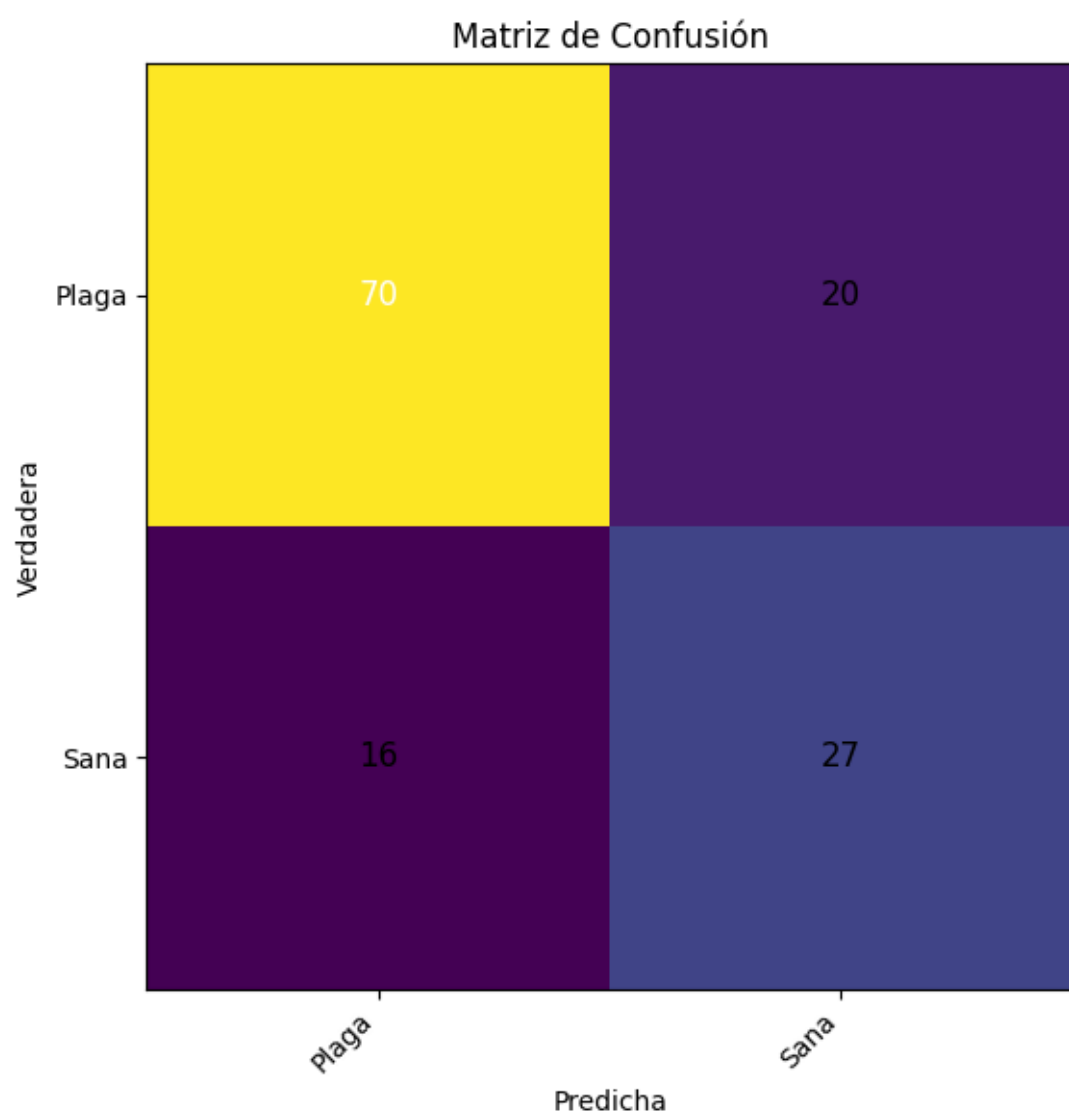


Figure 32: alt text

2.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 16:31:20
- **Modelo:** report_best_model_RANDOM_FOREST_MULTIESPECTRAL_20260325_1630.md

```
[[70 20]
 [16 27]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multiespectral\TTADDA-dataset\TTADDA_NARO
```

3.2 Consola:

```
2026-03-25 16:32:50.536965: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 16:32:52.101914: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1632] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando RANDOM FOREST MULTIESPECTRAL
```

```
--- [Fecha/hora inicio=20260325_1632] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn Multiespectral
```

```
--- [Fecha/hora inicio=20260325_1632] ---
```

```
--- [T=0.04s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1632] ---
```

```
--- [T=0.04s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```
--- [Fecha/hora inicio=20260325_1632] ---
```

```

--- [T=0.04s] ---
2.2. Archivos seleccionados: ['red.tif', 'red_edge.tif', 'nir.tif', 'blue.tif', 'green.tif']

--- [Fecha/hora inicio=20260325_1632] ---

--- [T=61.02s] ---
7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 665

Resumen de Datos Multiespectrales:
Total de parches extraídos: 665
X Train/Val Split: 532 / 133
Y Train/Val Split: 532 / 133
Forma X_train: (532, 224, 224, 5)
Forma Y_train: (532,)
Distribución y_val: {np.int64(0): np.int64(90), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1632] ---

--- [T=59.50s] ---
2. Cargando modelo...
2026-03-25 16:33:52.276349: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized with a variety of CPU instructions available on the current platform. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- DEBUG RF ---
X_val shape: (133, 224, 224, 5)
X_val min: -3020207.8
X_val max: 1.0

--- [Fecha/hora inicio=20260325_1632] ---

--- [T=59.84s] ---
3. Extrayendo features...
5/5          1s 177ms/step
Features shape: (133, 64)

--- [Fecha/hora inicio=20260325_1632] ---

--- [T=61.05s] ---
4. Prediciendo...

--- DEBUG PROB ---

```

MIN: 0.0
MAX: 1.0
STD: 0.3479576397721457

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

```
[[79 11]
 [25 18]]
```

--- REPORTE ---

	precision	recall	f1-score	support
Plaga	0.76	0.88	0.81	90
Sana	0.62	0.42	0.50	43
accuracy			0.73	133
macro avg	0.69	0.65	0.66	133
weighted avg	0.71	0.73	0.71	133

Reporte guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_mod

--- [Fecha/hora inicio=20260325_1632] ---

--- [T=69.54s] ---

4. Evaluación y Reporte de Random Forest...

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.76	0.88	0.81	90
Sana	0.62	0.42	0.50	43
accuracy			0.73	133
macro avg	0.69	0.65	0.66	133
weighted avg	0.71	0.73	0.71	133

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```

{
  "Plaga": {
    "precision": 0.7596153846153846,
    "recall": 0.8777777777777778,
    "f1-score": 0.8144329896907216,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.6206896551724138,
    "recall": 0.4186046511627907,
    "f1-score": 0.5,
    "support": 43.0
  },
  "accuracy": 0.7293233082706767,
  "macro avg": {
    "precision": 0.6901525198938991,
    "recall": 0.6481912144702843,
    "f1-score": 0.6572164948453608,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.7146995472766797,
    "recall": 0.7293233082706767,
    "f1-score": 0.7127742035501125,
    "support": 133.0
  }
}

```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_MULTIESPECTRAL_20260325_

- **Fecha:** 2026-03-25 16:34:02
- **Modelo:** RANDOM_FOREST_MULTIESPECTRAL_20260325_1632
- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.76	0.88	0.81	90
Sana	0.62	0.42	0.50	43

accuracy			0.73	133
macro avg	0.69	0.65	0.66	133
weighted avg	0.71	0.73	0.71	133

Generado automáticamente por el sistema de detección de plagas.

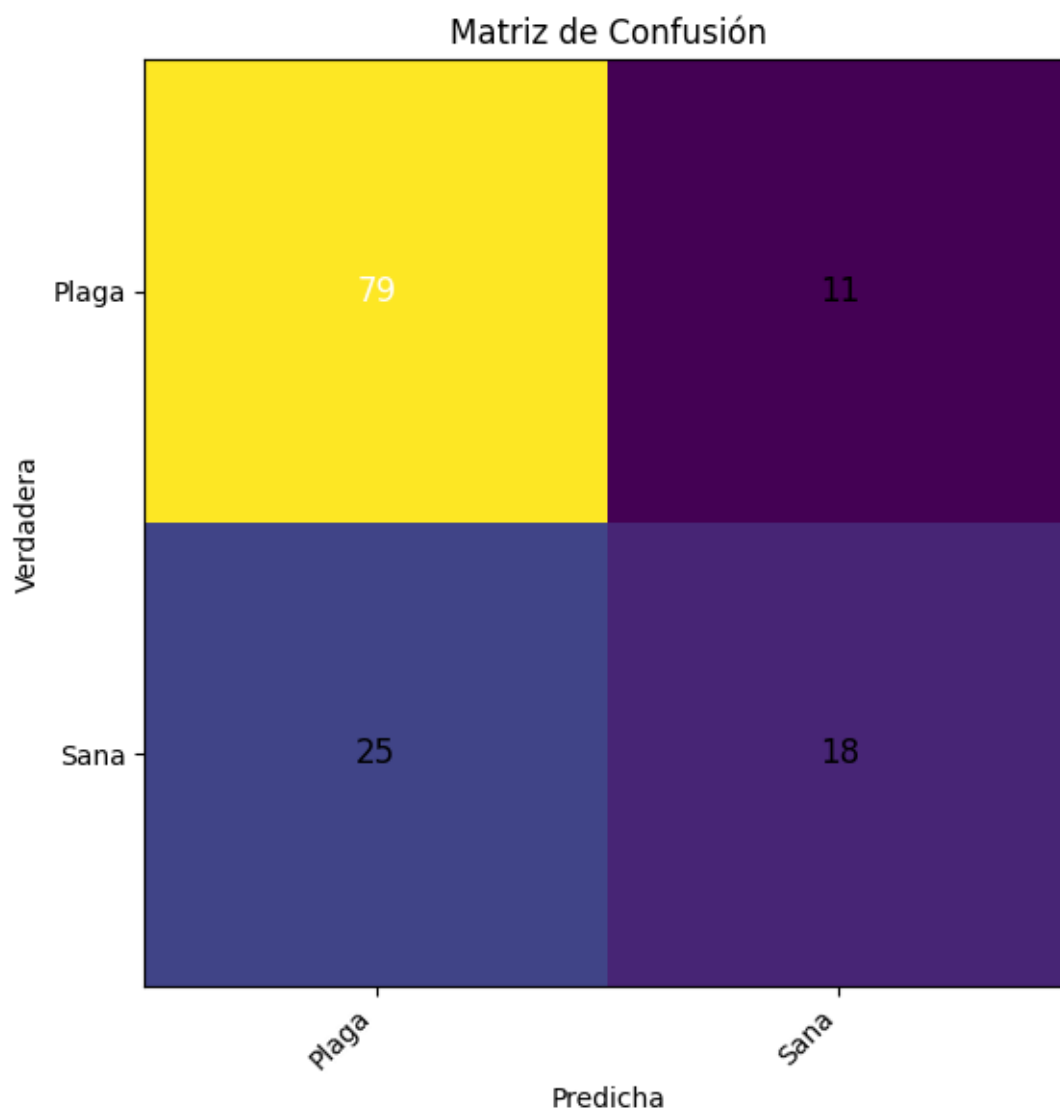


Figure 33: alt text

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 16:34:02
- **Modelo:** report_best_model_RANDOM_FOREST_MULTIESPECTRAL_20260325_1632.md

```
[[79 11]
 [25 18]]
```

Generado automáticamente por el sistema de detección de plagas.

CNN + RANDOM FOREST RGB

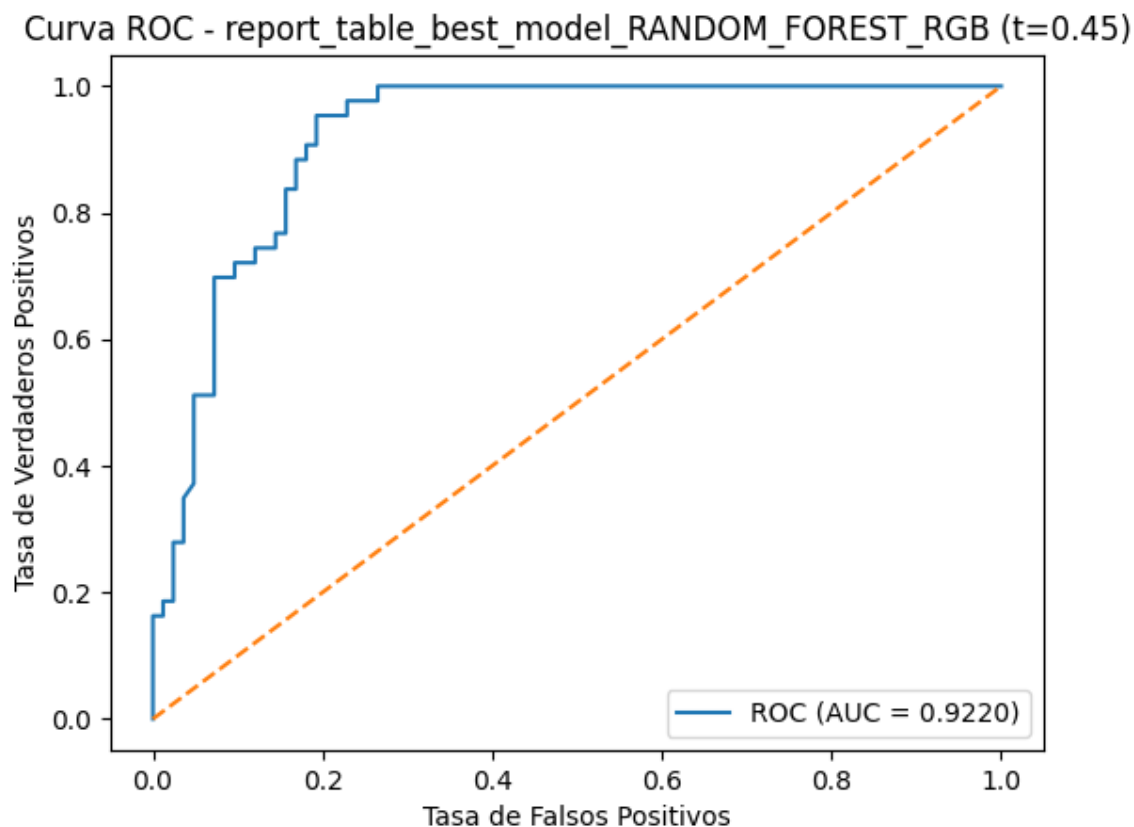


Figure 34: alt text

1 UMBRAL 0.45

1.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

1.2 Consola:

```
2026-03-25 16:38:44.481041: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 16:38:45.737168: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1638] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando RANDOM FOREST RGB
```

```
--- [Fecha/hora inicio=20260325_1638] ---
```



```
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
Imagen inválida: 4 canales (esperado 3)
```

```
--- [Fecha/hora inicio=20260325_1638] ---
```

```
--- [T=29.26s] ---
```

7. Extracción de imágenes completada.

Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

```
--- [Fecha/hora inicio=20260325_1638] ---
```

```
--- [T=27.95s] ---
```

2. Cargando modelo...

2026-03-25 16:39:14.220507: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

```
--- DEBUG RF ---
```

X_val shape: (126, 224, 224, 3)

X_val min: 0.0

X_val max: 1.0

```
--- [Fecha/hora inicio=20260325_1638] ---
```

```
--- [T=28.18s] ---
```

3. Extrayendo features...

4/4 1s 224ms/step

Features shape: (126, 64)

--- [Fecha/hora inicio=20260325_1638] ---

--- [T=29.22s] ---

4. Prediciendo...

--- DEBUG PROB ---

MIN: 0.0

MAX: 1.0

STD: 0.3760748937006677

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

[[69 14]

[6 37]]

--- REPORTE ---

	precision	recall	f1-score	support
Plaga	0.92	0.83	0.87	83
Sana	0.73	0.86	0.79	43
accuracy			0.84	126
macro avg	0.82	0.85	0.83	126
weighted avg	0.85	0.84	0.84	126

Reporte guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_mod

--- [Fecha/hora inicio=20260325_1638] ---

--- [T=33.37s] ---

4. Evaluación y Reporte de Random Forest...

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.92	0.83	0.87	83
Sana	0.73	0.86	0.79	43
accuracy			0.84	126
macro avg	0.82	0.85	0.83	126
weighted avg	0.85	0.84	0.84	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

1.3 Resultados:

1.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.92,
    "recall": 0.8313253012048193,
    "f1-score": 0.8734177215189873,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.7254901960784313,
    "recall": 0.8604651162790697,
    "f1-score": 0.7872340425531915,
    "support": 43.0
  },
  "accuracy": 0.8412698412698413,
  "macro avg": {
    "precision": 0.8227450980392157,
    "recall": 0.8458952087419445,
    "f1-score": 0.8303258820360895,
    "support": 126.0
  },
  "weighted avg": {
    "precision": 0.8536196700902584,
    "recall": 0.8412698412698413,
    "f1-score": 0.8440058310782792,
    "support": 126.0
  }
}
```

1.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_RGB_20260325_1638

- **Fecha:** 2026-03-25 16:39:19
- **Modelo:** RANDOM_FOREST_RGB_20260325_1638

- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.92	0.83	0.87	83
Sana	0.73	0.86	0.79	43
accuracy			0.84	126
macro avg	0.82	0.85	0.83	126
weighted avg	0.85	0.84	0.84	126

Generado automáticamente por el sistema de detección de plagas.

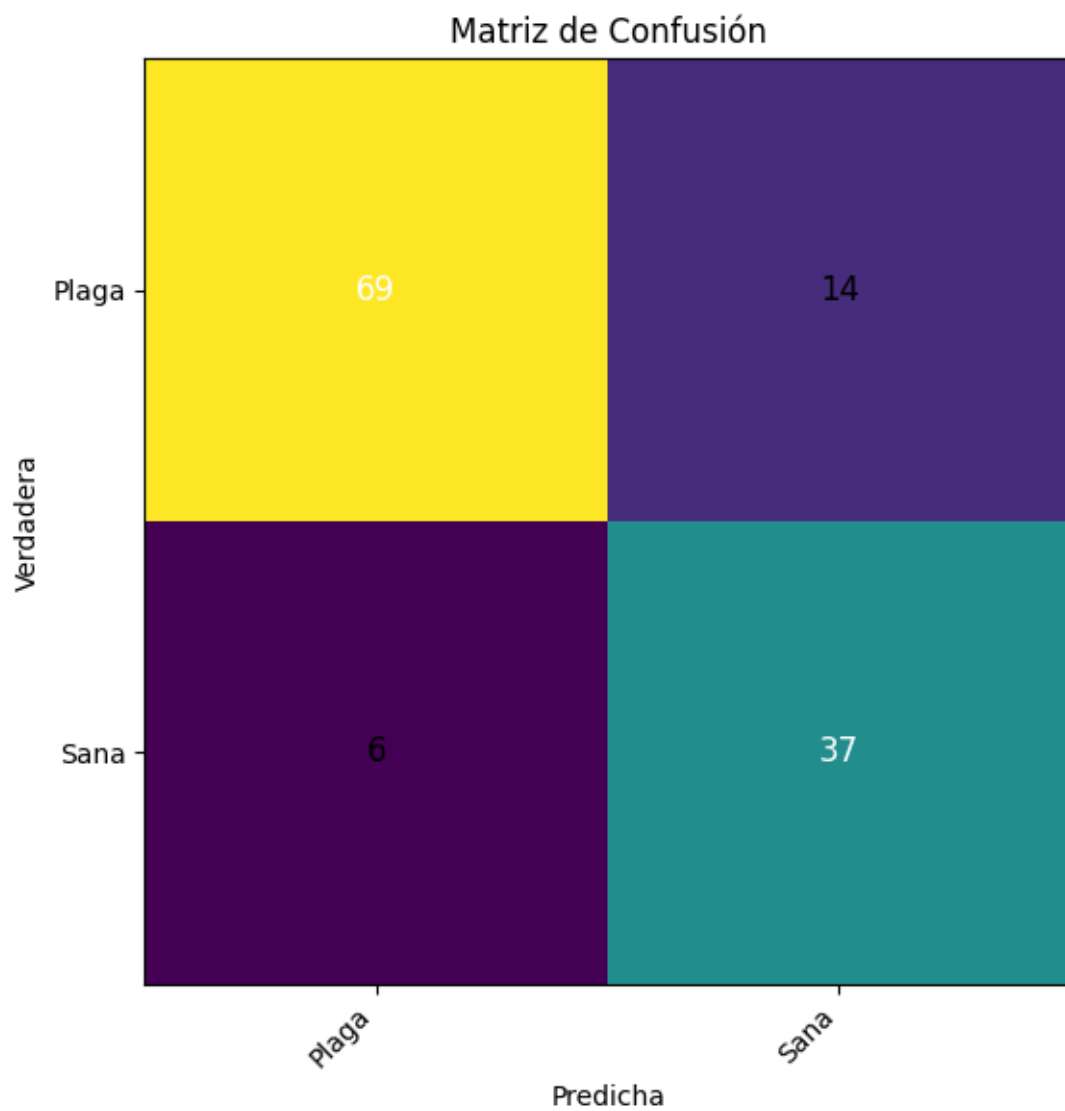


Figure 35: alt text

1.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 16:39:19
- **Modelo:** report_best_model_RANDOM_FOREST_RGB_20260325_1638.md

```
[[69 14]
 [ 6 37]]
```

Generado automáticamente por el sistema de detección de plagas.

2 UMBRAL 0.5214

2.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO
```

2.2 Consola:

```
2026-03-25 16:41:08.598837: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
2026-03-25 16:41:09.830692: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1641] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando RANDOM FOREST RGB
```

```
--- [Fecha/hora inicio=20260325_1641] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn RGB
```

```
--- [Fecha/hora inicio=20260325_1641] ---
```

```
--- [T=0.04s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1641] ---
```

```
--- [T=0.04s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```
--- [Fecha/hora inicio=20260325_1641] ---
```


Datos listos para entrenamiento. Total de muestras: 630

Resumen de Datos Multiespectrales:

Total de parches extraídos: 630

X Train/Val Split: 504 / 126

Y Train/Val Split: 504 / 126

Forma X_train: (504, 224, 224, 3)

Forma Y_train: (504,)

Distribución y_val: {np.int64(0): np.int64(83), np.int64(1): np.int64(43)}

--- [Fecha/hora inicio=20260325_1641] ---

--- [T=29.30s] ---

2. Cargando modelo...

2026-03-25 16:41:39.702051: I tensorflow/core/platform/cpu_feature_guard.cc:210] This Tensor

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other op

--- DEBUG RF ---

X_val shape: (126, 224, 224, 3)

X_val min: 0.0

X_val max: 1.0

--- [Fecha/hora inicio=20260325_1641] ---

--- [T=29.53s] ---

3. Extrayendo features...

4/4 1s 234ms/step

Features shape: (126, 64)

--- [Fecha/hora inicio=20260325_1641] ---

--- [T=30.64s] ---

4. Prediciendo...

--- DEBUG PROB ---

MIN: 0.0

MAX: 1.0

STD: 0.3760748937006677

Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_

Matriz de confusión:

[[70 13]

[9 34]]

--- REPORTE ---

	precision	recall	f1-score	support
Plaga	0.89	0.84	0.86	83
Sana	0.72	0.79	0.76	43
accuracy			0.83	126
macro avg	0.80	0.82	0.81	126
weighted avg	0.83	0.83	0.83	126

Reporte guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_model\reporte.json'

--- [Fecha/hora inicio=20260325_1641] ---

--- [T=35.17s] ---

4. Evaluación y Reporte de Random Forest...

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.89	0.84	0.86	83
Sana	0.72	0.79	0.76	43
accuracy			0.83	126
macro avg	0.80	0.82	0.81	126
weighted avg	0.83	0.83	0.83	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\reporte_md\reporte_md.md

Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\roc\roc.png

ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\roc\roc.png

2.3 Resultados:

2.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.8860759493670886,
    "recall": 0.8433734939759037,
    "f1-score": 0.8641975308641975,
    "support": 83.0
  },
  "Sana": {
    "precision": 0.72,
    "recall": 0.79,
    "f1-score": 0.76,
    "support": 43.0
  }
}
```

```

"Sana": {
  "precision": 0.723404255319149,
  "recall": 0.7906976744186046,
  "f1-score": 0.7555555555555555,
  "support": 43.0
},
"accuracy": 0.8253968253968254,
"macro avg": {
  "precision": 0.8047401023431188,
  "recall": 0.8170355841972541,
  "f1-score": 0.8098765432098765,
  "support": 126.0
},
"weighted avg": {
  "precision": 0.830561006160252,
  "recall": 0.8253968253968254,
  "f1-score": 0.8271213011953753,
  "support": 126.0
}
}

```

2.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_RGB_20260325_1641

- **Fecha:** 2026-03-25 16:41:45
- **Modelo:** RANDOM_FOREST_RGB_20260325_1641
- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.89	0.84	0.86	83
Sana	0.72	0.79	0.76	43
accuracy			0.83	126
macro avg	0.80	0.82	0.81	126
weighted avg	0.83	0.83	0.83	126

Generado automáticamente por el sistema de detección de plagas.

2.3.3 Matriz de confusión

Matriz de confusión

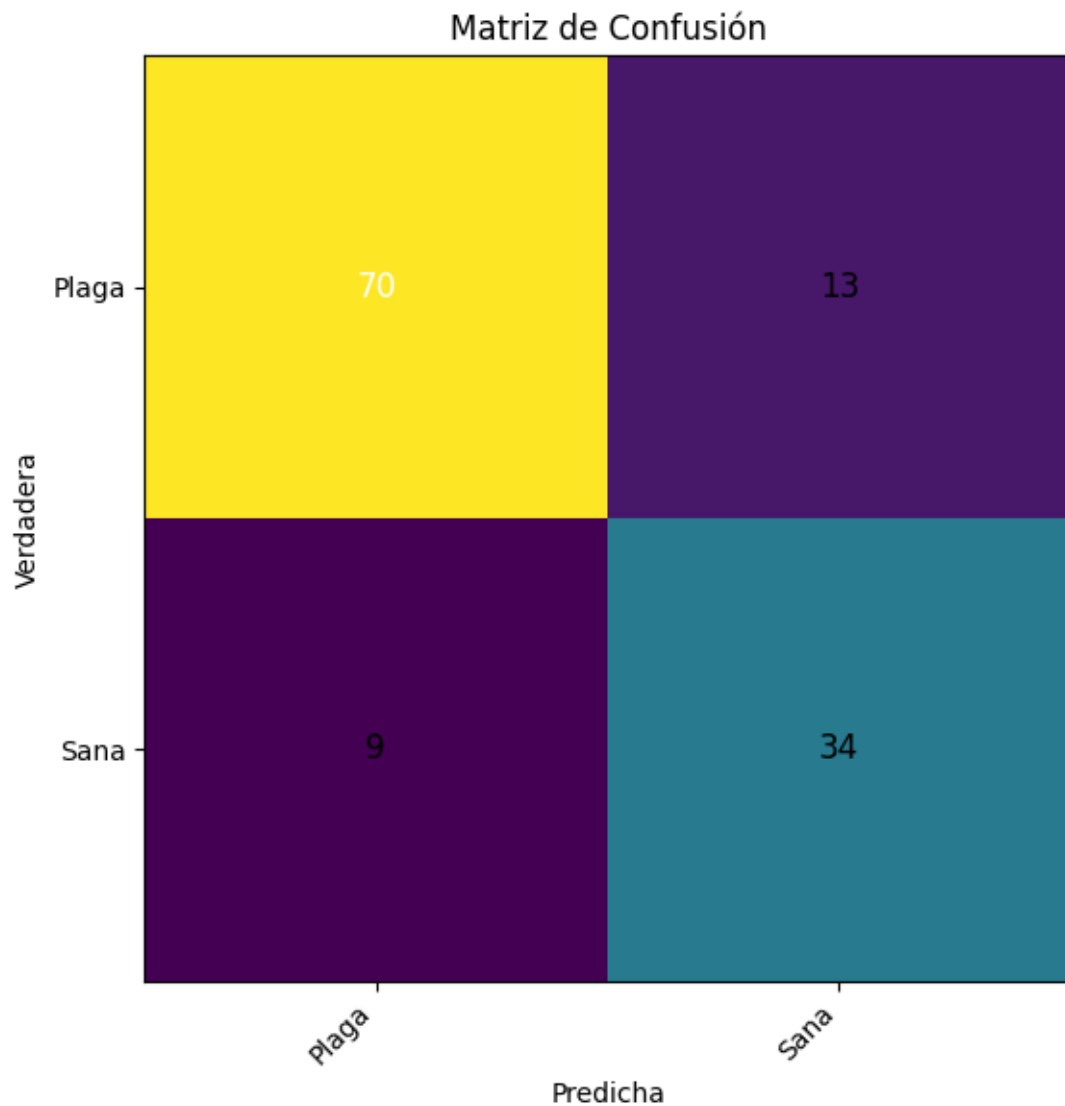


Figure 36: alt text

- **Fecha:** 2026-03-25 16:41:45
- **Modelo:** report_best_model_RANDOM_FOREST_RGB_20260325_1641.md

```
[[70 13]  
 [ 9 34]]
```

Generado automáticamente por el sistema de detección de plagas.

3 UMBRAL 0.70

3.1 Comando:

```
python src\pest_detection_models\evaluate.py data\multispectral\TTADDA-dataset\TTADDA_NARO.
```

3.2 Consola:

```
2026-03-25 16:42:42.739415: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2026-03-25 16:42:44.205156: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
```

```
--- [Fecha/hora inicio=20260325_1642] ---
```

```
--- [T=0.00s] ---
```

```
init. Evaluando RANDOM FOREST RGB
```

```
--- [Fecha/hora inicio=20260325_1642] ---
```

```
--- [T=0.00s] ---
```

```
1. Carga de datos cnn RGB
```

```
--- [Fecha/hora inicio=20260325_1642] ---
```

```
--- [T=0.04s] ---
```

```
2. Filtrado y extracción de datos
```

```
--- [Fecha/hora inicio=20260325_1642] ---
```

```
--- [T=0.04s] ---
```

```
2.1. Iniciando procesamiento de 665 imágenes etiquetadas...
```

```
--- [Fecha/hora inicio=20260325_1642] ---
```

```
--- [T=0.04s] ---
```

```
2.2. Archivos seleccionados: ['RGB.tif']
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```

```
Imagen inválida: 4 canales (esperado 3)
```



```

--- DEBUG RF ---
X_val shape: (126, 224, 224, 3)
X_val min: 0.0
X_val max: 1.0

--- [Fecha/hora inicio=20260325_1642] ---

--- [T=31.57s] ---
3. Extrayendo features...
4/4          1s 250ms/step
Features shape: (126, 64)

--- [Fecha/hora inicio=20260325_1642] ---

--- [T=32.67s] ---
4. Prediciendo...

--- DEBUG PROB ---
MIN: 0.0
MAX: 1.0
STD: 0.3760748937006677
Matriz de Confusión guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_
Matriz de confusión:
[[77  6]
 [17 26]]

--- REPORTE ---

```

	precision	recall	f1-score	support
Plaga	0.82	0.93	0.87	83
Sana	0.81	0.60	0.69	43
accuracy			0.82	126
macro avg	0.82	0.77	0.78	126
weighted avg	0.82	0.82	0.81	126

```

Reporte guardado en 'C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_mod

--- [Fecha/hora inicio=20260325_1642] ---

--- [T=34.59s] ---

```

4. Evaluación y Reporte de Random Forest...

--- RESUMEN DEL REPORTE DE CLASIFICACIÓN ---

	precision	recall	f1-score	support
Plaga	0.82	0.93	0.87	83
Sana	0.81	0.60	0.69	43
accuracy			0.82	126
macro avg	0.82	0.77	0.78	126
weighted avg	0.82	0.82	0.81	126

Reporte Markdown (Tabla) guardado en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\
Curva ROC guardada en: C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_
ROC guardada en C:\workspace\hd-image-pest-detection-cnn-and-rf\src\pest_detection_models\ev

3.3 Resultados:

3.3.1 Reporte de clasificación json:

```
{
  "Plaga": {
    "precision": 0.7596153846153846,
    "recall": 0.8777777777777778,
    "f1-score": 0.8144329896907216,
    "support": 90.0
  },
  "Sana": {
    "precision": 0.6206896551724138,
    "recall": 0.4186046511627907,
    "f1-score": 0.5,
    "support": 43.0
  },
  "accuracy": 0.7293233082706767,
  "macro avg": {
    "precision": 0.6901525198938991,
    "recall": 0.6481912144702843,
    "f1-score": 0.6572164948453608,
    "support": 133.0
  },
  "weighted avg": {
    "precision": 0.7146995472766797,
    "recall": 0.7293233082706767,
```

```

    "f1-score": 0.7127742035501125,
    "support": 133.0
  }
}

```

3.3.2 Reporte de clasificación tabla

Reporte de Clasificación - RANDOM_FOREST_MULTIESPECTRAL_20260325_1632

- **Fecha:** 2026-03-25 16:34:02
- **Modelo:** RANDOM_FOREST_MULTIESPECTRAL_20260325_1632
- **Umbral de decisión:** por defecto 0.5

Métricas por Clase

	precision	recall	f1-score	support
Plaga	0.76	0.88	0.81	90
Sana	0.62	0.42	0.50	43
accuracy			0.73	133
macro avg	0.69	0.65	0.66	133
weighted avg	0.71	0.73	0.71	133

Generado automáticamente por el sistema de detección de plagas.

3.3.3 Matriz de confusión

Matriz de confusión

- **Fecha:** 2026-03-25 16:34:02
- **Modelo:** report_best_model_RANDOM_FOREST_MULTIESPECTRAL_20260325_1632.md

```

[[79 11]
 [25 18]]

```

Generado automáticamente por el sistema de detección de plagas.

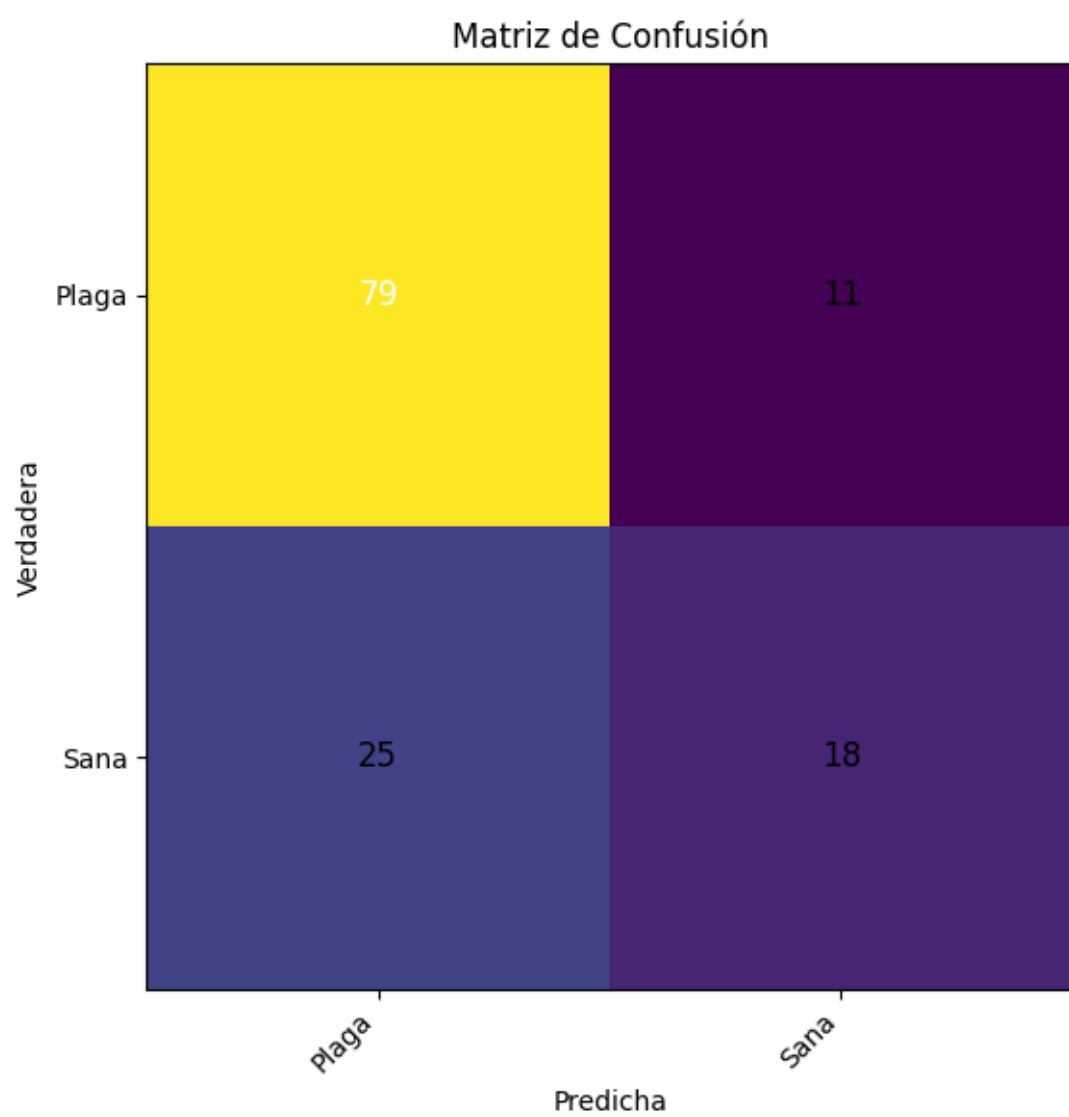


Figure 37: alt text